

Intrinsic Quantum Geometric Tensor on the Grassmannian and Spectral Lower Bounds on Dissipation

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ABSTRACT

Reviewer advisory: this paper was created with AI assistance.

We construct a fully coordinate-free formulation of the Quantum Geometric Tensor on the Grassmannian manifold $\mathrm{Gr}(k, \mathcal{H})$ of rank-k orthogonal projectors associated with an isolated spectral subspace of a self-adjoint operator on a separable Hilbert space $\mathcal{H}$. Starting from Kato's analytic perturbation theory and the Riesz resolvent formula, we define an intrinsic Hermitian form $Q_\Pi(V, W) = \mathrm{Tr}(\Pi V W)$ on the tangent bundle of the Grassmannian. Its symmetric real part is the intrinsic Hilbert–Schmidt quantum metric and its skew-symmetric imaginary part is the Berry curvature 2-form. Every tangent deformation is generated by transitions across the spectral gap, and both components admit exact spectral decompositions in terms of inter-band matrix elements weighted by inverse gaps. We prove the sharp metric–curvature inequality in operator form: the determinant inequality $4 \det g \geq \Omega^2$ of band geometry, extended to any pair of tangent directions on the Grassmannian, with its extremal partner given explicitly. Every bound in the paper is sharp and rigid: equality in the gap bound, the speed limit, the dissipation floor, the curvature bound and the trace

condition each forces one recognisable configuration, and the curvature deficit is an exact quadratic form whose kernel is the complex line of the given direction.

For dissipation, the metric is the quantum Fisher information of the moving subspace, and under an explicit bath model the friction on projector rotations is the metric reweighted by the spectrum, which bounds the excess work of finite-time driving from below by the squared intrinsic length over the time allowed; the friction weight is checked against slow-driving simulations of two different baths. Transport tensors, including the quantum metric dipole, the nonlinear Hall response and the orbital moment, arise as pullbacks of one cross-gap pair sum read with different powers of the gap. Across degenerate strata, geodesics of the pullback metric obey a refraction law whose index is the rate at which the subspace rotates: strata where the subspace stops rotating reflect trajectories, while lines where the gap narrows guide them. A stratum on which the metric collapses along itself is a single projector; a fold is silent only at first order and moves at second, and the bending of every geodesic is a quantum metric dipole. The construction extends to mixed states through the symmetric logarithmic derivative. Applied to public data from a superconducting heat engine and a Bose–Einstein condensate, a three-level standard for extracting angles shows that single eigenvectors trade identity where their gap is unresolved while the subspace keeps it. These data test the stability of the subspace, the reason for building the geometry on the projector, rather than the tensor itself. The Davis–Kahan bound, read as a rule for what can be looked at separately and what only as a whole, ties every result together.

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1 Introduction and Foundational Underpinnings

The variation of eigenstates under deformations of a self-adjoint operator is a central theme in mathematical physics. The rigorous framework for tracking such variations was established by Tosio Kato through his development of analytic perturbation theory for linear operators, building on the functional calculus and resolvent methods of Frigyes Riesz [Kato 1995]. Kato showed that when a finite subset of the spectrum remains isolated from the rest by a contour in the complex plane, the corresponding spectral projector varies analytically with the operator. This projector-valued formulation provides a globally smooth description of spectral subspaces and naturally circumvents the technical difficulties associated with individual, possibly degenerate, eigenvectors.

In modern condensed matter and quantum information physics, the geometric consequences of this spectral sensitivity are captured by the Quantum Geometric Tensor [Provost–Vallee 1980; Berry 1984]. Conventionally, the tensor is introduced parametrically on a manifold of external parameters, such as crystal momentum in the Brillouin zone. Its imaginary part yields the Berry curvature, whose integral over closed manifolds produces topological invariants, while its real part defines a

Riemannian quantum metric that governs fidelity susceptibility, orbital magnetism and Wannier localisation.

Despite its success, the parametric formulation inherently mixes two distinct geometric structures: the intrinsic geometry of the state manifold and the arbitrary choice of coordinates on the parameter space. To disentangle them, we develop a coordinate-free formulation in which the full tensor is defined intrinsically on the tangent bundle of the Grassmannian manifold $\mathrm{Gr}(k, \mathcal{H})$ of rank-k orthogonal projectors. Using the Riesz–Kato resolvent identity, we show that both the quantum metric and the Berry curvature emerge directly from a single canonical Hermitian form on this manifold.

A central result of this approach is that every admissible tangent deformation of a spectral projector is generated exclusively by virtual transitions across the spectral gap. This structural constraint leads to exact spectral decompositions of both the metric and the curvature in terms of interband matrix elements weighted by inverse energy gaps. It also implies a sharp inequality relating the Berry curvature to the quantum metric, establishing that geometric phase effects are quantitatively bounded by the strength of cross-gap susceptibility.

Furthermore, we show that the intrinsic quantum metric controls the cost of non-adiabatic driving. The metric is the quantum Fisher information of the moving subspace; a bath converts that speed into dissipation, and the friction carried by the pairs that rotate the projector is the intrinsic metric reweighted pair by pair by the spectrum and the bath. The excess work generated by finite-rate driving is then bounded from below by the thermodynamic length measured with this metric, with an explicit spectral prefactor. This provides a direct geometric origin for dissipative costs in driven quantum systems.

By anchoring the construction in operator theory rather than band parameters, the framework reveals that physical parameter spaces do not generate quantum geometry; they select finite-dimensional trajectories through a pre-existing Grassmannian geometry. Two consequences follow that are invisible in coordinates. Transport tensors of every order are pullbacks of one cross-gap structure, so that the quantum metric dipole, the nonlinear Hall response and the orbital moment are the same pair sum weighted differently. And where the pullback degenerates, the intrinsic geometry decides what happens: geodesics of the pullback metric refract with an index equal to the rate at which the subspace rotates, is reflected where that rotation stops, and is guided where the gap narrows instead. The kinds of stratum are told apart by order: where the metric has a kernel, the second derivative of the projector is itself a tangent vector, and it decides whether the kernel ever opens.

Every inequality in the paper is sharp, and we determine when each is met. Equality is rigid in every case: it is reached only by couplings of one strength across one gap, by a rigid rotation at fixed gap, by motion at constant intrinsic speed, on a complex line of the Grassmannian, or, for permutations, by involutions. The equality cases are proved alongside their inequalities, and their algebraic cores are checked in Lean.

Several statements here are coordinate-free recastings of established results, and we say so where they appear: the metric–curvature inequality is the determinant inequality of band geometry and its saturation the Kähler or ideal-band condition; the pullback description of the band metric and curvature is standard; the stability of subspaces against perturbation is the Davis–Kahan theorem; the redistribution flow is Brockett's double-bracket flow. Each is included because the projector formulation makes its proof, or its scope, transparent, and because later sections rest on it.

The paper is organised as follows. Section 2 introduces the intrinsic quantum geometric tensor on the Grassmannian. Section 3 derives its explicit spectral realisation. Section 4 shows how conventional parametric expressions arise as pullbacks and where the pullback degenerates. Section 5 establishes the spectral lower bound on dissipation. Section 6 derives the metric–curvature inequality and its geometric consequences, and determines when every bound in the paper is met. Section 7 builds the transport tensors, Section 8 treats geodesic refraction across degenerate strata, and Section 9 extends the construction to mixed states. Section 10 sets out a standard for extracting angles from raw data and applies it to two experiments. Section 11 discusses the results, Section 12 draws them together as one law read at two grains, separately and as a whole, and Section 13 concludes. Technical details and explicit examples are collected in the appendices.

2 Mathematical Framework and Intrinsic Geometric Structures

Let $\mathcal{H}$ be a separable complex Hilbert space and let $\mathrm{Gr}(k, \mathcal{H})$ denote the Grassmannian manifold of rank-k orthogonal projectors on $\mathcal{H}$; it is a real-analytic Banach manifold. An element $\Pi \in \mathrm{Gr}(k, \mathcal{H})$ is a bounded self-adjoint operator satisfying

$$\Pi^\dagger = \Pi = \Pi^2, \qquad \mathrm{Tr}\,\Pi = k,$$

with $k < \infty$ throughout.

2.1 The tangent space

The tangent space $T_\Pi \mathrm{Gr}(k, \mathcal{H})$ at a point Π consists of the self-adjoint operators $V \in \mathcal{B}(\mathcal{H})$ satisfying the linearised constraint obtained by differentiating the projector identity $\Pi^2 = \Pi$. This yields the relation

$$V = \Pi V + V \Pi.$$

Multiplying on both sides by Π immediately gives $\Pi V \Pi = 0$. The same operation with the complementary projector forces $(I - \Pi) V (I - \Pi) = 0$. Consequently, every tangent vector is strictly off-diagonal with respect to the orthogonal decomposition $\mathcal{H} = \mathrm{Ran}(\Pi) \oplus \mathrm{Ran}(I - \Pi)$:

$$V = \Pi V (I - \Pi) + (I - \Pi) V \Pi = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad X = \Pi V (I - \Pi). \tag{2.1}$$

This structural constraint is fundamental: all geometric features of the tangent space arise exclusively from mixing between the occupied spectral subspace and its complement. There are no intra-subspace contributions to the geometry.

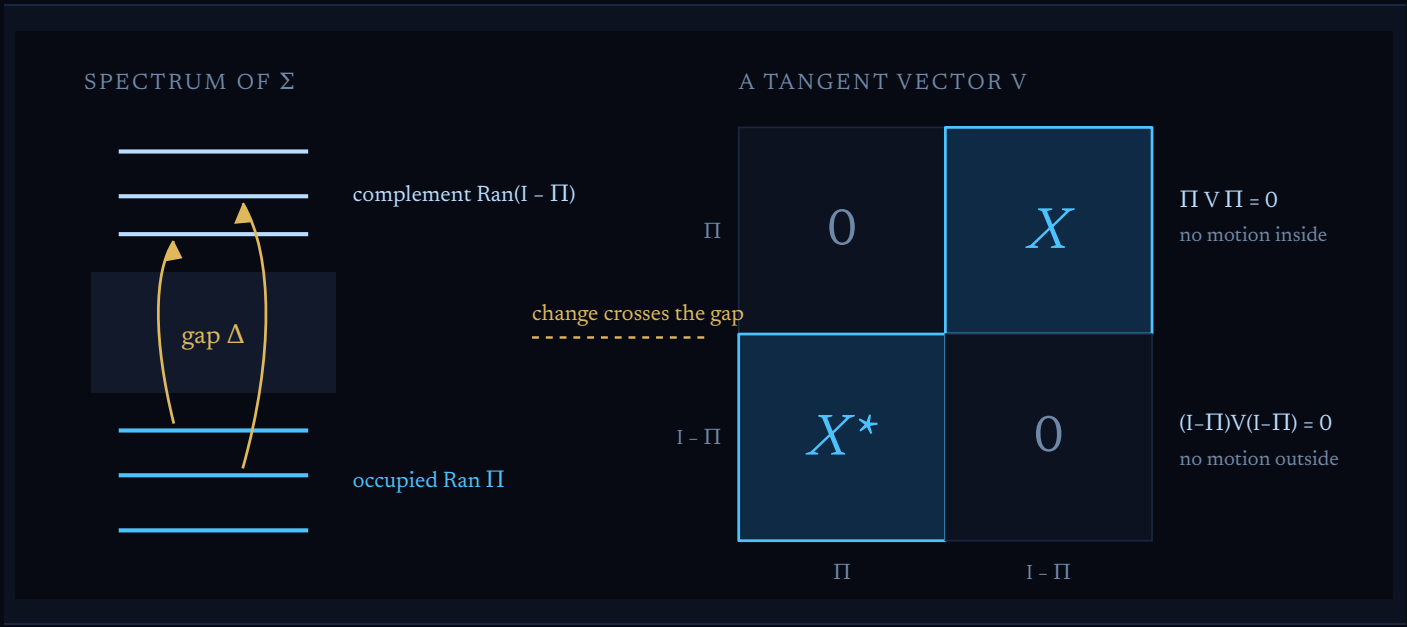


FIGURE 1. Change crosses the gap. Every tangent vector of the Grassmannian is block off-diagonal: the occupied subspace and its complement never deform internally, so all geometry is carried by the cross-gap block X.

LEMMA 2.1 *(Exact characterisation of the tangent space). Let $\Pi \in \mathrm{Gr}(k, \mathcal{H})$ and let V be bounded, self-adjoint and of the form (2.1) for some bounded $X : \mathrm{Ran}(I - \Pi) \to \mathrm{Ran}(\Pi)$. Then V is the velocity at $t = 0$ of a curve in $\mathrm{Gr}(k, \mathcal{H})$ through Π. Consequently $T_\Pi \mathrm{Gr}(k, \mathcal{H})$ is exactly the set of bounded self-adjoint operators of the form (2.1).*

PROOF. Set

$$A = \begin{pmatrix} 0 & iX \\ -iX^* & 0 \end{pmatrix},$$

which is bounded and self-adjoint, and let $\Pi(t) = e^{itA}\,\Pi\,e^{-itA}$. Each $\Pi(t)$ is a rank-k orthogonal projector, and $\dot\Pi(0) = i[A,\Pi] = V$. In finite dimensions the dimension count agrees: an arbitrary complex X of size $k \times (n-k)$ has real dimension $2k(n-k) = \dim_{\mathbb{R}} \mathrm{Gr}(k, \mathbb{C}^n)$. ∎

REMARK 2.2 (Convergence is automatic). $X = \Pi V(I-\Pi)$ factors through $\mathrm{Ran}(\Pi)$ and therefore has rank at most k. A bounded finite-rank operator is trace class, so every trace appearing below converges for any bounded V, with no additional hypothesis. This is the only place finiteness of k is used.

2.2 The Intrinsic Quantum Geometric Tensor

The tangent space is a real vector space: if V is self-adjoint, iV is not. We therefore define the intrinsic Quantum Geometric Tensor, the canonical Hermitian form on the Grassmannian, on the complexification $T_\Pi^{\mathbb{C}} = \{A + iB : \ A, B \in T_\Pi \mathrm{Gr}(k, \mathcal{H})\}$:

$$Q_\Pi(A,B) := \mathrm{Tr}\big(\Pi A^\dagger B\big). \tag{2.2}$$

This form is sesquilinear, reduces to $\mathrm{Tr}(\Pi V W)$ on self-adjoint arguments, and is positive semi-definite:

$$Q_\Pi(A,A) = \mathrm{Tr}\big(\Pi A^\dagger A\,\Pi\big) = \|A\Pi\|_{\mathrm{HS}}^2 \ \geq 0.$$

On the real tangent space it is definite, since there the two off-diagonal blocks are adjoint to one another; on the complexification it is only semi-definite, because a complex tangent vector carrying only its upper block has $A\Pi = 0$.

For $V, W \in T_\Pi \mathrm{Gr}(k, \mathcal{H})$ with blocks X, Y one computes $VW = \mathrm{diag}(XY^*, X^*Y)$, hence $Q_\Pi(V,W) = \mathrm{Tr}(XY^*)$. This complex-valued form decomposes uniquely into a symmetric real part and a skew-symmetric imaginary part,

$$Q_\Pi(V,W) = g_\Pi(V,W) - \tfrac{i}{2}\,\Omega_\Pi(V,W), \tag{2.3}$$

$$g_\Pi(V,W) = \tfrac{1}{2}\,\mathrm{Tr}(VW) = \mathrm{Re}\,\mathrm{Tr}(XY^*), \qquad \Omega_\Pi(V,W) = i\,\mathrm{Tr}\big(\Pi[V,W]\big) = -2\,\mathrm{Im}\,\mathrm{Tr}(XY^*),$$

where g_Π is the intrinsic Hilbert–Schmidt Riemannian metric and Ω_Π is the intrinsic Berry curvature 2-form. These are the signs of the standard parametric convention $Q = g - \frac{i}{2}F$: pulled back along a

family of projectors, Ω_Π is the ordinary Berry curvature $F = dA$, $A = i\langle u|du\rangle$, for $k = 1$, and the trace of the non-abelian curvature for $k > 1$. The metric g_Π quantifies the infinitesimal distinguishability between nearby projectors, while the curvature Ω_Π encodes the geometric phase accumulated along closed paths in the Grassmannian. Both quantities derive from the same underlying form and are therefore linked through the off-diagonal structure of the tangent space.

REMARK 2.3 (Where the Π-dependence lives). The metric $g_\Pi(V, W) = \frac{1}{2}\mathrm{Tr}(VW)$ appears to have forgotten Π. It has not. The identity $\mathrm{Tr}(\Pi VW) + \mathrm{Tr}(\Pi WV) = \mathrm{Tr}(VW)$ is false for general operators; it holds precisely because V and W are off-diagonal with respect to this Π. The dependence has migrated from the formula into its domain.

REMARK 2.4 (The factor $\frac{1}{2}$ is forced). On the Bloch sphere, $\Pi_- = \frac{1}{2}(I - \hat{n}\cdot\boldsymbol{\sigma})$ gives $\frac{1}{2}\mathrm{Tr}(V^2) = \frac{1}{4}|\delta\hat{n}|^2$, hence $g_{\theta\theta} = 1/4$: the Fubini–Study value. The alternative normalisation $\mathrm{Tr}(VW)$ would give $1/2$ and disagree with the parametric literature. The factor is fixed by that agreement, not chosen.

2.3 Riesz–Kato Spectral Projectors

Let Σ be a self-adjoint operator (bounded or unbounded) possessing an isolated spectral subspace of dimension k separated from the remainder of the spectrum by a positive gap $\Delta > 0$. Let Γ_k be a simple closed contour lying in the resolvent set that encloses precisely the eigenvalues associated with this subspace. Write $\mathcal{I}$ for the set of spectral indices inside Γ_k and $\mathcal{I}^c$ for those outside. The corresponding spectral projector is given by the Riesz integral

$$\Pi_k = \frac{1}{2\pi i}\oint_{\Gamma_k} (zI - \Sigma)^{-1}\, dz. \tag{2.4}$$

This projector is independent of the particular choice of contour within the gap and depends analytically on Σ as long as the gap remains open; a contour that crosses the gap encloses a different spectral subspace and returns a different projector. Appendix A treats the unbounded case. All subsequent geometric constructions are formulated directly in terms of Π_k and its tangent variations.

3 Spectral Realization of Tangent Deformations

For clarity, we first derive the explicit spectral formulas in the case of an isolated discrete spectrum; the operator statements extend to the general setting without change (Remark 3.8). To evaluate the intrinsic geometric tensor and its components, the abstract tangent vectors must be resolved into

explicit, computable spectral identities. This section evaluates the contour integrals that define the first-order response of a spectral projector under an arbitrary operator perturbation.

LEMMA 3.1 *(Spectral response identity). Let $\delta\Sigma$ be bounded and self-adjoint, and let $V = \delta\Pi_k$ be the tangent vector it generates through the linear variation of the Riesz formula,*

$$V = \frac{1}{2\pi i} \oint_{\Gamma_k} (zI - \Sigma)^{-1}\, \delta\Sigma\, (zI - \Sigma)^{-1}\, dz. \tag{3.1}$$

Then, in an orthonormal eigenbasis $\{|m\rangle\}$ of Σ,

$$\langle m|V|n\rangle = \frac{\langle m|\delta\Sigma|n\rangle}{\lambda_{\mathrm{in}} - \lambda_{\mathrm{out}}} \quad \text{if exactly one of } m, n \text{ lies in } \mathcal{I}, \qquad \langle m|V|n\rangle = 0 \quad \text{otherwise,} \tag{3.2}$$

where λ_{in} is the eigenvalue of the index inside Γ_k and λ_{out} that of the index outside.

PROOF. Sandwiching the integral between two eigenvectors and pulling the scalar denominators out of the inner product gives

$$\langle m|V|n\rangle = \frac{1}{2\pi i} \oint_{\Gamma_k} \frac{\langle m|\delta\Sigma|n\rangle}{(z - \lambda_m)(z - \lambda_n)}\, dz.$$

The value is fixed by where the two poles sit relative to Γ_k, which encloses precisely the eigenvalues of Π_k. If both lie inside and $\lambda_m \neq \lambda_n$, the residues $1/(\lambda_m - \lambda_n)$ and $1/(\lambda_n - \lambda_m)$ cancel; if $\lambda_m = \lambda_n$ the pole is of order two and its residue vanishes. If both lie outside, the integrand is holomorphic inside Γ_k and the integral vanishes. If exactly one lies inside, its residue is $1/(\lambda_{\mathrm{in}} - \lambda_{\mathrm{out}})$, which gives (3.2). ∎

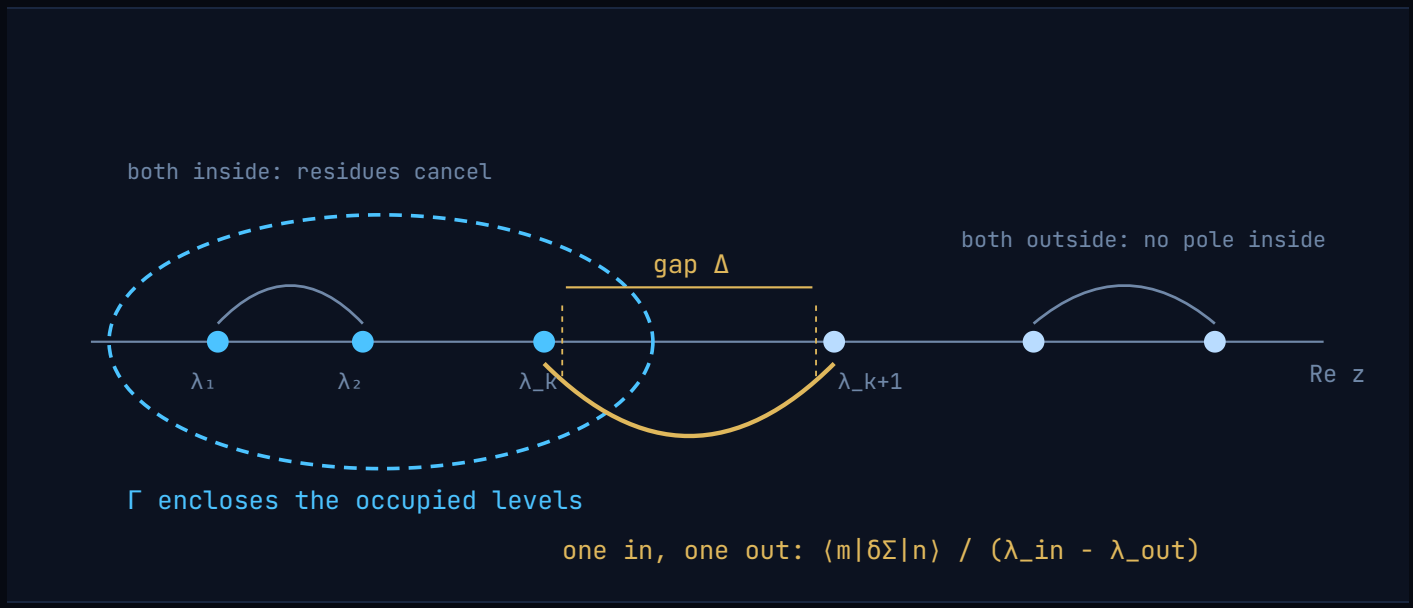


FIGURE 2. Where the tangent vector comes from. The first variation of the Riesz projector is a contour integral with two poles per matrix element. Pairs on the same side of the gap give nothing; each pair straddling it contributes its coupling divided by its own gap.

REMARK 3.2 (Degeneracy). Inside a degenerate block the eigenbasis is not unique, but (3.2) depends only on eigenvalues and on $\boldsymbol{\Pi_k}$, so it is unaffected by the choice. Equal eigenvalues necessarily lie on the same side of $\boldsymbol{\Gamma_k}$, so no cross-gap pair can have a vanishing denominator. This is precisely the work done by the gap hypothesis.

REMARK 3.3 (Self-adjointness). For a cross-gap pair, $\langle n|V|m\rangle = \langle n|\delta\Sigma|m\rangle/(\lambda_{\text{in}} - \lambda_{\text{out}}) = \overline{\langle m|V|n\rangle}$, since the denominator is real. Hence $\boldsymbol{V}$ is self-adjoint, as a tangent vector must be.

COROLLARY 3.4 *(Off-diagonal structure of tangent vectors). The first-order variation $\boldsymbol{V = \delta\Pi_k}$ possesses no matrix elements internal to either spectral sector: $\boldsymbol{\Pi_k V \Pi_k = 0}$ and $\boldsymbol{(I - \Pi_k)V(I - \Pi_k) = 0}$. Consequently $\boldsymbol{V}$ has the block form (2.1), and the entire tangent geometry is generated by virtual transitions across the spectral gap.*

Corollary 3.4 recovers by residue calculus what Section 2 obtained from the projector identity. The two statements are logically distinct: the earlier one covers all tangent vectors, this one those generated by a perturbation $\boldsymbol{\delta\Sigma}$. Every admissible tangent direction is generated exclusively by mixing between the spectral subspace of $\boldsymbol{\Pi_k}$ and its complement, and consequently both the metric and the curvature derived below depend on cross-gap transition amplitudes.

PROPOSITION 3.5 *(Gap bound).* $\boldsymbol{\|V\|_{\text{HS}} \leq \dfrac{\sqrt{2k}\,\|\delta\Sigma\|}{\Delta}}$.

PROOF. From (2.1), $\|V\|_{\mathrm{HS}}^2 = 2\|X\|_{\mathrm{HS}}^2$. By (3.2), in the eigenbasis X is $\Pi\,\delta\Sigma\,(I-\Pi)$ divided entrywise by gaps $\lambda_m - \lambda_n$ of modulus at least Δ. The Hilbert–Schmidt norm is the square root of the sum of the squared moduli of the entries, so $\|X\|_{\mathrm{HS}}^2 = \sum_{m\in\mathcal{I},\,n\in\mathcal{I}^c} |\langle m|\delta\Sigma|n\rangle|^2/(\lambda_m-\lambda_n)^2 \le \|\Pi\,\delta\Sigma\,(I-\Pi)\|_{\mathrm{HS}}^2/\Delta^2$. The operator $\Pi\,\delta\Sigma\,(I-\Pi)$ has rank at most k and operator norm at most $\|\delta\Sigma\|$, so $\|\Pi\,\delta\Sigma\,(I-\Pi)\|_{\mathrm{HS}} \le \sqrt{k}\,\|\delta\Sigma\|$. Hence $\|X\|_{\mathrm{HS}} \le \sqrt{k}\,\|\delta\Sigma\|/\Delta$, for any spectral set $\mathcal{I}$, adjacent or not. ∎

The bound is sharp: a two-level system pushed by a purely off-diagonal $\delta\Sigma$ attains it, and Theorem 6.3(i) gives every case of equality.

THEOREM 3.6 *(Spectral decomposition of the metric and curvature forms). Let $V = \delta_1\Pi_k$ and $W = \delta_2\Pi_k$ be generated by self-adjoint perturbations $\delta_1\Sigma$ and $\delta_2\Sigma$. Then*

$$g_{\Pi_k}(V,W) = \sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c} \frac{\mathrm{Re}\big[\langle m|\delta_1\Sigma|n\rangle\langle n|\delta_2\Sigma|m\rangle\big]}{(\lambda_n-\lambda_m)^2}, \tag{3.3}$$

$$\Omega_{\Pi_k}(V,W) = i\sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c} \frac{\langle m|\delta_1\Sigma|n\rangle\langle n|\delta_2\Sigma|m\rangle - \langle m|\delta_2\Sigma|n\rangle\langle n|\delta_1\Sigma|m\rangle}{(\lambda_n-\lambda_m)^2}. \tag{3.4}$$

In particular $g_{\Pi_k}(V,V) = \sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c} |\langle m|\delta_1\Sigma|n\rangle|^2/(\lambda_n-\lambda_m)^2$.

PROOF OF THE SYMMETRIC METRIC COMPONENT. From the block decomposition $V = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}$, $W = \begin{pmatrix} 0 & Y \\ Y^* & 0 \end{pmatrix}$ we have $g_{\Pi_k}(V,W) = \mathrm{Re}\,\mathrm{Tr}(XY^*) = \mathrm{Re}\sum_{m\in\mathcal{I},\,n\in\mathcal{I}^c} X_{mn}\overline{Y_{mn}}$. By (3.2) both entries carry the denominator $\lambda_m-\lambda_n$, whose square is $(\lambda_n-\lambda_m)^2$, and $\overline{\langle m|\delta_2\Sigma|n\rangle} = \langle n|\delta_2\Sigma|m\rangle$. ∎

PROOF OF THE SKEW-SYMMETRIC CURVATURE COMPONENT. Inserting a resolution of the identity into $\Omega_{\Pi_k}(V,W) = i\,\mathrm{Tr}(\Pi_k[V,W])$ and restricting to cross-gap terms,

$$\Omega_{\Pi_k}(V,W) = i\sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c}\Big(\langle m|V|n\rangle\langle n|W|m\rangle - \langle m|W|n\rangle\langle n|V|m\rangle\Big).$$

Each product carries the same denominator twice, $(\lambda_m-\lambda_n)(\lambda_m-\lambda_n) = (\lambda_n-\lambda_m)^2$, which gives (3.4). ∎

REMARK 3.7 (The numerator is gap-weighted). The antisymmetric numerator in (3.4) is not the commutator $[\delta_1\Sigma,\delta_2\Sigma]$ projected across the gap: each term carries its own weight $(\lambda_n-\lambda_m)^{-2}$, and without the weights it would be. The Berry curvature measures the oriented phase accumulated by successive virtual transitions between the spectral subspace of Π_k and its complement, each suppressed by its own energy denominator.

REMARK 3.8 (Beyond discrete spectrum). The operator statements — (2.1), Lemma 2.1, (2.3), $g_\Pi(V,V) = \mathrm{Tr}(XX^*)$ and Proposition 3.5 — hold verbatim with no discreteness assumption. Only the explicit index sums (3.3)–(3.4) require it.

REMARK 3.9 (Which perturbations generate which tangent vectors). In finite dimensions the map $\delta\Sigma \mapsto V$ is onto $T_\Pi \mathrm{Gr}(k,\mathcal{H})$: given X, take $\langle m|\delta\Sigma|n\rangle = (\lambda_m - \lambda_n)X_{mn}$ on cross-gap pairs. In infinite dimensions this inverse may require an unbounded $\delta\Sigma$, since the gaps are unbounded above; Lemma 2.1, which constructs a unitary flow directly, is unaffected and is the statement to use.

REMARK 3.10 (The many-body form of Lemmas 2.1 and 3.1). In a system of N interacting particles the same construction is known as quasi-adiabatic continuation, or spectral flow [Hastings–Wen 2005; Bachmann–Michalakis–Nachtergaele–Sims 2012]. One builds an anti-Hermitian generator Y with $\langle a|Y|b\rangle = F(E_a - E_b)\langle a|\delta\Sigma|b\rangle$, where $F(\omega) = -1/\omega$ for $|\omega|$ at least the gap and F is smoothly cut off below it; then $\partial\Pi = [Y,\Pi]$. On cross-gap pairs this is exactly (3.2), and Lemma 2.1 is its unitary flow. The smooth cut-off is what the many-body setting adds: it keeps Y local. It also marks where Proposition 3.5 weakens as N grows. For a sum of local terms $\|\delta\Sigma\|$ grows with the size of the system, so the bound $\sqrt{2k}\|\delta\Sigma\|/\Delta$ grows with it, even when the gap stays uniform; a bound in terms of the local strength of $\delta\Sigma$ is what locality supplies, and it is not derived here.

4 Parametric Realization (Pullback Structure)

The intrinsic quantum geometric tensor Q_{Π_k} constructed on the Grassmannian is independent of any auxiliary parameter space. In physical applications, however, one is typically given a smooth family of Hamiltonians $\Sigma(a)$ parameterized by points of a manifold M. This section shows how the familiar coordinate expressions for the quantum metric and Berry curvature arise as the pullback of the intrinsic tensor under the map induced by the family.

Let M be a smooth manifold and $a \mapsto \Sigma(a)$ a smooth family of self-adjoint operators. Let $M_\Delta = \{a \in M : \text{ the isolating contour } \Gamma_k \text{ persists at } a\}$ be the open set on which the spectral gap stays open. By the Riesz–Kato construction the map

$$\gamma : M_\Delta \longrightarrow \mathrm{Gr}(k,\mathcal{H}), \qquad a \longmapsto \Pi_k(a)$$

is smooth. Smoothness is a consequence of the gap hypothesis, and γ need not extend to the boundary of M_Δ; for unbounded Σ it is read in the norm-resolvent sense (Appendix A). If $\{\partial/\partial a^i\}$ is the coordinate basis of $T_a M$, then

$$V_i := \gamma_*\Big(\frac{\partial}{\partial a^i}\Big) = \partial_{a^i}\Pi_k(a)$$

is a tangent vector at $\Pi_k(a)$, and by Corollary 3.4 each V_i is strictly off-diagonal with respect to the decomposition induced by $\Pi_k(a)$.

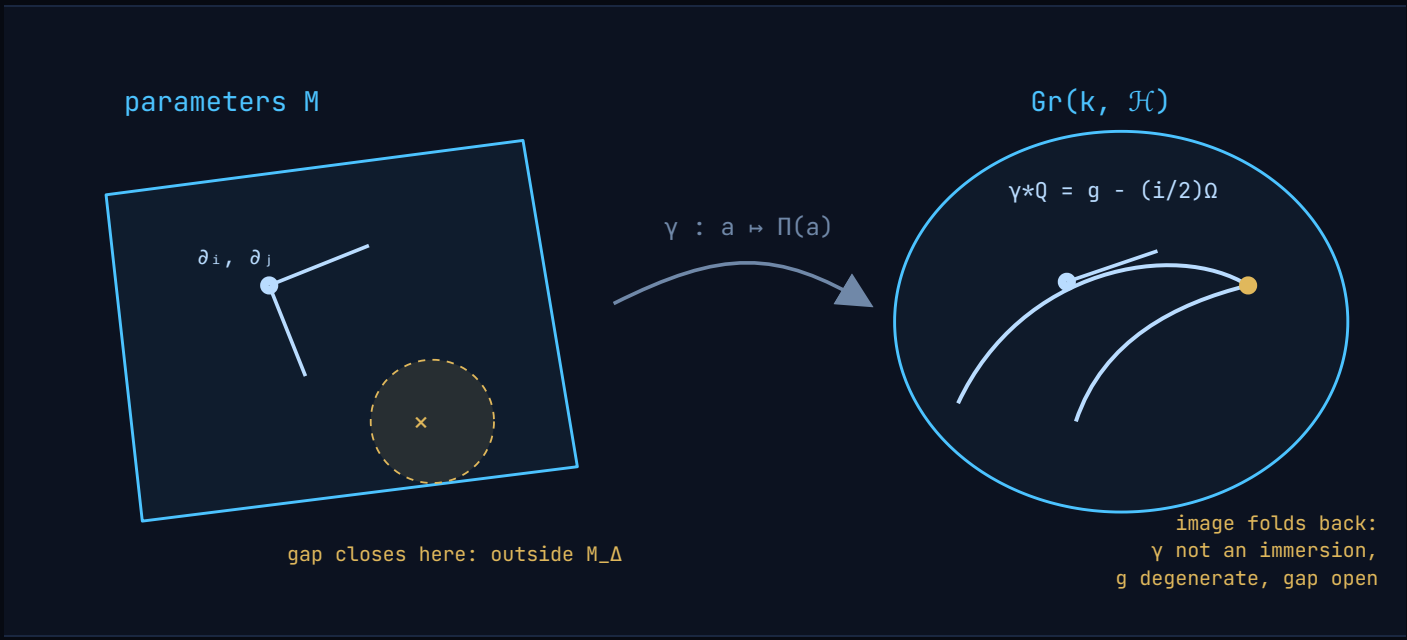


FIGURE 3. A family of operators is a map into the Grassmannian. The intrinsic tensor is sampled along the image. Two opposite things can go wrong: the gap closes (the map leaves the region where it is defined, and the metric grows without bound), or the image folds back on itself with the gap wide open (the pulled-back metric becomes degenerate).

4.1 Pullback of the metric and the curvature

The components $g_{ij}(a) = g_{\Pi_k}(\partial_i\Pi_k, \partial_j\Pi_k)$ and $\Omega_{ij}(a) = \Omega_{\Pi_k}(\partial_i\Pi_k, \partial_j\Pi_k)$ follow from Theorem 3.6 applied pointwise:

$$g_{ij}(a) = \sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c} \frac{\operatorname{Re}\big[\langle m|\partial_i\Sigma|n\rangle\langle n|\partial_j\Sigma|m\rangle\big]}{\big(\lambda_n(a)-\lambda_m(a)\big)^2}, \tag{4.1}$$

$$\Omega_{ij}(a) = i\sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c} \frac{\langle m|\partial_i\Sigma|n\rangle\langle n|\partial_j\Sigma|m\rangle - \langle m|\partial_j\Sigma|n\rangle\langle n|\partial_i\Sigma|m\rangle}{\big(\lambda_n(a)-\lambda_m(a)\big)^2}. \tag{4.2}$$

Expanding in blocks gives the band-traced forms used in the literature,

$$g_{ij} = \operatorname{Re}\operatorname{Tr}\big(\Pi\,\partial_i\Pi\,\partial_j\Pi\big), \qquad \Omega_{ij} = -2\operatorname{Im}\operatorname{Tr}\big(\Pi\,\partial_i\Pi\,\partial_j\Pi\big). \tag{4.3}$$

For $k = 1$ the diagonal component $g_{ii} = \sum_n |\langle m|\partial_i \Sigma|n\rangle|^2/(\lambda_n - \lambda_m)^2$ is the fidelity susceptibility of the state. For $k > 1$ it is the corresponding subspace quantity.

PROPOSITION 4.1 *(Two ways the pullback degenerates).* $g_{ij}(a) = \mathrm{Re}\,\mathrm{Tr}(X_i X_j^*)$ *is a Gram matrix and is therefore positive semi-definite. It is degenerate at* a *precisely when the* $\partial_i \Pi$ *are linearly dependent, that is, when* γ *fails to be an immersion at* a*. This is a different mechanism from a closing gap: a closing gap makes the denominators in (4.1) small and the metric large, while failure of immersion makes the metric degenerate with the gap wide open. The two are opposite limits and are kept distinct throughout.*

4.2 Structural statement

Collecting (4.1)–(4.2), the pullback of the intrinsic tensor is

$$\gamma^* Q = g_{ij}\, da^i \otimes da^j - \tfrac{i}{2}\, \Omega_{ij}\, da^i \wedge da^j. \tag{4.4}$$

The conventional parametric quantum geometric tensor on M is therefore not a primitive object: it is the coordinate representation of the pullback $\gamma^* Q_{\Pi_k}$ of the single intrinsic tensor living on the Grassmannian. The parameter manifold M does not generate the geometry; it merely selects a finite-dimensional trajectory through the much larger tangent bundle of $\mathrm{Gr}(k, \mathcal{H})$. This makes the coordinate independence of the underlying structures transparent and explains why different parameterizations or local frames yield equivalent physical results.

REMARK 4.2 (Trace sector). For $k > 1$ the Berry curvature of the occupied subspace is non-abelian: F_{ij} is a $k \times k$ matrix and $\Omega_{ij} = \mathrm{tr}\, F_{ij}$. The traceless part of F_{ij} is invisible on $\mathrm{Gr}(k, \mathcal{H})$, since Π retains no frame information; the non-abelian connection lives on the Stiefel bundle $V_k(\mathcal{H}) \to \mathrm{Gr}(k, \mathcal{H})$, a principal $U(k)$ bundle. Statement (4.4) is a statement about the trace sector.

PROPOSITION 4.3 *(First-order factorization through the gap). Every first-order geometric quantity of the family – V_i, g_{ij}, Ω_{ij} – depends on* $\partial_i \Sigma$ *only through its cross-gap block* $\Pi\, \partial_i \Sigma\, (I - \Pi)$ *and the eigenvalues. A direction* B *whose cross-gap block vanishes is silent:* $\partial_B \Pi = 0$*.*

This is immediate from (3.2). It does not persist to second order. A silent direction leaves Π fixed but can still change the metric, because moving along it rotates the eigenvectors inside each sector and shifts the gaps; in a three-parameter family of six-level operators with $k = 2$ and random complex Hermitian coefficients (fixed in the Section 4 check script), the derivative of g_{11} along a direction B that is block-diagonal in the eigenbasis is 0.113, while $\partial_B \Pi$ vanishes to 10^{-12}. One nonzero example is all the statement needs: it shows the second-order dependence exists. Second-order quantities therefore see the intra-sector couplings as well.

4.3 Second-order response

Differentiating $g_{ij} = \tfrac{1}{2}\mathrm{Tr}(\partial_i\Pi\,\partial_j\Pi)$ gives the variation of the metric along the parameter manifold,

$$\partial_k g_{ij} = \tfrac{1}{2}\,\mathrm{Tr}\big(\partial_k\partial_i\Pi\;\partial_j\Pi + \partial_i\Pi\;\partial_k\partial_j\Pi\big), \tag{4.5}$$

which is exact and coordinate-free. Differentiating $V_i = \Pi V_i + V_i\Pi$ gives

$$\partial_k V_i = V_k V_i + \Pi(\partial_k V_i) + (\partial_k V_i)\Pi + V_i V_k.$$

Since V_j is purely off-diagonal, $\mathrm{Tr}\big((\partial_k V_i)V_j\big)$ receives contributions only from the off-diagonal blocks of the second variation; the diagonal blocks, fixed algebraically by products of first derivatives, drop out. The second variation itself is given exactly by the second-order Riesz formula,

$$\partial_k\partial_i\Pi = \frac{1}{2\pi i}\oint_{\Gamma_k}\Big[R\,\partial_k\partial_i\Sigma\,R + R\,\partial_k\Sigma\,R\,\partial_i\Sigma\,R + R\,\partial_i\Sigma\,R\,\partial_k\Sigma\,R\Big]\,dz, \qquad R = (zI-\Sigma)^{-1}. \tag{4.6}$$

Inserted into (4.5), this gives the metric gradient (the quantum metric dipole) with no truncation. Resolved in the eigenbasis, (4.6) contains three-state terms, in which one coupling crosses the gap and the other stays within a sector, and the gap derivatives $\partial_k(\lambda_n - \lambda_m) = \langle n|\partial_k\Sigma|n\rangle - \langle m|\partial_k\Sigma|m\rangle$ (Hellmann–Feynman). Any spectral form of the dipole that treats $|m\rangle, |n\rangle$ as parameter-independent omits the three-state terms, which live in exactly the off-diagonal block that survives the trace, and is therefore incomplete; (4.6) carries them automatically.

4.4 The redistribution operator and its flow

For any operator Ω define its redistribution relative to Π,

$$F(\Omega,\Pi) = (I-\Pi)\,\Omega\,\Pi + \Pi\,\Omega\,(I-\Pi) = \Omega\Pi + \Pi\Omega - 2\Pi\Omega\Pi = \big[[\Omega,\Pi],\Pi\big]. \tag{4.7}$$

F is the part of Ω that couples the subspace to its complement, so $F(\Omega,\Pi) = 0$ exactly when $[\Omega,\Pi] = 0$, and $\|F\|_{\mathrm{HS}} = \|[\Omega,\Pi]\|_{\mathrm{HS}}$ for every Ω. For self-adjoint Ω, F is a tangent vector of the form (2.1), and it is the gradient of $\Pi \mapsto \mathrm{Tr}(\Omega\Pi)$ with respect to the intrinsic metric: $g_\Pi(F,V) = \tfrac{1}{2}\mathrm{Tr}(\Omega V)$ for all $V \in T_\Pi\mathrm{Gr}(k,\mathcal{H})$. The gradient flow

$$\dot\Pi = F(\Omega,\Pi) = \big[[\Omega,\Pi],\Pi\big] \tag{4.8}$$

is Brockett's double-bracket flow [Brockett 1991; Helmke–Moore 1994]. It stays on $\mathrm{Gr}(k,\mathcal{H})$ and carries Π to the spectral projector onto the top k eigenvectors of Ω; the reversed flow carries it to the bottom k. It is the motion through the Grassmannian, in the intrinsic metric, by which a subspace

finds the leading spectral subspace of an operator — the operation every eigen-analysis of measured data performs implicitly.

5 Spectral Lower Bounds on Dissipation

One of the most direct physical applications of the intrinsic quantum metric is a lower bound on the cost of non-adiabatic driving. The metric measures how fast a spectral subspace is being moved; a bath charges for that motion. This section makes both halves precise. The first half is kinematic and needs nothing beyond Sections 2–4: the metric is the quantum Fisher information of the moving subspace, and the gap limits how quickly the subspace can be moved. The second half needs a declared bath, and there the metric reappears, reweighted pair by pair by the spectrum, as a lower bound on the excess work.

Let $\Sigma(a)$ be a family of self-adjoint operators driven along a smooth path $a(t)$ over the time interval $[t_i, t_f]$, with the gap open throughout, and write $\dot{\Pi} = \partial_t \Pi_k(a(t))$, so that $g_\Pi(\dot{\Pi}, \dot{\Pi}) = \dot{a}^i \dot{a}^j g_{ij}(a)$.

5.1 The quantum Fisher information of a moving subspace

LEMMA 5.1 *(Fisher information of the subspace). Let $\rho(t) = \Pi_k(t)/k$ be the normalised state supported on the moving subspace. Its symmetric logarithmic derivative is $L = 2\dot{\Pi}$, and its quantum Fisher information is*

$$F_Q(t) = \operatorname{Tr}\big(\rho L^2\big) = \frac{4}{k}\, g_\Pi(\dot{\Pi}, \dot{\Pi}). \tag{5.1}$$

In particular, for $k = 1$, $\tfrac{1}{4} F_Q = g_\Pi(\dot{\Pi}, \dot{\Pi})$.

PROOF. The symmetric logarithmic derivative is defined by $\partial_t \rho = \tfrac{1}{2}(L\rho + \rho L)$. Differentiating the projector identity $\Pi_k^2 = \Pi_k$ with respect to time gives the linearised relation $\dot{\Pi} = \dot{\Pi}\,\Pi_k + \Pi_k\,\dot{\Pi}$, which is satisfied identically by the choice $L = 2\dot{\Pi}$. Substituting into the Fisher information gives $F_Q = \tfrac{4}{k}\operatorname{Tr}(\Pi_k \dot{\Pi}^2)$. Because $\dot{\Pi}$ is a tangent vector, it is strictly off-diagonal, $\dot{\Pi} = \Pi_k \dot{\Pi}(I - \Pi_k) + (I - \Pi_k)\dot{\Pi}\,\Pi_k$; squaring isolates the two diagonal blocks, which contribute equally to the trace, so $\operatorname{Tr}(\Pi_k \dot{\Pi}^2) = \tfrac{1}{2}\operatorname{Tr}(\dot{\Pi}^2) = g_\Pi(\dot{\Pi}, \dot{\Pi})$. ∎

The intrinsic metric is therefore the information-geometric speed of the subspace: it measures how distinguishable the state becomes per unit time, and so how much the driving must change. It does not by itself measure dissipation. Moving Π_k unitarily leaves the spectrum of ρ fixed, so its von Neumann entropy is constant and no entropy is produced; the Fisher metric here is the symmetric-

logarithmic-derivative metric, while near-equilibrium thermodynamic length uses the Kubo–Mori metric and a relaxation time that no purely geometric quantity supplies. Turning speed into cost requires a bath (Section 5.3).

5.2 The gap limits the driving speed

Let $\mathcal{L} = \int_{t_i}^{t_f} \sqrt{g_\Pi(\dot\Pi, \dot\Pi)}\, dt$ be the length of the path traced by $\Pi_k(t)$ in the intrinsic metric, and let the gap satisfy $\Delta(t) \geq \Delta_{\min} > 0$.

THEOREM 5.2 *(Minimum driving time).* $t_f - t_i \;\geq\; \dfrac{\Delta_{\min}\,\mathcal{L}}{\sqrt{k}\,\sup_t \|\partial_t \Sigma(t)\|}.$

PROOF. By (2.3), $\sqrt{g_\Pi(\dot\Pi,\dot\Pi)} = \|\dot\Pi\|_{\mathrm{HS}}/\sqrt{2}$. Proposition 3.5 applied pointwise gives $\|\dot\Pi\|_{\mathrm{HS}} \leq \sqrt{2k}\,\|\partial_t\Sigma\|/\Delta(t)$, so the integrand is at most $\sqrt{k}\,\|\partial_t\Sigma\|/\Delta_{\min}$. Integrating over $[t_i, t_f]$ and rearranging gives the claim. ∎

Theorem 5.2 bounds the time needed to move a spectral subspace a given intrinsic distance, in terms of the available driving strength and the gap. It is a statement about kinematics, proved from the machinery of Sections 2–3 alone.

5.3 Friction on projector rotations

Let the driven system, with instantaneous spectrum $\Sigma = \sum_n E_n |n\rangle\langle n|$ (nondegenerate), be in contact with a bath at inverse temperature β, with Gibbs state $\omega = e^{-\beta\Sigma}/Z$ and populations p_n. Declare the relaxation model

$$\dot\rho = -i\chi\,[\Sigma, \rho] + \frac{\omega(\Sigma(t)) - \rho}{\tau}, \qquad \chi \in \{0, 1\},$$

pure relaxation to the instantaneous Gibbs state ($\chi = 0$) or relaxation with coherent precession ($\chi = 1$). Work is $W = \int \mathrm{Tr}(\rho\,\dot\Sigma)\,dt$, and the excess work is $W_{\mathrm{ex}} = W - \Delta F$.

Derivation. Write $\rho = \omega + \delta\rho$. Because $[\Sigma, \omega] = 0$, $\frac{d}{dt}\delta\rho = -i\chi[\Sigma, \delta\rho] - \delta\rho/\tau - \dot\omega$. For slow driving $\delta\rho = O(\dot a)$ and its time derivative is $O(\dot a^2)$, so in the instantaneous eigenbasis

$$\delta\rho_{mn} = -\frac{\tau\,\dot\omega_{mn}}{1 + i\chi\tau(E_m - E_n)} + O(\dot a^2).$$

The Gibbs state changes on the diagonal as $\dot\omega_{nn} = -\beta p_n(\dot\Sigma_{nn} - \langle\dot\Sigma\rangle)$, because $\dot E_n = \dot\Sigma_{nn}$, and off the diagonal as $\dot\omega_{mn} = \dot\Sigma_{mn}(p_n - p_m)/(E_n - E_m)$, because $\langle m|\dot n\rangle = \dot\Sigma_{mn}/(E_n - E_m)$. Since $\dot F = \mathrm{Tr}(\omega\dot\Sigma)$,

the excess power is $\mathrm{Tr}(\delta\rho\,\dot\Sigma)$; pairing the terms (m,n) and (n,m), whose denominators are complex conjugates, gives $\dot W_{\mathrm{ex}} = \zeta$ with

$$\zeta = \tau\beta\,\mathrm{Var}_p(\dot\Sigma_{\mathrm{diag}}) + \sum_{m\neq n} c_{mn}\,|\dot\Sigma_{mn}|^2\,\frac{p_m - p_n}{E_n - E_m}, \qquad c_{mn} = \frac{\tau}{1+\chi^2\tau^2(E_m-E_n)^2}. \tag{5.2}$$

Every term is nonnegative, since p decreases as E increases. For $\chi = 0$, ζ is the friction tensor of Sivak and Crooks for a single relaxation time, in the quantum Kubo–Mori form of Scandi and Perarnau-Llobet [Sivak–Crooks 2012; Scandi–Perarnau-Llobet 2019].

The metric inside the friction. For the spectral projector of a cluster $\mathcal{I}$, Theorem 3.6 gives $g = \sum_{m\in\mathcal{I},\,n\notin\mathcal{I}} g_{mn}$ with $g_{mn} = |\dot\Sigma_{mn}|^2/(E_n-E_m)^2$. The friction of the same pair, counted in both orders, is therefore

$$\zeta_{mn} = 2c_{mn}\,(p_m - p_n)(E_n - E_m)\,g_{mn}. \tag{5.3}$$

Define $\zeta_{\mathrm{rot}} = \sum_{m\in\mathcal{I},\,n\notin\mathcal{I}} \zeta_{mn}$, the part of ζ carried by the pairs that move Π. It is the intrinsic metric reweighted pair by pair, the weight being twice the population difference times the gap times the relaxation factor. The rest of ζ — the diagonal term and the pairs inside $\mathcal{I}$ or inside its complement — moves the Gibbs state without moving Π, so $\zeta \geq \zeta_{\mathrm{rot}}$.

THEOREM 5.3 *(Spectral lower bound on dissipation). Under the declared model, to first order in the driving rate:*

(i) Two levels. With gap Δ, $\zeta_{\mathrm{rot}} = 2c(\Delta)\,\Delta\tanh(\beta\Delta/2)\;g_\Pi(\dot\Pi,\dot\Pi)$.

(ii) Zero temperature. For a nondegenerate ground state and $\mathcal{I}=\{0\}$, with $\Delta_n = E_n - E_0$,

$$\min_n 2c(\Delta_n)\Delta_n\;g_\Pi(\dot\Pi,\dot\Pi) \;\leq\; \zeta_{\mathrm{rot}} \;\leq\; \max_n 2c(\Delta_n)\Delta_n\;g_\Pi(\dot\Pi,\dot\Pi), \tag{5.4}$$

and consequently, for a protocol of duration T whose path has intrinsic length $\mathcal{L}$,

$$W_{\mathrm{ex}} \;\geq\; \Big(\min_{t,n} 2c(\Delta_n)\Delta_n\Big)\int_{t_i}^{t_f} g_\Pi(\dot\Pi,\dot\Pi)\,dt \;\geq\; \Big(\min_{t,n} 2c(\Delta_n)\Delta_n\Big)\frac{\mathcal{L}^2}{T}. \tag{5.5}$$

PROOF. (i) For two levels $p_- - p_+ = \tanh(\beta\Delta/2)$ in (5.3). (ii) At zero temperature pairs of excited levels carry no population difference, so $\zeta_{\mathrm{rot}} = \sum_n 2c(\Delta_n)\Delta_n\,g_{0n}$, a convex combination of the weights $2c(\Delta_n)\Delta_n$ times $g = \sum_n g_{0n}$; this is (5.4). Integrating $W_{\mathrm{ex}} = \int \zeta\,dt \geq \int \zeta_{\mathrm{rot}}\,dt$ gives the first inequality of (5.5), and Cauchy–Schwarz, $\mathcal{L}^2 = (\int\sqrt{g}\,dt)^2 \leq T\int g\,dt$, gives the second. ∎

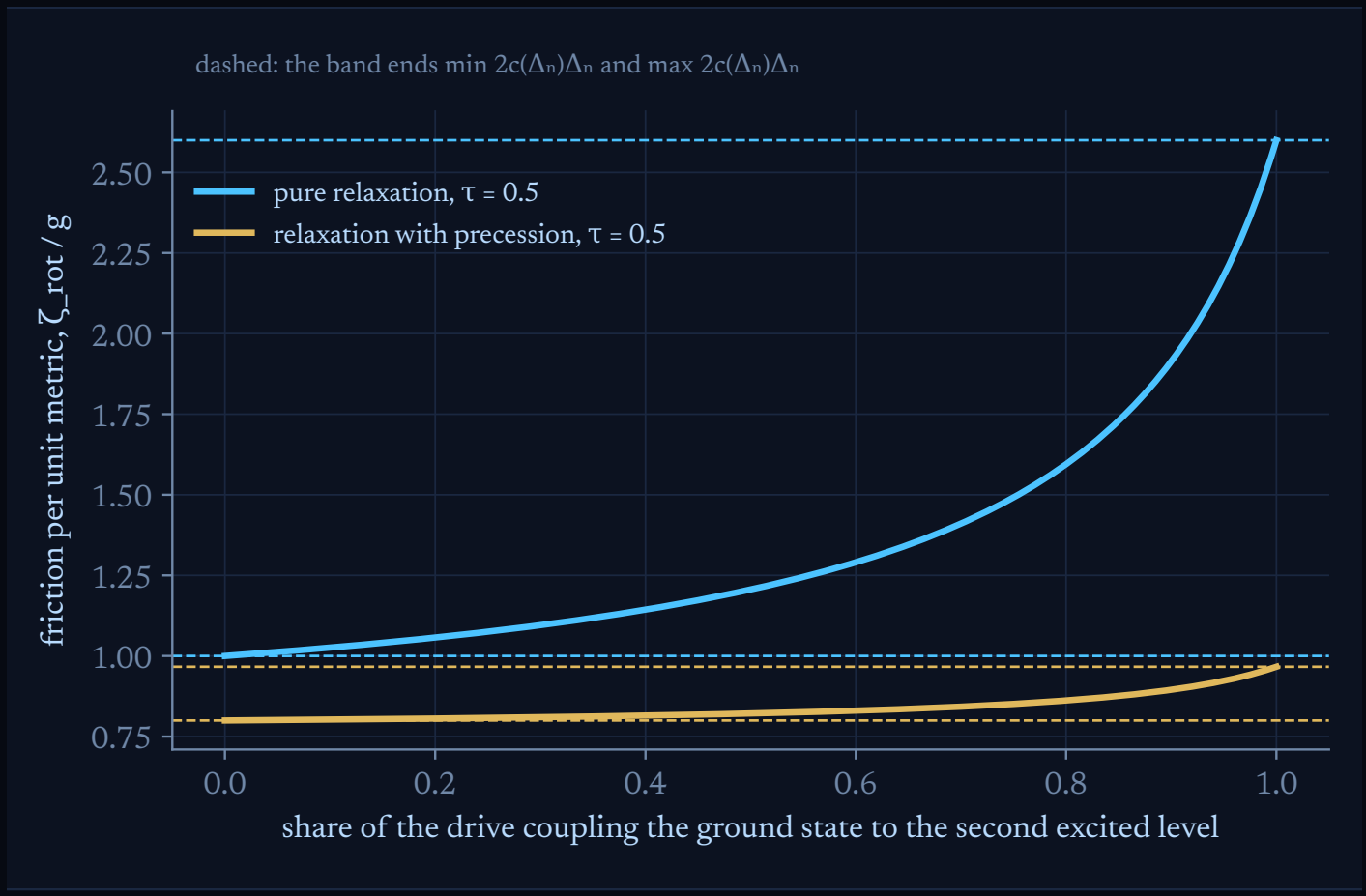


FIGURE 4. The zero-temperature band of Theorem 5.3 for three levels with gaps 1 and 2.6. As the drive moves its weight from the first to the second excited level, the friction per unit of intrinsic metric slides between the two ends of the band; it never leaves it. With precession the faster pair is penalised and the band narrows.

The bound (5.5) is the thermodynamic-length form of the statement that finite-rate motion of a spectral subspace costs work: the cost is at least the squared intrinsic length divided by the time allowed, with a prefactor set by the spectrum and the bath. The two metrics, ζ_{rot} and g, agree up to one constant exactly when the drive couples the retained projector only to levels sharing one value of $c(\Delta)\Delta$, for instance a single gap; otherwise they are different metrics on the same tangent space, each bounding the other.

Work and entropy. For a process that starts in the Gibbs state, the entropy production $\Sigma_{\mathrm{irr}} = \Delta S - \beta Q$ satisfies $\beta W_{\mathrm{ex}} = \Sigma_{\mathrm{irr}} + D(\rho_f \,\|\, \omega_f)$, because $\beta(W - \Delta F)$ differs from Σ_{irr} exactly by the relative entropy of the final state to the final Gibbs state. That term is second order in the driving rate and vanishes once the system re-equilibrates, so to leading order $\Sigma_{\mathrm{irr}} = \beta \int \zeta \, dt \geq \beta \int \zeta_{\mathrm{rot}} \, dt$. At zero temperature the statement stays a bound on work. For two levels the friction splits with no cross term into a scale part and a rotation part,

$$\zeta = \tau\beta \,\mathrm{sech}^2(\beta\Delta/2)\, \dot{r}^2 + 2c(\Delta)\,\Delta \tanh(\beta\Delta/2)\, g_{\Pi}(\dot{\Pi}, \dot{\Pi}),$$

the two blocks of $|\dot{\mathbf{d}}|^2 = \dot{r}^2 + \Delta^2 g$ for $\Sigma = \mathbf{d} \cdot \boldsymbol{\sigma}$, $r = |\mathbf{d}|$, $\Delta = 2r$, with thermodynamic weights. As $T \to 0$ the scale block freezes and all slow-driving dissipation is projector rotation.

A second bath. The same structure holds for a detailed-balance (Davies) bath, with each pair's own coherence lifetime in place of τ: $c_{mn} = \Gamma_{mn}/(\Gamma_{mn}^2 + \Delta_{mn}^2)$, and the zero-temperature band (5.4) with $c_n = \Gamma_{0n}/(\Gamma_{0n}^2 + \Delta_n^2)$.

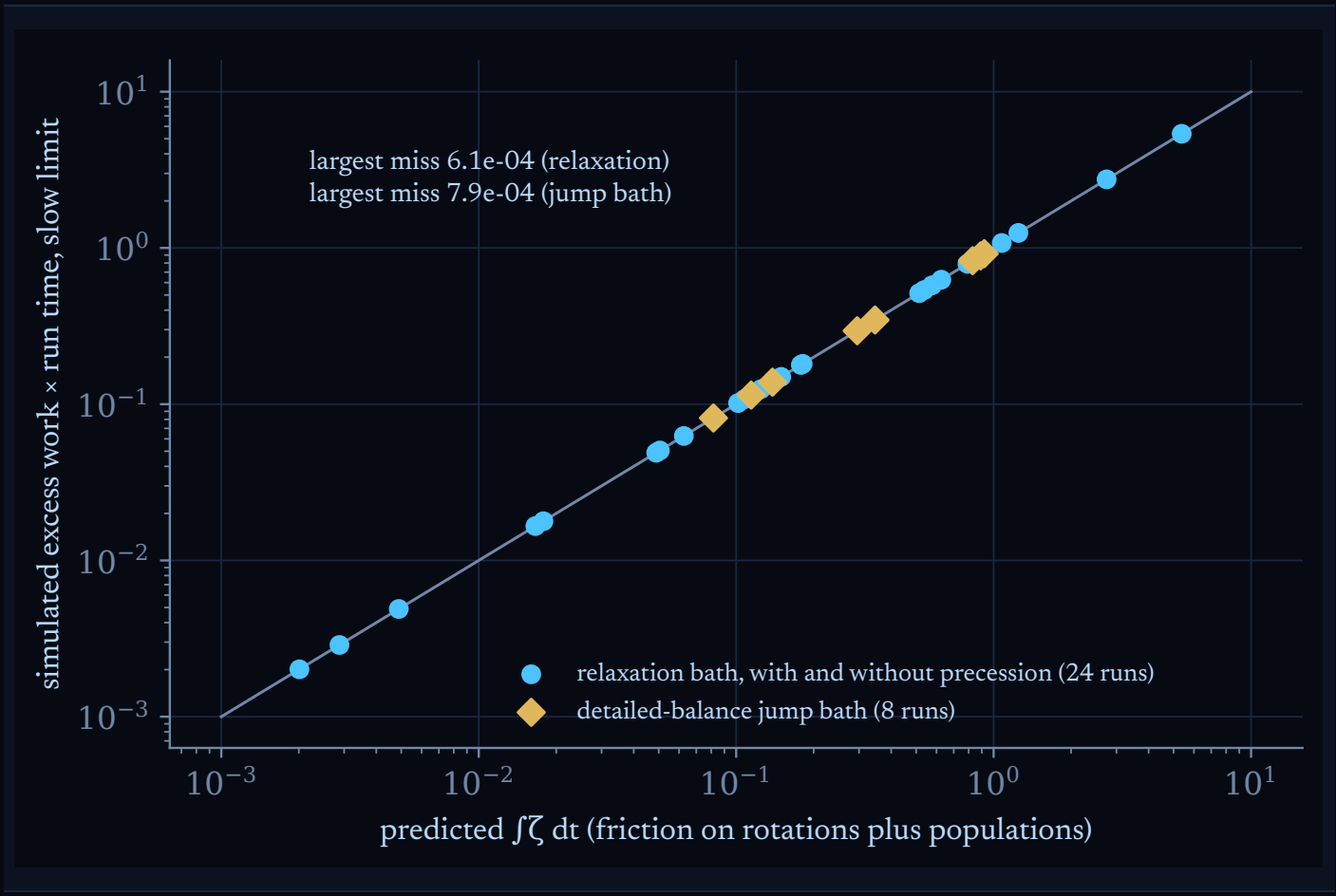


FIGURE 5. Friction on projector rotations, tested. Each point is one exact simulation of a slowly driven system in contact with a bath, taken to the slow limit from four run lengths, against the prediction ∫ζ dt. Circles: the relaxation bath, with and without precession. Diamonds: a detailed-balance jump bath the law was not built on.

None of this is universal: the weight depends on the bath model, the gaps and the populations. What is universal is the shape — the intrinsic metric of the moving subspace, multiplied by a spectral weight that the bath supplies. In both baths the prediction $\int \zeta \, dt$ matches exact simulation of slow driving to within 10^{-3} relative (Figure 5), the zero-temperature band holds in all 2,000 random draws for each bath, and the integrated entropy production approaches βW_{ex} at the slowest runs.

6 Metric–Curvature Inequalities, Equality and Rigidity

The preceding sections established that the intrinsic Quantum Geometric Tensor admits a canonical decomposition into its symmetric metric component and antisymmetric curvature component, $Q_\Pi = g_\Pi - \frac{i}{2}\Omega_\Pi$. While these objects are traditionally presented as independent geometric quantities in parametric band theory, the coordinate-free projector formulation reveals a structural constraint: the generation of a geometric phase is quantitatively bounded by the intrinsic metric of the Grassmannian tangent bundle. In coordinates the bound below is the determinant inequality $4\det g \geq \Omega^2$, standard in the band-geometry literature [Roy 2014; Mera–Ozawa 2021]; the projector formulation makes both its proof and its saturation transparent, and no priority is claimed for it.

THEOREM 6.1 *(Metric control of Berry curvature). Let $V, W \in T_\Pi \mathrm{Gr}(k, \mathcal{H})$. The intrinsic Berry curvature 2-form satisfies the sharp inequality*

$$|\Omega_\Pi(V,W)| \;\leq\; 2\sqrt{g_\Pi(V,V)\,g_\Pi(W,W) - g_\Pi(V,W)^2}. \tag{6.1}$$

PROOF. By (2.2), Q_Π is a positive semi-definite Hermitian form on the complexified tangent space. Consequently, for any two tangent vectors the 2×2 Gram matrix of their inner products has non-negative determinant:

$$\det \begin{pmatrix} Q_\Pi(V,V) & Q_\Pi(V,W) \\ Q_\Pi(W,V) & Q_\Pi(W,W) \end{pmatrix} = Q_\Pi(V,V)\,Q_\Pi(W,W) - |Q_\Pi(V,W)|^2 \;\geq\; 0.$$

Using the off-diagonal constraint $\Pi V \Pi = 0$, $Q_\Pi(V,V) = \frac{1}{2}\mathrm{Tr}(V^2) = g_\Pi(V,V)$, and likewise for W. Separating the cross term into its real and imaginary parts by (2.3), $|Q_\Pi(V,W)|^2 = g_\Pi(V,W)^2 + \frac{1}{4}\Omega_\Pi(V,W)^2$. Substituting, $g_\Pi(V,V)g_\Pi(W,W) - g_\Pi(V,W)^2 - \frac{1}{4}\Omega_\Pi(V,W)^2 \geq 0$; rearranging and taking the square root completes the proof. ∎

Dropping the subtracted term gives the weaker envelope $|\Omega_\Pi(V,W)| \leq 2\sqrt{g_\Pi(V,V)\,g_\Pi(W,W)}$, which also follows directly from the triangle inequality and Cauchy–Schwarz applied to $\mathrm{Tr}(\Pi V W)$ and $\mathrm{Tr}(\Pi W V)$.

COROLLARY 6.2 *(Sharpness, with an explicit extremal pair). Equality holds in (6.1) if and only if $Y = cX$ for some $c \in \mathbb{C}$, where X, Y are the blocks of V, W. In particular, given any $V \neq 0$ with block X, the tangent vector W whose block is iX (which exists by Lemma 2.1) satisfies $g_\Pi(V,W) = 0$ and $|\Omega_\Pi(V,W)| = 2\|X\|_{\mathrm{HS}}^2 = 2\sqrt{g_\Pi(V,V)g_\Pi(W,W)}$, so the bound is attained, in every rank k.*

PROOF. Equality in the Cauchy–Schwarz step holds exactly when X and Y are linearly dependent over $\mathbb{C}$. Writing $Y = cX$ gives $g_\Pi(V,V) = \|X\|_{\mathrm{HS}}^2$, $g_\Pi(W,W) = |c|^2\|X\|_{\mathrm{HS}}^2$, $g_\Pi(V,W) = \mathrm{Re}(\bar{c})\|X\|_{\mathrm{HS}}^2$ and $\Omega_\Pi(V,W) = -2\,\mathrm{Im}(\bar{c})\|X\|_{\mathrm{HS}}^2$; both sides of (6.1) equal $2|\mathrm{Im}\,c|\,\|X\|_{\mathrm{HS}}^2$. Setting $c = i$ gives the stated pair. ∎

In coordinate form, saturation of the determinant inequality is known to occur exactly when the parametric tensor acquires a null vector, and bands saturating it everywhere are called Kähler, ideal or vortexable. Corollary 6.2 is that criterion in block language: $Y = cX$ is the statement that the Gram matrix of V, W is singular, and the extremal partner can be written down explicitly, in every rank, as the tangent vector whose block is iX. Pulled back along a two-parameter family, (6.1) is $4 \det g_{ij} \geq \Omega_{12}^2$.

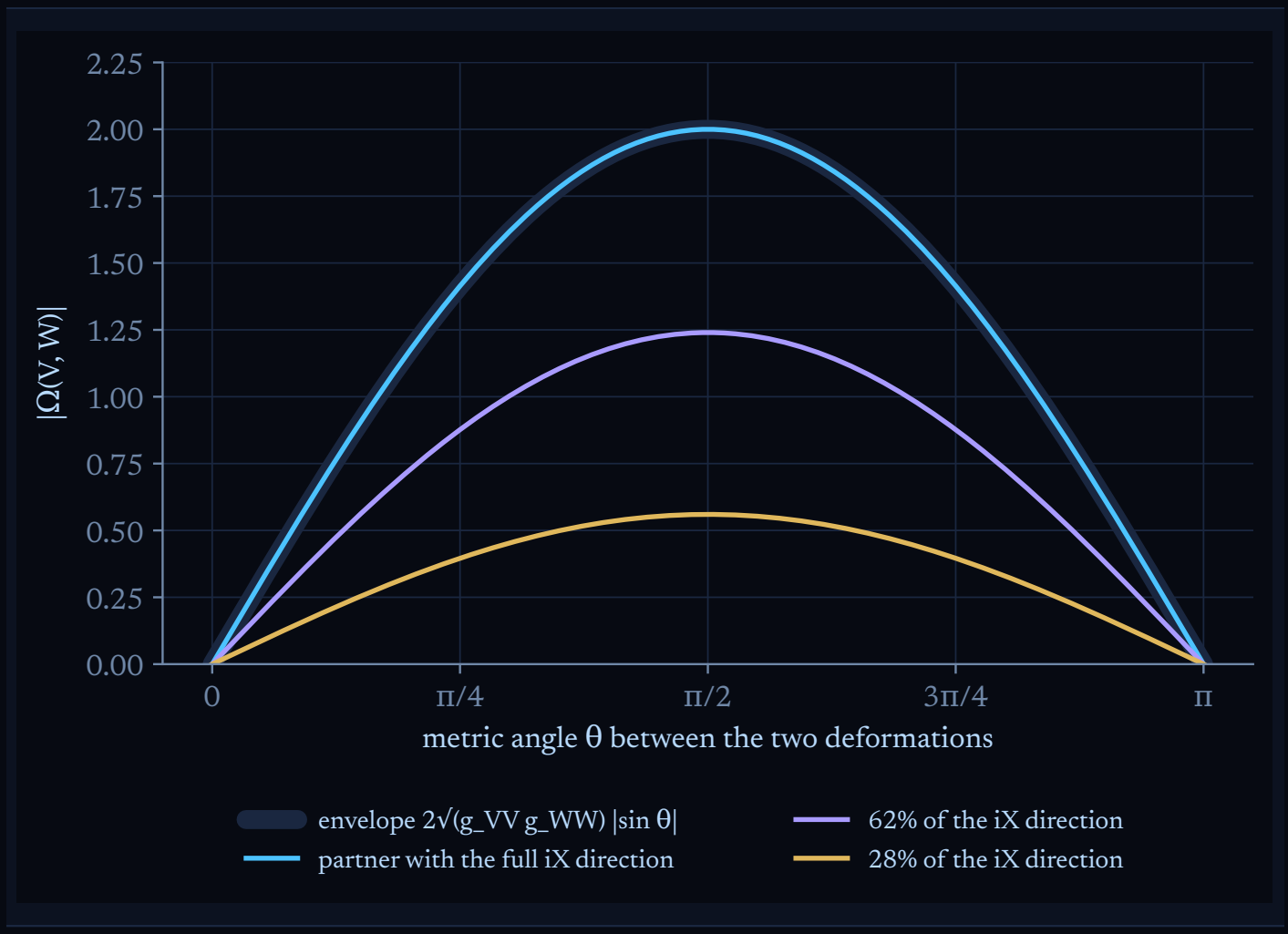


FIGURE 6. The envelope of Theorem 6.1, normalised to g(V,V) = g(W,W) = 1. The partner whose block is iX lies on the envelope at every angle; partners carrying only part of that direction fall inside.

6.1 The geometric angle safeguard

Expressing the metric inner product as $g_\Pi(V,W) = \sqrt{g_\Pi(V,V)g_\Pi(W,W)}\,\cos\theta_G$, where θ_G is the geometric angle between the two tangent deformations, turns (6.1) into

$$|\Omega_\Pi(V,W)| \;\le\; 2\sqrt{g_\Pi(V,V)\,g_\Pi(W,W)}\,|\sin\theta_G|. \tag{6.2}$$

This $\sin\theta_G$ dependence is a directional safeguard: the curvature a pair of deformations can generate is limited by how far apart they point in the intrinsic metric. The control it gives is relative, not absolute. Near a degeneracy both sides of (6.2) diverge at the same rate — a two-level system saturates the bound at every gap, so the ratio stays one as the gap closes — and the inequality is not a regularisation. The suppression at $\sin\theta_G = 0$ is a property of antisymmetry, since any 2-form vanishes on proportional arguments.

6.2 Control hierarchy

Both geometric quantities are driven by the same interband transition amplitudes $\langle m|\delta\Sigma|n\rangle$ across the spectral cut (Theorem 3.6). The quantum metric quantifies the total capacity of the transport channels; the Berry curvature tracks the phase interference among them; and the metric is the quantitative envelope for every admissible curvature. Metric rigidity therefore enforces curvature rigidity: if $g_\Pi(V,V) = 0$ then $V = 0$ and $\Omega_\Pi(V,\cdot) = 0$, and if a perturbation has no cross-gap coupling, $\langle m|\delta\Sigma|n\rangle = 0$ for every $m \in \mathcal{I},\ n \in \mathcal{I}^c$, then it moves nothing and carries no curvature. The ordering is one of control, not of generation: cross-gap transitions generate both g and Ω, and g bounds Ω.

6.3 When the bounds are met

Every inequality in this paper is sharp, and in every case the equality is rigid: it holds only in one recognisable configuration, which equality alone forces. This subsection collects the equality cases. Each is proved from the proof of its inequality, by asking where each step loses nothing.

THEOREM 6.3 *(Equality cases). Let* $B = \Pi\,\delta\Sigma\,(I - \Pi)$.

1. *Gap bound.* $\|V\|_{\mathrm{HS}} = \sqrt{2k}\,\|\delta\Sigma\|/\Delta$ *in Proposition 3.5 if and only if (a)* $\delta\Sigma$ *couples the two sectors only through pairs* (m,n) *with* $|\lambda_m - \lambda_n| = \Delta$ *exactly, and (b)* B *has* k *nonzero singular values, all equal to* $\|\delta\Sigma\|$.
2. *Minimum driving time. Theorem 5.2 holds with equality if and only if, at almost every time, the gap bound holds with equality at the gap* $\Delta(t) = \Delta_{\min}$*, with drive strength* $\|\partial_t\Sigma(t)\| = \sup_s \|\partial_s\Sigma(s)\|$.
3. *Dissipation. The chain (5.5) holds with equality if and only if* $\zeta = \zeta_{\mathrm{rot}}$ *(no part of the drive moves the Gibbs state without moving* Π*), the weight* $2c(\Delta_n)\Delta_n$ *equals its minimum over all pairs and times on every pair that carries metric, at every time, and the intrinsic speed* $g_\Pi(\dot\Pi,\dot\Pi)$ *is constant in time.*
4. *Curvature. Equality in (6.1) holds if and only if* W *lies in the complex line of* V *(Corollary 6.2). Quantitatively, for* $V \neq 0$*,*

$$g_\Pi(V,V)\,g_\Pi(W,W) - g_\Pi(V,W)^2 - \tfrac{1}{4}\Omega_\Pi(V,W)^2 = g_\Pi(V,V)\;g_\Pi(W_\perp,W_\perp), \tag{6.3}$$

where $W_\perp$ *is the tangent vector with block* $Y - cX$*,* $c = \mathrm{Tr}(YX^*)/\mathrm{Tr}(XX^*)$.

PROOF. (i) The proof of Proposition 3.5 is a chain of three steps: $\sum |B_{mn}|^2/(\lambda_m-\lambda_n)^2 \le \|B\|_{\mathrm{HS}}^2/\Delta^2$; $\|B\|_{\mathrm{HS}} \le \sqrt{k}\,\|B\|$; and $\|B\| \le \|\delta\Sigma\|$. The first is a sum of termwise inequalities, each an equality exactly when $B_{mn} = 0$ or $|\lambda_m - \lambda_n| = \Delta$. The second compares the root sum of the squares of at most k singular values with $\sqrt{k}$ times the largest, with equality exactly when all k are equal. The third is then an equality exactly when that common value is $\|\delta\Sigma\|$. (ii) The proof of Theorem 5.2 bounds the integrand pointwise by (i) and then replaces $\Delta(t)$ by $\Delta_{\min}$ and $\|\partial_t\Sigma\|$ by its supremum; the integral equals the bound exactly when each replacement loses nothing almost everywhere. (iii) The three inequalities of (5.5) are $\zeta \ge \zeta_{\mathrm{rot}}$, the convex-combination bound (5.4), and Cauchy–Schwarz $\mathcal{L}^2 \le T\int g\,dt$, which is an equality exactly when $\sqrt{g}$ is constant. (iv) With $g_\Pi(V,V) = \|X\|^2$, $g_\Pi(W,W) = \|Y\|^2$ and $g_\Pi(V,W)^2 + \tfrac{1}{4}\Omega_\Pi(V,W)^2 = |Q_\Pi(V,W)|^2 = |\mathrm{Tr}(XY^*)|^2$, the left side of (6.3) is $\|X\|^2\|Y\|^2 - |\mathrm{Tr}(XY^*)|^2$. Writing $Y = cX + (Y - cX)$ with the two parts orthogonal, $\|Y\|^2 = |c|^2\|X\|^2 + \|Y - cX\|^2$ and $|\mathrm{Tr}(XY^*)|^2 = |c|^2\|X\|^4$, which gives (6.3). ∎

Each equality case is a picture. In (i) the drive acts as k couplings of one strength across one gap; the simplest instance is k copies of a two-level system, and in the checks of this subsection that configuration meets the bound to 10^{-8} for $k = 1, 2, 3$. Moving one coupled partner from the gap Δ to 1.5Δ drops the ratio to 0.67–0.90, and unequal coupling strengths drop it to 0.79–0.87. In (ii) the equality is a rigid rotation: $\Sigma(t) = \frac{\Delta}{2}\,\mathbf{n}(t)\cdot\boldsymbol{\sigma}$ with $\mathbf{n}$ moving along a great circle at constant rate meets Theorem 5.2 exactly (ratio 1.000000), while the same rotation with a breathing gap reaches only 0.667. In (iii), with one gap and zero temperature, $\int g\,dt = \mathcal{L}^2/T$ exactly at constant speed (ratio 1.00000000), and a speed modulated by 60% overspends by 19%. The cheapest protocol therefore

moves the subspace along a geodesic at constant intrinsic speed, the thermodynamic-length result of Salamon and Berry in this setting [Salamon–Berry 1983].

Equation (6.3) says more than the equality case. For fixed V, the deficit of the curvature bound is a nonnegative quadratic form in W, and its kernel is exactly the plane spanned by V and its partner JV, the tangent vector with block iX. The map $J : X \mapsto iX$ is the complex structure of the Grassmannian, so equality and rigidity are one statement: two directions saturate the bound exactly when they span a complex line. (Identity (6.3) holds on 300 random pairs to 10^{-6} relative.)

PROPOSITION 6.4 *(The trace condition and its equality). For a two-parameter family and any Riemannian metric h on its parameter surface,*

$$\operatorname{tr}_h g \;\geq\; 2\sqrt{\det g / \det h} \;\geq\; \frac{|\Omega_{12}|}{\sqrt{\det h}}. \tag{6.4}$$

The first inequality is an equality exactly when g is conformal to h; the second exactly when the curvature bound is saturated.

PROOF. If $\mu_1, \mu_2 \geq 0$ are the eigenvalues of $h^{-1}g$, then $\operatorname{tr}_h g = \mu_1 + \mu_2 \geq 2\sqrt{\mu_1 \mu_2} = 2\sqrt{\det g / \det h}$, with equality exactly when $\mu_1 = \mu_2$, that is $g = \mu h$. The second inequality is $4 \det g \geq \Omega_{12}^2$, the pullback of (6.1). ∎

With h Euclidean on a Brillouin zone, (6.4) is the trace condition $\operatorname{tr} g \geq |\Omega|$ of ideal bands [Roy 2014]. With h the round metric of the parameter sphere, both bands of Appendices B and C meet it with equality at every point: the two-level band has $g = \tfrac{1}{4} h$ and $|\Omega_{\theta\phi}| = \tfrac{1}{2}\sin\theta$, the lowest spin-1 band $g = \tfrac{1}{2} h$ and $|\Omega_{\theta\phi}| = \sin\theta$ (both to 6×10^{-16}). Breaking the spin-1 symmetry with $0.8\, S_z^2$ lifts it strictly above, by 0.04 to 0.46 over the sample. The same equality holds for the lowest band in every spin j, which is the lowest Landau level on a sphere (Appendix C). Equality in both steps means the map is conformal and its image is a complex line at every point, which is what a holomorphic band is.

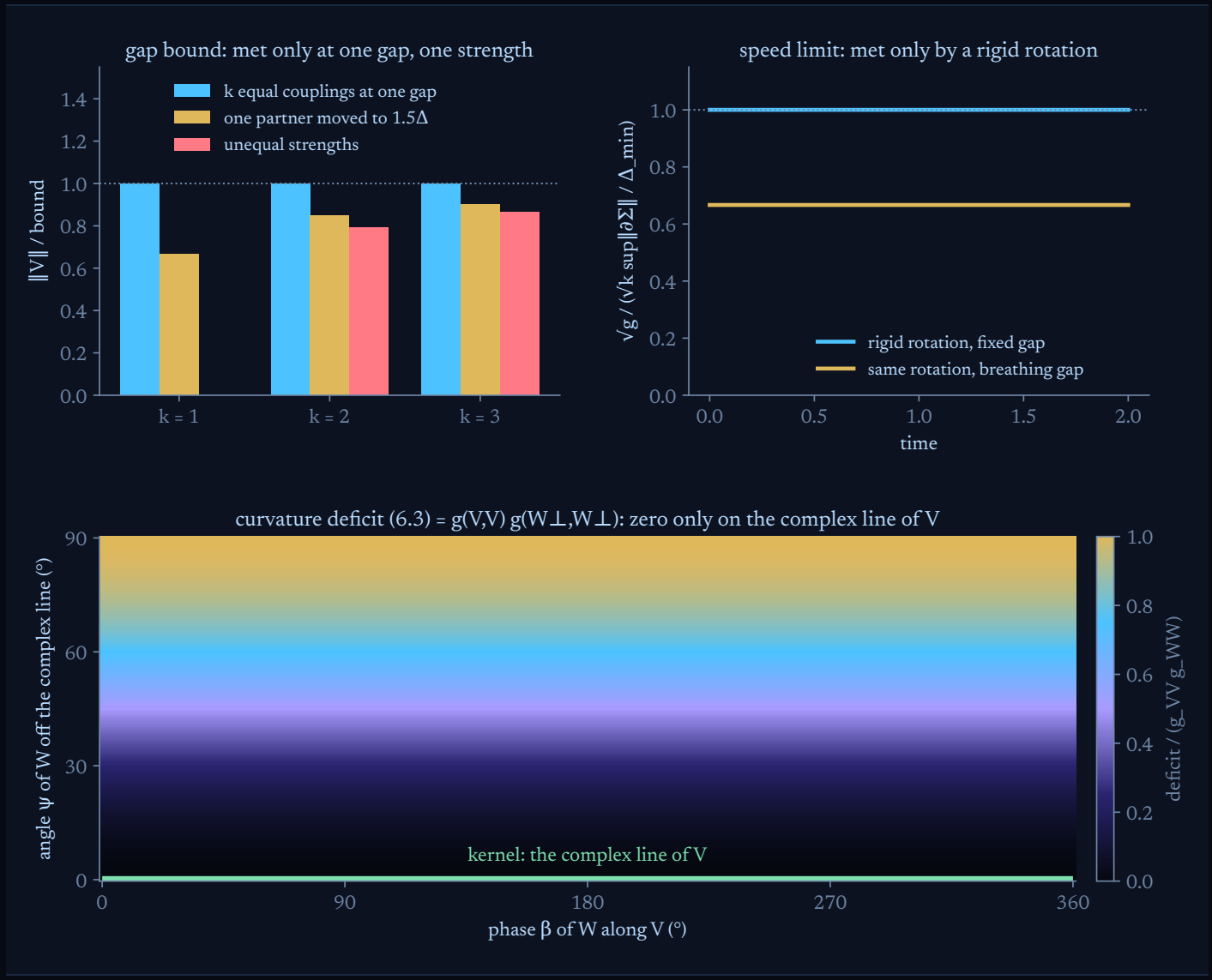


FIGURE 7. When the bounds are met. Top left: the gap bound is reached by k equal couplings across one gap, for k = 1, 2, 3; moving one coupled partner to 1.5Δ, or making the strengths unequal, falls short. Bottom: the deficit of the curvature bound, identity (6.3), as W turns away from the complex line of V; it vanishes on that whole line, at every phase β, and nowhere else. Top right: the speed limit of Theorem 5.2 is met at every instant by a rigid rotation at fixed gap; the same rotation with a breathing gap stays at 0.667 of it.

EXAMPLE 6.5 (A loop that comes back still remembers). Carry the lower two-level band once around the circle of polar angle $\theta_0 = 60°$. The projector returns exactly to where it started (to 10^{-16}), so no measurement of the final subspace can tell that the loop happened. The state vector does not return: it comes back multiplied by the Berry phase $-\frac{1}{2}\times$(enclosed solid angle) $= -\pi(1-\cos\theta_0) = -\pi/2$, computed from overlaps around the loop as -1.570797. The phase is the curvature of this section integrated over the cap, $\iint \Omega$. The subspace is back where it began; the history of the path is kept only in the phase, and only the curvature can write it there.

7 Geometric Transport Tensors as Pullbacks

Quantities such as the quantum metric dipole, the nonlinear Hall conductivity and the orbital magnetisation are usually computed in a band basis on a chosen parameter space. In the present framework each of them is read off the same cross-gap data that define Q_Π: the pairs $\langle m|\partial\Sigma|n\rangle$ with $m \in \mathcal{I},\ n \in \mathcal{I}^c$, weighted by powers of their gaps, together – at second order – with the rotation of the eigenvectors inside each sector. This section makes that statement exact for each tensor.

7.1 The quantum metric dipole

The quantum metric dipole measures the variation of the quantum metric across the parameter manifold, $G_{ij,k}(a) \equiv \partial_{a^k} g_{ij}(a)$. Differentiating the pullback $g_{ij} = \frac{1}{2}\mathrm{Tr}(\partial_i \Pi_k\, \partial_j \Pi_k)$ gives (4.5),

$$G_{ij,k} = \tfrac{1}{2}\,\mathrm{Tr}\big[(\partial_k\partial_i\Pi_k)\,\partial_j\Pi_k + \partial_i\Pi_k\,(\partial_k\partial_j\Pi_k)\big], \tag{7.1}$$

and differentiating the constraint $V_i = \Pi_k V_i + V_i \Pi_k$ gives the second-order response equation $\partial_k V_i = V_k V_i + \Pi_k(\partial_k V_i) + (\partial_k V_i)\Pi_k + V_i V_k$. With the exact second variation (4.6), (7.1) is the complete dipole. To resolve it in the eigenbasis, write $A^i = \partial_i \Sigma$ in the instantaneous eigenbasis and $\Delta_{mn} = \lambda_n - \lambda_m$. Holding the eigenvectors fixed and differentiating only the operator and the gaps gives the two-index form

$$G^{(2)}_{ij,k} = \sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c}\Bigg[\frac{\mathrm{Re}\big(\langle m|\partial_k\partial_i\Sigma|n\rangle A^j_{nm} + A^i_{mn}\langle n|\partial_k\partial_j\Sigma|m\rangle\big)}{\Delta_{mn}^2} - 2\,\frac{\mathrm{Re}\big(A^i_{mn}A^j_{nm}\big)\,\partial_k\Delta_{mn}}{\Delta_{mn}^3}\Bigg], \tag{7.2}$$

with $\partial_k\Delta_{mn} = A^k_{nn} - A^k_{mm}$ by Hellmann–Feynman. The eigenvectors also turn: $\partial_k\langle m|\partial_i\Sigma|n\rangle$ contains, besides $\langle m|\partial_k\partial_i\Sigma|n\rangle$, the rotation terms

$$\mathcal{R}^{ik}_{mn} = \sum_{l\neq m}\frac{A^k_{ml}A^i_{ln}}{\lambda_m - \lambda_l} + \sum_{l\neq n}\frac{A^i_{ml}A^k_{ln}}{\lambda_n - \lambda_l},$$

and for a spectrum that is nondegenerate within each sector the dipole is exactly

$$G_{ij,k} = G^{(2)}_{ij,k} + \sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c}\frac{\mathrm{Re}\big(\mathcal{R}^{ik}_{mn}A^j_{nm} + A^i_{mn}\mathcal{R}^{jk}_{nm}\big)}{\Delta_{mn}^2}. \tag{7.3}$$

The rotation sum runs over every level l, including those in the same sector as m or n: these are the three-state terms, and they live in exactly the off-diagonal block that survives the trace in (7.1). On

forty random families (7.3) reproduces the contour value of (7.1) to 4×10^{-12}, while the two-index form (7.2) alone can be off by more than a factor of a hundred where its terms cancel. The dipole is therefore controlled by the cross-gap channels, the gap variations and the rotation of the eigenvectors within each sector — the last being the second-order content that Proposition 4.3 isolates.

7.2 One pair sum, many weights

Take a nondegenerate ground state $|0\rangle$, gaps $\Delta_n = E_n - E_0$, and for each excited level the pair metric and pair curvature

$$g_n = \frac{|\langle 0|\partial_x \Sigma|n\rangle|^2}{\Delta_n^2}, \qquad \Omega_n = -\frac{2\,\mathrm{Im}\big(\langle 0|\partial_x \Sigma|n\rangle\langle n|\partial_y \Sigma|0\rangle\big)}{\Delta_n^2},$$

so that $g = \sum_n g_n$ and $\Omega = \sum_n \Omega_n$ by Theorem 3.6. The responses of transport theory are the same pair sums with different weights in the gap:

QUANTITY	PAIR SUM	WEIGHT	CHECKED AGAINST
quantum metric	$\sum_n g_n$	1	Theorem 3.6
static response $-\partial_x^2 E_0$	$\sum_n 2\Delta_n g_n$	2Δ	finite differences of E_0, 7×10^{-6}
Berry connection polarizability G_{xx}	$-\sum_n 2g_n/\Delta_n$	$-2/\Delta$	interband Berry connections, 2×10^{-7}
zero-temperature rotation friction	$\sum_n 2c(\Delta_n)\Delta_n g_n$	$2c\Delta$	Theorem 5.3
Berry curvature	$\sum_n \Omega_n$	1	Theorem 3.6
orbital-moment kernel $\mathrm{Im}\langle\partial_x u_0\|(\Sigma - E_0)\|\partial_y u_0\rangle$	$-\frac{1}{2}\sum_n \Delta_n \Omega_n$	Δ	gauge-fixed finite differences, 4×10^{-8}

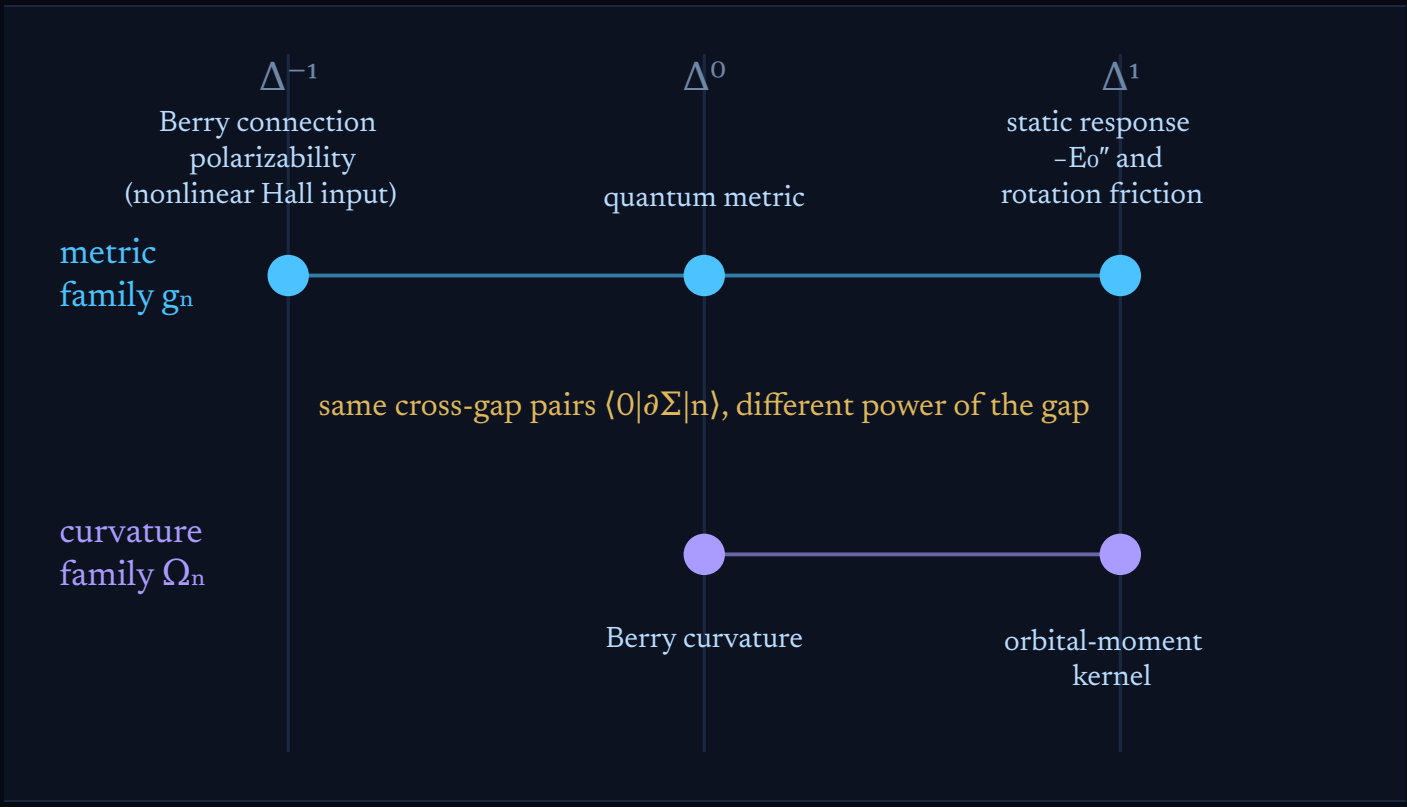


FIGURE 8. One pair sum, many weights. The same cross-gap pairs, read with different powers of the gap, give the quantum metric, the Berry connection polarizability that drives the intrinsic nonlinear Hall effect, the static response and the rotation friction; in the curvature family, the Berry curvature and the orbital moment.

The metric family and the curvature family each run through the powers of the gap. The nonlinear Hall input sits one power below the metric, the static response one power above, and the orbital moment one power above the curvature. Which power a measurement selects tells which part of the cross-gap geometry it reads, and fixes how it behaves at a closing gap (Section 7.5). With $\alpha(x) = \sum_n 2\Delta_n^3 g_n/(\Delta_n^2 - x^2)$, the static response function, the friction of Section 5 is read directly from it: $\zeta_{\rm rot} = \tau\,\alpha(0)$ for pure relaxation and $\zeta_{\rm rot} = \tau\big[\alpha(0) - \alpha(i/\tau)\big]$ with precession (to 10^{-13} on 200 random systems) — the familiar statement that a friction coefficient is a response function evaluated at the bath's rate, written in the spectral weights of this paper.

7.3 Nonlinear Hall conductivity

The second-order current $j_a = \sigma^{ab;c} E_b E_c$ generated in the absence of a magnetic field has an intrinsic, scattering-time-independent part fixed by quantum geometry. Its input is not the quantum metric itself but the Berry connection polarizability, the band-normalised metric

$$G_n^{ab} = 2\,\mathrm{Re}\sum_{m\neq n} \frac{\mathcal{A}_{nm}^a \mathcal{A}_{mn}^b}{\varepsilon_n - \varepsilon_m}, \qquad \mathcal{A}_{nm}^a = i\langle u_n|\partial_a u_m\rangle,$$

which for an isolated band is the pair sum with weight $-2/\Delta$ of the table above. The intrinsic conductivity is built from its dipoles [Gao–Yang–Niu 2014; Kaplan–Holder–Yan 2024]:

$$\sigma^{ab;c}_{\mathrm{int}} = -\frac{e^3}{\hbar}\sum_n \int_{\mathbf{k}} f_n \Big[2\,\partial_{k_c} G^{ab}_n - \tfrac{1}{2}\left(\partial_{k_a} G^{bc}_n + \partial_{k_b} G^{ac}_n\right)\Big]. \tag{7.4}$$

In the language of this paper, (7.4) is the dipole of a pullback: the same cross-gap pairs as g, weighted one power of the gap lower, differentiated along the Brillouin zone. The second-order machinery of Section 7.1 — the exact second variation and the rotation terms — is what its spectral evaluation requires; a form that differentiates only the operator and the gaps misses the same three-state terms as (7.2).

7.4 Orbital magnetisation

The orbital moment of a Bloch state is the curvature-family pair sum weighted by one power of the gap, $m_n \propto \mathrm{Im}\langle \partial u_n| \times (\Sigma - \varepsilon_n)|\partial u_n\rangle = -\frac{1}{2}\sum_m \Delta_{nm}\Omega_{nm}$ (table above), so orbital magnetism belongs to the antisymmetric half of the cross-gap geometry. A nonlinear orbital response $M_i = \alpha_{ijk} E_j E_k$ is the field derivative of that kernel; its full second-order expression, which needs the second-order machinery of Section 7.1 applied to the curvature family, is not derived here.

7.5 Singular nodes and regularisation

When the gap closes the denominators diverge, each tensor at the rate set by its weight. Following each quantity toward the charge-neutral point with everything else fixed (Figure 9), the metric and the curvature diverge as Δ^{-2}, the dipole one power faster as Δ^{-3}, the zero-temperature rotation friction as Δ^{-1}, and the finite-temperature friction not at all, since the population difference $\tanh(\beta\Delta/2)$ closes with the gap. The measured slopes are $-2.000, -2.000, -2.997, -0.996, +0.001$.

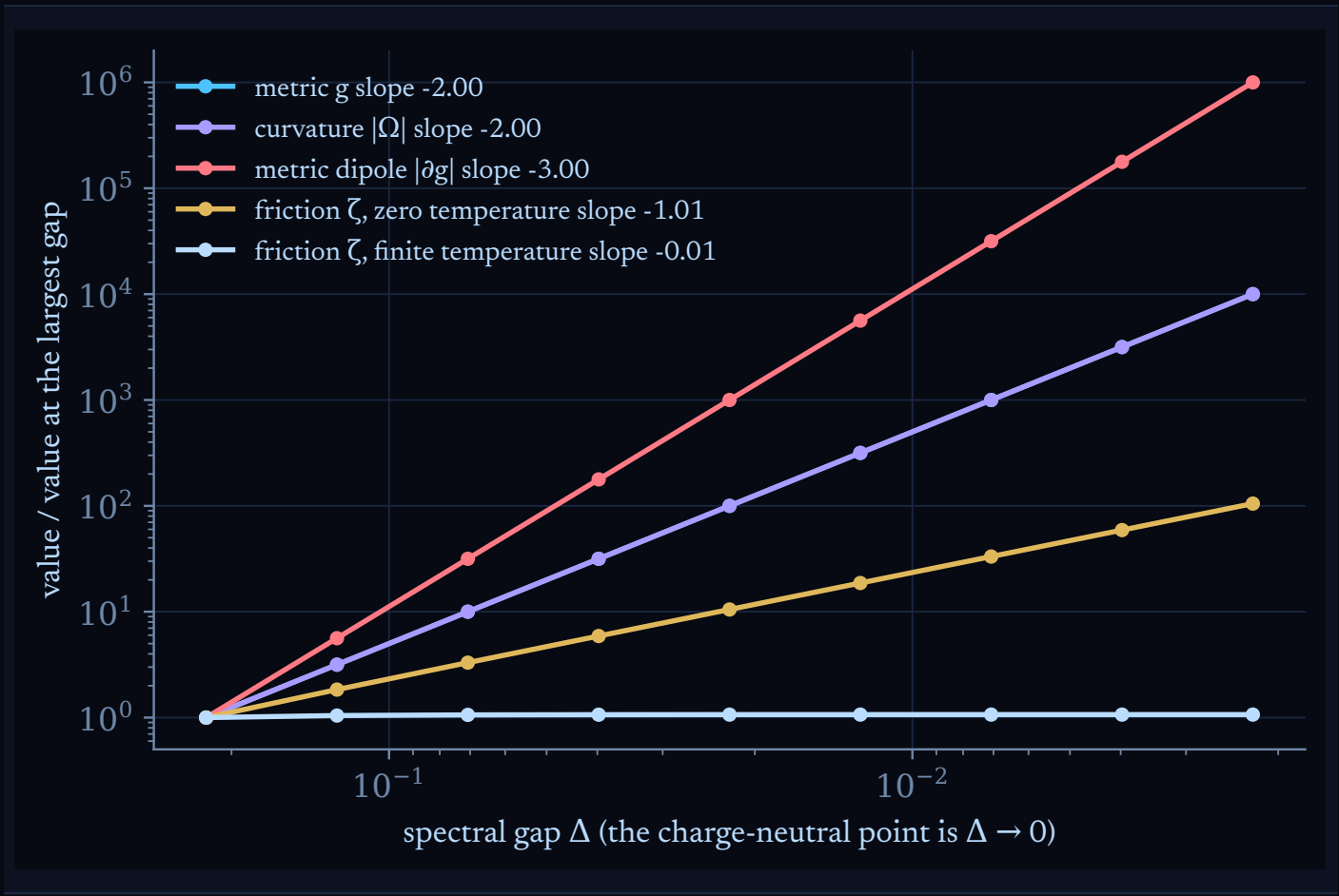


FIGURE 9. Closing the gap. Each quantity is normalised at the largest gap and followed toward the charge-neutral point. Metric and curvature diverge together and the dipole one power faster, the zero-temperature friction one power slower, while the finite-temperature cost of rotating the projector stays flat.

A finite quasiparticle lifetime enters through a complex regularisation $i\eta$, $\eta > 0$, in the interband denominators,

$$\langle m|V_\eta|n\rangle = \frac{\langle m|\delta\Sigma|n\rangle}{\lambda_m - \lambda_n + i\eta}, \tag{7.5}$$

which replaces the poles by Lorentzian profiles. V_η is not self-adjoint – $\langle n|V_\eta|m\rangle$ carries $\lambda_n - \lambda_m - i\eta$ where $\langle m|V_\eta|n\rangle$ carries $\lambda_m - \lambda_n + i\eta$ – so it is not a tangent vector, and the metric, Gram positivity and real angles of Sections 2–6 do not apply to it. What the prescription defines is a retarded response, and the modulus-squared weight $\left((\lambda_n - \lambda_m)^2 + \eta^2\right)^{-1}$ that follows from it is the correct regularised weight for response calculations. The regularised pair sum $\sum |\langle m|\delta\Sigma|n\rangle|^2/(\Delta_{mn}^2 + \eta^2)$ stays below $\sum |\langle m|\delta\Sigma|n\rangle|^2/\eta^2$ even at a closed gap and returns the metric as $\eta \to 0$ on an open one. Applied to the two-index form (7.2), the regularisation gives

$$G^{(2),\eta}_{ij,k} = \sum_{m\in\mathcal{I}}\sum_{n\in\mathcal{I}^c}\Bigg[\frac{\mathrm{Re}\big(\langle m|\partial_k\partial_i\Sigma|n\rangle A^j_{nm} + A^i_{mn}\langle n|\partial_k\partial_j\Sigma|m\rangle\big)}{\Delta_{mn}^2+\eta^2} - 2\,\frac{\mathrm{Re}\big(A^i_{mn}A^j_{nm}\big)\,\Delta_{mn}\,\partial_k\Delta_{mn}}{\big(\Delta_{mn}^2+\eta^2\big)^2}\Bigg],$$

whose gap-derivative term is the derivative of the regularised weight $1/(\Delta_{mn}^2+\eta^2)$, exactly as the term in (7.2) is the derivative of $1/\Delta_{mn}^2$. It vanishes with the gap, and the $1/\Delta^3$ singularity becomes a peak at $\Delta_{mn}=\eta/\sqrt{3}$ of height $(3\sqrt{3}/8)/\eta^3$ times $|\mathrm{Re}(A^i_{mn}A^j_{nm})\,\partial_k\Delta_{mn}|$; the rotation terms of (7.3) are regularised the same way. Lifetimes near nodes proper require a non-Hermitian or Lindblad formulation, which is a different theory.

8 Geodesic Refraction Across Degenerate Strata

The metric–curvature inequality governs the magnitude of geometric phase effects but does not prescribe how geometric transport continues across loci where the effective rank of the pulled-back tensor changes. By Proposition 4.1 the pullback metric may degenerate on codimension-one strata of M_Δ without the spectral gap closing. The established band-geometry literature assumes this does not happen — the map into the Grassmannian is taken to be an immersion, so that $\det g \neq 0$ throughout. This section asks what governs the geodesics of the pullback metric when that assumption fails. Geodesics are the object studied throughout: they are the paths a transport calculation would follow, not a measurement of transport.

8.1 The refraction law

Suppose g admits a Killing field tangent to a family of hypersurfaces, the interfaces; let x be the coordinate along the Killing direction and y transverse to it, so that $g = g_\parallel(y)\,dx^2 + g_\perp(y)\,dy^2$. Let θ be the angle, measured in g, between a trajectory and the interface normal.

PROPOSITION 8.1 *(Variational refraction). Along any geodesic of g,*

$$\sqrt{g_\parallel}\,\sin\theta = \text{const}, \qquad \text{equivalently} \qquad g_\parallel \sin^2\theta = \text{const}. \tag{8.1}$$

PROOF. The Lagrangian $g_\parallel \dot{x}^2 + g_\perp \dot{y}^2$ is independent of x, so $p_x = g_\parallel \dot{x}$ is conserved. For a unit-speed geodesic the along-interface component of the velocity is $\sin\theta = \sqrt{g_\parallel}\,\dot{x}$, whence $p_x = \sqrt{g_\parallel}\sin\theta$. Squaring gives the second form. ∎

Across two sides of an interface, (8.1) is the sine-squared refraction law $R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2$, with rigidity $R = g_{\parallel}$. It is Clairaut's relation, and therefore Snell's law with refractive index $n = \sqrt{g_{\parallel}}$. Three consequences fix the reading. The sine-squared form is the square of ordinary Snell, not an independent law. There is no separate index to introduce: Fermat's principle with g as the metric yields (8.1) directly. And the quantity playing the role of the index is $g_{\parallel}$, the metric component along the interface — not the smallest eigenvalue $\lambda_{\min}$ of g, and not its reciprocal. Near a degenerate stratum the collapsing direction is the along-interface direction, so $\lambda_{\min}$ and $g_{\parallel}$ coincide there; a rule written with $\lambda_{\min}$ is correct in the regime where it is usually applied and fails away from it (Figure 11).

8.2 The index is a subspace rotation rate

PROPOSITION 8.2. *For subspaces $\mathrm{Ran}(\Pi_1), \mathrm{Ran}(\Pi_2)$ with principal angles ϑ_i, $\|\Pi_1 - \Pi_2\|_{\mathrm{HS}}^2 = 2\sum_i \sin^2\vartheta_i$. Since $\|\delta\Pi\|_{\mathrm{HS}}^2 = 2g_{\Pi}(\delta\Pi, \delta\Pi)$, $\sqrt{g} = \sin\vartheta$ to leading order for a single principal angle, and the refractive index $n = \sqrt{g_{\parallel}}$ is the rate at which the occupied subspace rotates as one moves along the interface.*

Refraction in this setting is therefore not an optical analogy imported from outside. The trajectory bends because the subspace rotates at different rates on the two sides of the stratum, and $\sin^2$ appears because the Hilbert–Schmidt metric on $\mathrm{Gr}(k, \mathcal{H})$ is itself a sum of squared sines of principal angles.

8.3 Total internal reflection

PROPOSITION 8.3 *(Total internal reflection at a degenerate stratum). Let $C = g_{\parallel}\sin^2\theta$ be the conserved quantity of (8.1). Since $\sin^2\theta \le 1$, a geodesic can only visit points where $g_{\parallel} \ge C$. A trajectory with $C > 0$ cannot reach a stratum on which $g_{\parallel} \to 0$: it turns at the locus $g_{\parallel} = C$, where $\sin\theta = 1$ and the velocity is tangent to the interface. Only trajectories with $C = 0$, already exactly normal, reach the stratum.*

The degenerate stratum is repulsive: generic transport does not approach it, and is turned away before reaching it. This is the directional protection of the geometry, correctly stated; the mechanism is the caustic of a metric whose along-interface component vanishes, the same as total internal reflection in optics.

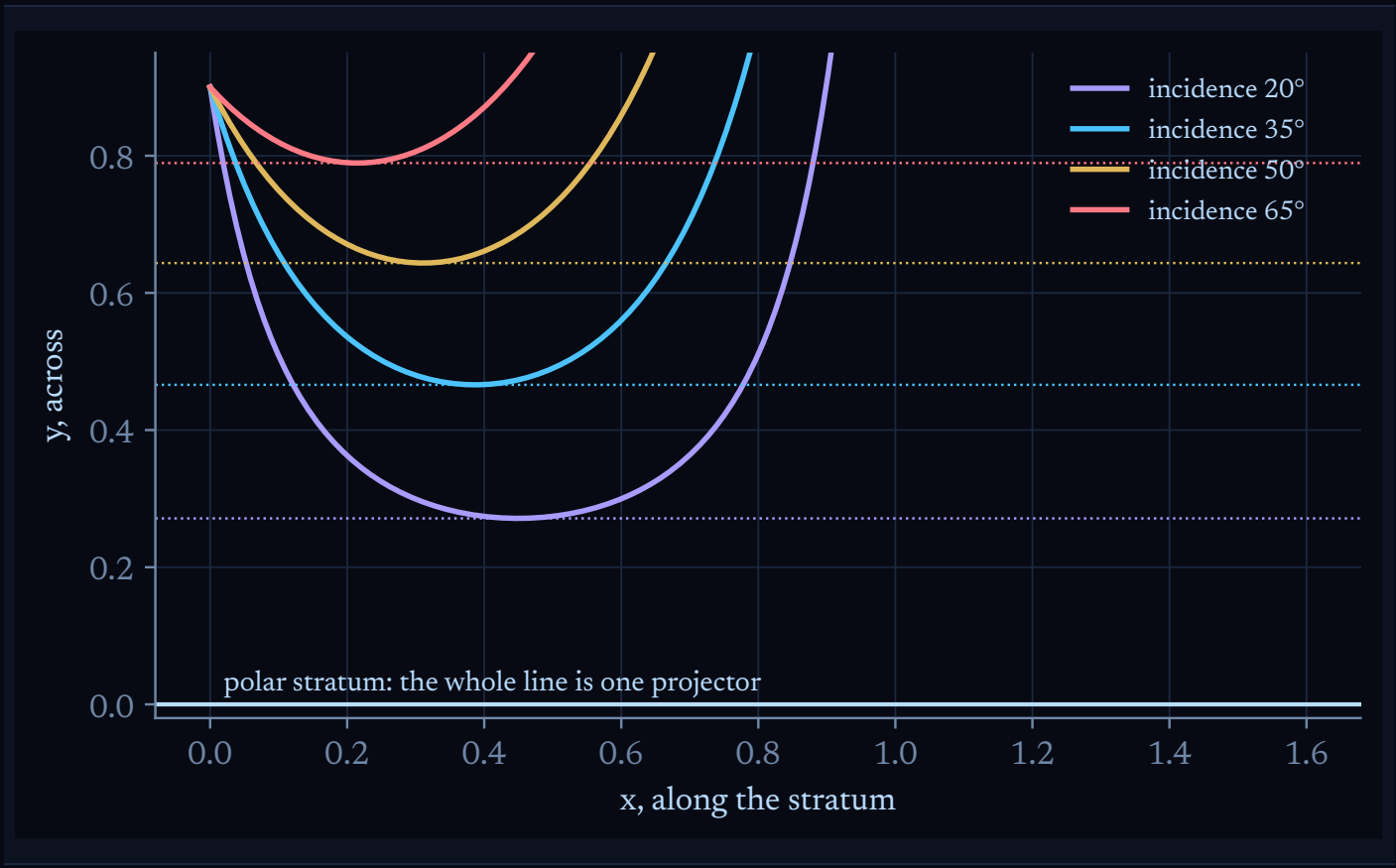


FIGURE 10. Total internal reflection at a polar stratum, with the gap held fixed. Geodesics launched toward y = 0 turn tangentially where g∥ falls to their conserved level C (dotted). None reaches the stratum.

8.4 A solvable test

Take the two-band operator $\Sigma = \hat{\mathbf{d}} \cdot \boldsymbol{\sigma}$ with $\hat{\mathbf{d}}(x, y) = (\sin y \cos kx,\ \sin y \sin kx,\ \cos y)$, $k = 3$, so that the spectral gap equals 2 everywhere: any degeneration of the pulled-back metric is failure of immersion, not a closing gap. The metric computed from Π by central differences is $g = \mathrm{diag}\left(\tfrac{1}{4}k^2 \sin^2 y,\ \tfrac{1}{4}\right)$ to 5×10^{-10}, and $\sqrt{g_{\parallel}}\, dx$ equals the principal angle between $\Pi(x, y)$ and $\Pi(x + dx, y)$, computed from eigenvector overlaps with no trace identity, to 10^{-8} (Proposition 8.2). The full geodesic system, integrated from $y = 0.15$ at $\theta = 60°$ with no conservation law imposed, crosses the branch switch of $\lambda_{\min}$ at $\sin y = 1/k$; along it $g_{\parallel} \sin^2 \theta$ drifts by 1.7×10^{-12}, while $\lambda_{\min} \sin^2 \theta$ drifts by 1.3 and $\sin^2 \theta / \lambda_{\min}$ by 3.3. A geodesic launched inward from $y = 0.60$ at $\theta = 35°$ has $C = 0.23600011$ and a predicted turning point $y = 0.32981248$; the integrated trajectory turns at $y = 0.32981248$ with $\sin \theta = 1.00000000$ — grazing incidence, not normal. Over incidences from $20°$ to $89°$ every trajectory turns before the stratum; the exactly normal one passes through it.

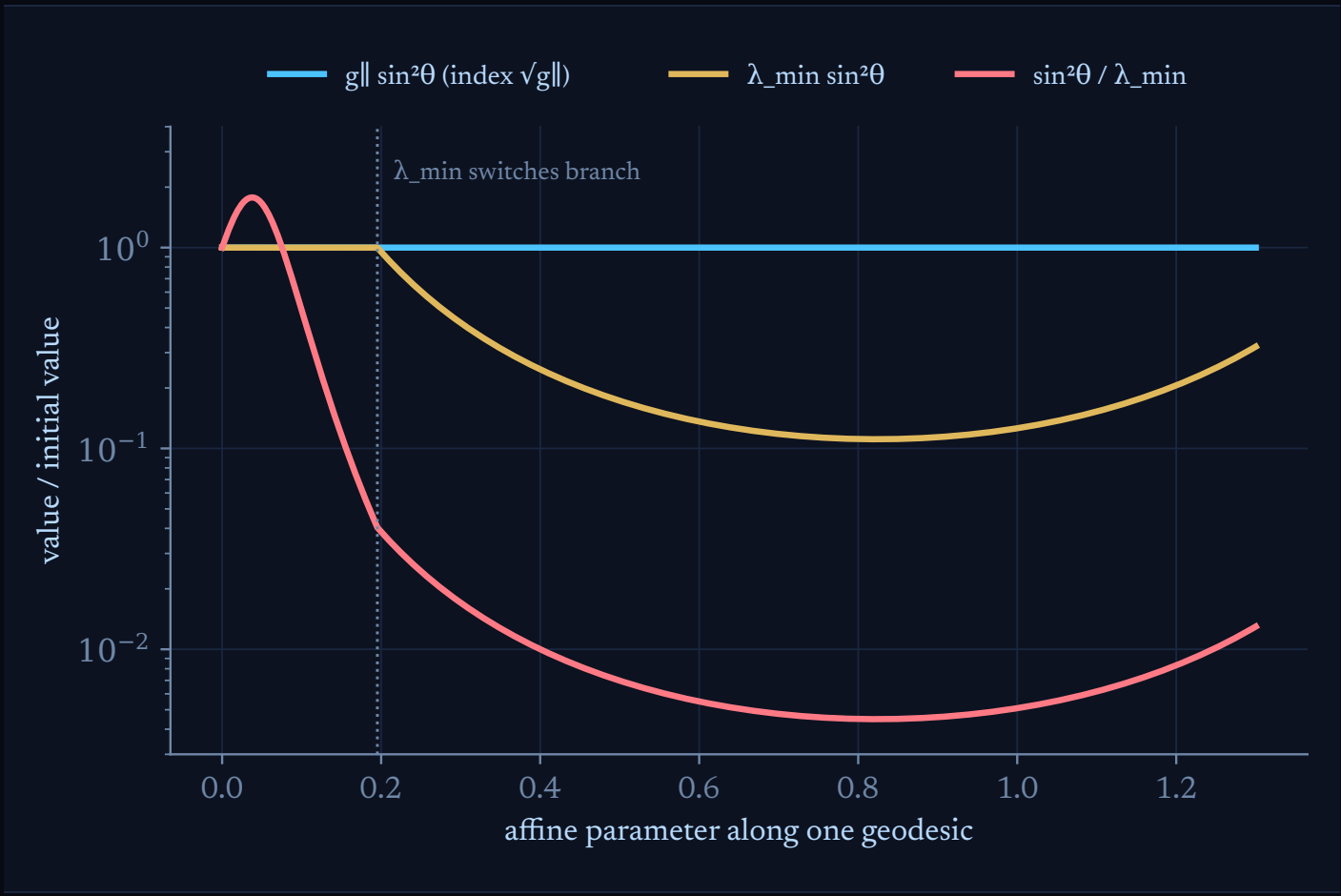


FIGURE 11. Three candidate refractive indices along one geodesic, each normalised to its initial value. The metric component along the interface gives a conserved quantity. Neither λ_min nor its reciprocal does; λ_min agrees with g∥ only until it switches branch (dotted).

REMARK 8.4 (What bends a geodesic is a metric dipole). The geodesic equation $\ddot{a}^k + \Gamma^k_{ij}\dot{a}^i\dot{a}^j = 0$ is driven by the Christoffel symbols, and those of the first kind are half-sums of derivatives of the metric:

$$\Gamma_{ij,k} = \tfrac{1}{2}\big(G_{ik,j} + G_{jk,i} - G_{ij,k}\big), \qquad G_{ij,k} = \partial_k g_{ij}. \tag{8.2}$$

$G_{ij,k}$ is the quantum metric dipole of Section 7.1. In the coordinates of Proposition 8.1 the only transverse force is $\Gamma_{xx,y} = -\tfrac{1}{2}\partial_y g_{\parallel}$, and in the test above, computed from the projector by finite differences, it agrees with the dipole to 3×10^{-8}. The bending of this section and the dipole responses of Section 7 are therefore the same derivative of the same metric, read once as a force on a path and once as a transport coefficient.

8.5 Two kinds of degenerate stratum

Let L be a connected interface on which g is degenerate, with x along L and y across it. Which direction collapses decides everything that follows.

LEMMA 8.5 *(A stratum on which nothing moves). On L, $g_{\parallel} \equiv 0$ if and only if $\partial_x \Pi \equiv 0$, if and only if Π is constant along L: the whole interface maps to one point of $\mathrm{Gr}(k, \mathcal{H})$.*

PROOF. $g_{\parallel} = g_\Pi(\partial_x \Pi, \partial_x \Pi) = \tfrac{1}{2}\|\partial_x \Pi\|_{\mathrm{HS}}^2$ vanishes only when $\partial_x \Pi = 0$. A curve with zero derivative on a connected interval is constant. The converse is immediate. ∎

So a degenerate stratum is one of two kinds. It is *polar* if the kernel of g is tangent to it: then, by Lemma 8.5, Π is constant along it, and the refraction law with its total internal reflection applies. The stratum of Section 8.4 is polar: the whole line $y = 0$ is one projector. It is a *fold* if the kernel points across it while Π keeps moving along it, as on the Bloch sphere with $\theta = y + 1.1,\ \phi = x^2/2$, where $\partial_x \Pi = 0$ on $x = 0$ while $|\partial_y \Pi| = 0.49$. The two are the null-hypersurface and caustic cases of the singularity theory of maps.

The first-order metric cannot tell the two apart beyond the direction of its kernel. The next order can.

PROPOSITION 8.6 *(The first order at which a silent direction moves). Let v be a silent direction at a point, $\partial_v \Pi = 0$, and let $m \geq 2$ be the first order with $\partial_v^m \Pi \neq 0$. Then $\partial_v^m \Pi$ is block off-diagonal, so it is itself a tangent vector, and along the line through the point in direction v*

$$d(\varepsilon) = \frac{|\varepsilon|^m}{m!}\sqrt{g_\Pi\big(\partial_v^m \Pi, \partial_v^m \Pi\big)} + O(\varepsilon^{m+1}), \tag{8.3}$$

where $d = \|\Pi(\varepsilon) - \Pi\|_{\mathrm{HS}}/\sqrt{2}$ is the chordal distance of Proposition 8.2. If m is even, $\Pi(\varepsilon)$ and $\Pi(-\varepsilon)$ agree to order ε^m: two sheets of the family meet, and the direction is a fold. If m is odd, they separate at order ε^m: the family passes through, and the reparametrisation $t = \varepsilon^m$ removes the silence.

PROOF. Differentiating $\Pi^2 = \Pi$ m times gives $\Pi^{(m)} = \sum_{i=0}^{m} \binom{m}{i} \Pi^{(i)} \Pi^{(m-i)}$. The derivatives of orders $1, \dots, m-1$ vanish, so this is $\Pi^{(m)} = \Pi^{(m)}\Pi + \Pi\Pi^{(m)}$, the tangent law of Section 2; multiplying by Π on both sides gives $\Pi\Pi^{(m)}\Pi = 2\Pi\Pi^{(m)}\Pi$, so $\Pi\Pi^{(m)}\Pi = 0$, and multiplying by $I - \Pi$ on both sides gives $(I - \Pi)\Pi^{(m)}(I - \Pi) = 0$. Taylor's theorem gives $\Pi(\varepsilon) - \Pi = \frac{\varepsilon^m}{m!}\Pi^{(m)} + O(\varepsilon^{m+1})$, and $\|\Pi^{(m)}\|_{\mathrm{HS}}^2 = 2 g_\Pi(\Pi^{(m)}, \Pi^{(m)})$. The leading term is even in ε exactly when m is even, which gives the last two statements. ∎

This sorts the strata by order. Along a polar stratum Π is silent at every order. Across a fold it is silent at first order and moves at second: the distance grows as ε^2, not ε, and the two sides meet. In the fold above, at $x = 0$, $\partial_x \Pi = 0$ exactly, the diagonal blocks of $\partial_x^2 \Pi$ vanish to 5×10^{-17}, and the ratio of $d(\varepsilon)$ to the leading term of (8.3) is 0.999999, 1.000000, 1.000000 at $\varepsilon = 0.1, 0.03, 0.01$; along the polar stratum the projector drifts by 10^{-15} over the whole line (Figure 12). On the Bloch sphere with

$\phi = \varepsilon^m/m!$, the same holds at every order checked, $m = 2, 3, 4, 5$: the ratio to (8.3) is 1.000000; for even m the two sides coincide, and for odd m they separate with $|\Pi(\varepsilon) - \Pi(-\varepsilon)| = 2|\Pi(\varepsilon) - \Pi|$ to leading order. Adding a third-order term to a fold, $\phi = \varepsilon^2/2 + 0.3\varepsilon^3$, makes the two sides separate only at the next order, with the ratio falling as $1.45\,\varepsilon$. The metric is a quadratic form with a kernel, and what happens in the kernel is decided by the first order that moves: even orders fold, odd orders are an artefact of the parametrisation, and a polar stratum never moves at all.

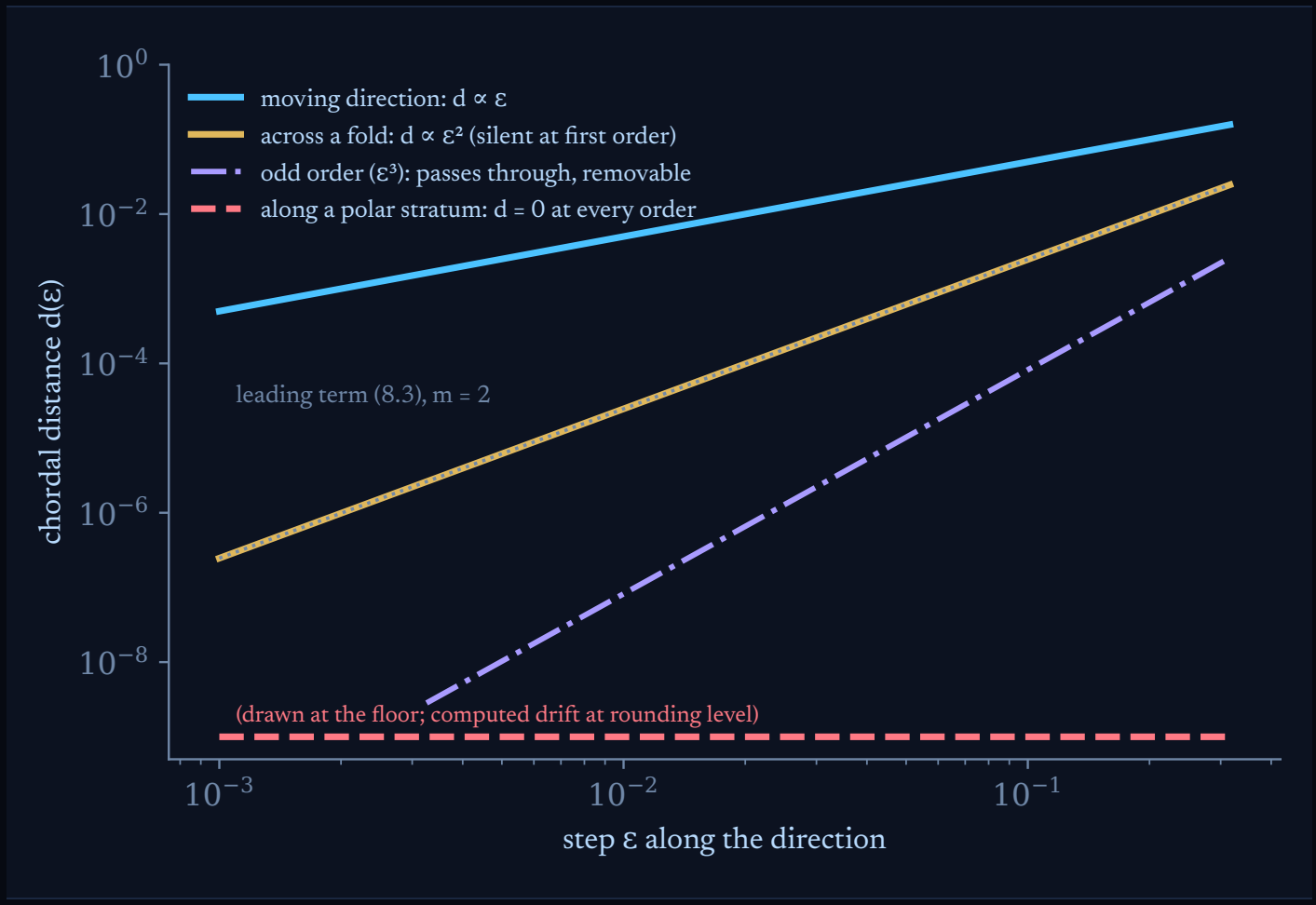


FIGURE 12. Silent directions sorted by order. Along a moving direction the chordal distance grows as ε; across a fold the projector is silent at first order and the distance grows as ε², on the leading term (8.3) of Proposition 8.6; at an odd first order (ε³) the family passes through and the silence is removable by reparametrisation; along a polar stratum it does not grow at any order (its computed drift is at rounding level, drawn at the floor).

REMARK 8.7 (What a fold needs). The Christoffel symbols (8.2) are contracted with g^{-1}, and both kinds of stratum make them diverge: at the polar stratum of Section 8.4, $\Gamma^x_{xy} = \cot y$; at the fold above, where $g = \operatorname{diag}\big(\tfrac{1}{4}\sin^2(y + 1.1)\,x^2,\ \tfrac{1}{4}\big)$, $\Gamma^x_{xx} = 1/x$. At a polar stratum this is harmless, because the conserved quantity of (8.1) keeps every trajectory with $C > 0$ away from it. At a fold nothing does: the Clairaut constant contains $g_{\parallel}$ only, $g_{\parallel}$ does not vanish there, trajectories reach the fold at every incidence, and the

geodesics of g alone are not determined across it. The geodesic problem must be supplemented by a choice of direction in the kernel, which the degenerate metric does not supply. The same structure is basic in Carrollian geometry, where a degenerate metric describes a null hypersurface only together with a vector field in its kernel [Ciambelli–Jai-akson 2025]. Only the polar case is governed by Proposition 8.3.

EXAMPLE 8.8 (Three bodies seen through their distances). Place three equal masses in the plane at $q_1, q_2, q_3 \in \mathbb{C}$, and form the Jacobi vectors $\rho_1 = (q_2 - q_1)/\sqrt{2}$, $\rho_2 = (2q_3 - q_1 - q_2)/\sqrt{6}$ and the vector $\psi = (\rho_1, \rho_2) \in \mathbb{C}^2$. The projector $\Pi = \psi\psi^*/|\psi|^2 \in \mathrm{Gr}(1, \mathbb{C}^2)$ forgets the overall size and the planar rotation, which is the phase of ψ, and keeps exactly the shape of the triangle: $\mathrm{Gr}(1, \mathbb{C}^2)$ is the shape sphere of the three-body problem [Montgomery 2015]. Its Bloch vector $w = (|\rho_1|^2 - |\rho_2|^2,\ 2\,\mathrm{Re}\,\rho_1\bar{\rho}_2,\ 2\,\mathrm{Im}\,\rho_1\bar{\rho}_2)/|\psi|^2$ has third component proportional to the signed area, so the collinear triangles form the great circle $w_3 = 0$.

Now record the shape only through the normalised squared distances $r_{ij}^2/\sum r^2$. They are linear in $|\rho_1|^2$, $|\rho_2|^2$ and $\mathrm{Re}\,\rho_1\bar{\rho}_2$, so they see w_1 and w_2 and not the sign of w_3: a triangle and its mirror image, $w_3 \mapsto -w_3$, have the same distances (difference exactly 0). The metric that the distances pull back to the sphere is degenerate exactly on the collinear circle, where the ratio of its singular values is 0 (it is 0.24 at latitude 0.4). Its kernel points straight across the circle (latitude component $\mathbf{1.000000}$), while Π keeps moving along it ($|\partial\Pi| \geq \mathbf{0.707}$). To an observer who sees only distances, the collinear triangles are a fold: the two mirror sheets of the sphere meet there, and the direction that observer cannot see is the orientation of the triangle. The three-body motion itself runs on the nondegenerate round metric of the shape sphere, inherited from the mass metric, and that metric supplies exactly the structure the fold lacks (Figure 13).

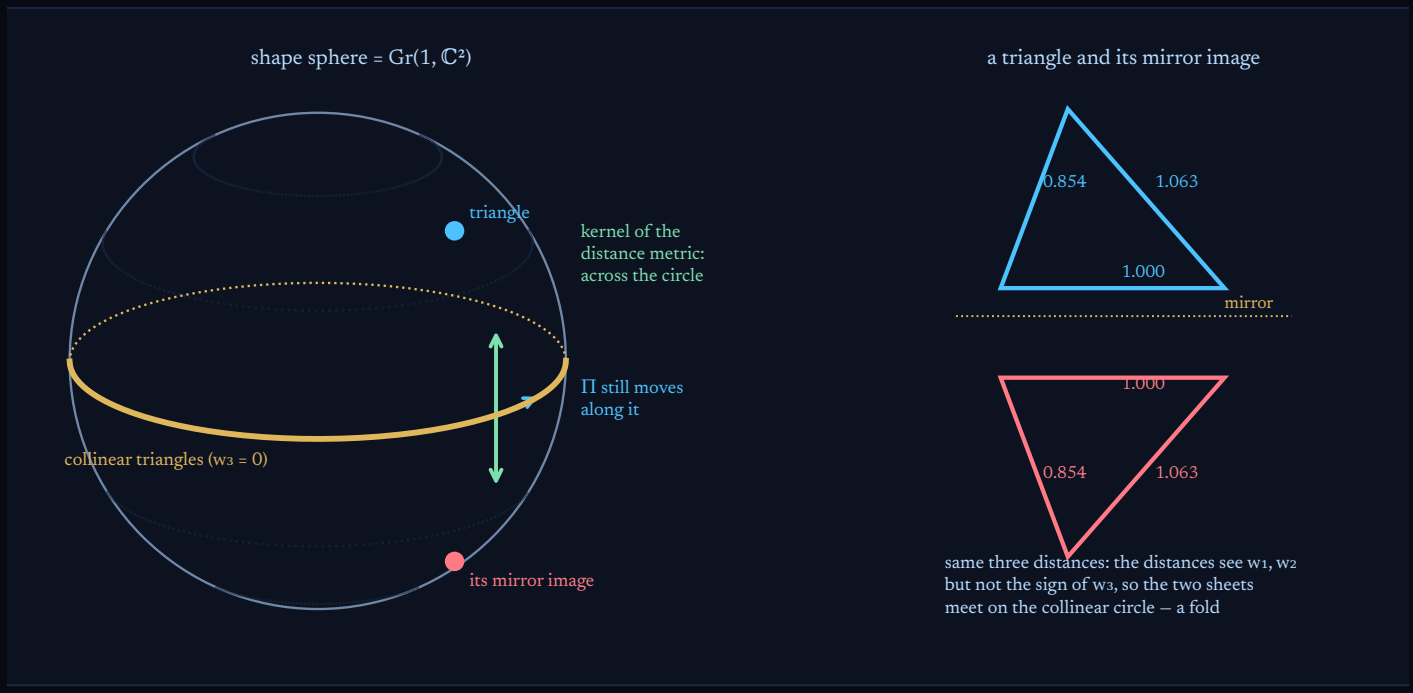


FIGURE 13. Three bodies seen through their distances. Left: the projector of the Jacobi vector is the shape of the triangle, a point of Gr(1, ℂ²); collinear triangles form the great circle $w_3 = 0$. The metric pulled back by the mutual distances has its kernel across that circle while the projector keeps moving along it, so the collinear circle is a fold. Right: a triangle and its mirror image have the same three distances; they are the two sheets that meet at the fold.

8.6 When the gap narrows instead

The refraction law asks only for the pullback metric, so it applies unchanged when the index varies because the gap varies rather than because the image of the family folds. What changes is the direction of the effect. By Lemma 3.1 a fixed coupling divided by a smaller gap moves the projector faster, so where the gap narrows $g_{\parallel}$ rises and the index rises with it. Trajectories then bend toward the normal, cross the narrow-gap line, and are turned back only where the index has fallen to their conserved level on either side: the narrow-gap line guides them as a graded-index fibre guides light.

In the family $\Sigma(x, y) = \cos(3x)\,\sigma_x + \sin(3x)\,\sigma_y + 2.5y\,\sigma_z$, the gap $2\sqrt{1 + 6.25y^2}$ is narrowest on $y = 0$ and the occupied projector never stops rotating: $g_{\parallel} = 9/\big(4(1 + 6.25y^2)\big)$. A geodesic launched from $y = 0.8$ at $50°$ conserves $g_{\parallel}\sin^2\theta$ to 1.1×10^{-11} (the $\lambda_{\min}$ candidate drifts by 4.4), crosses $y = 0$, and turns at $y = -1.0969$, where the conserved level predicts 1.0970 (Figure 14). Total internal reflection is therefore a property of strata on which the subspace stops rotating, not of places where the gap closes.

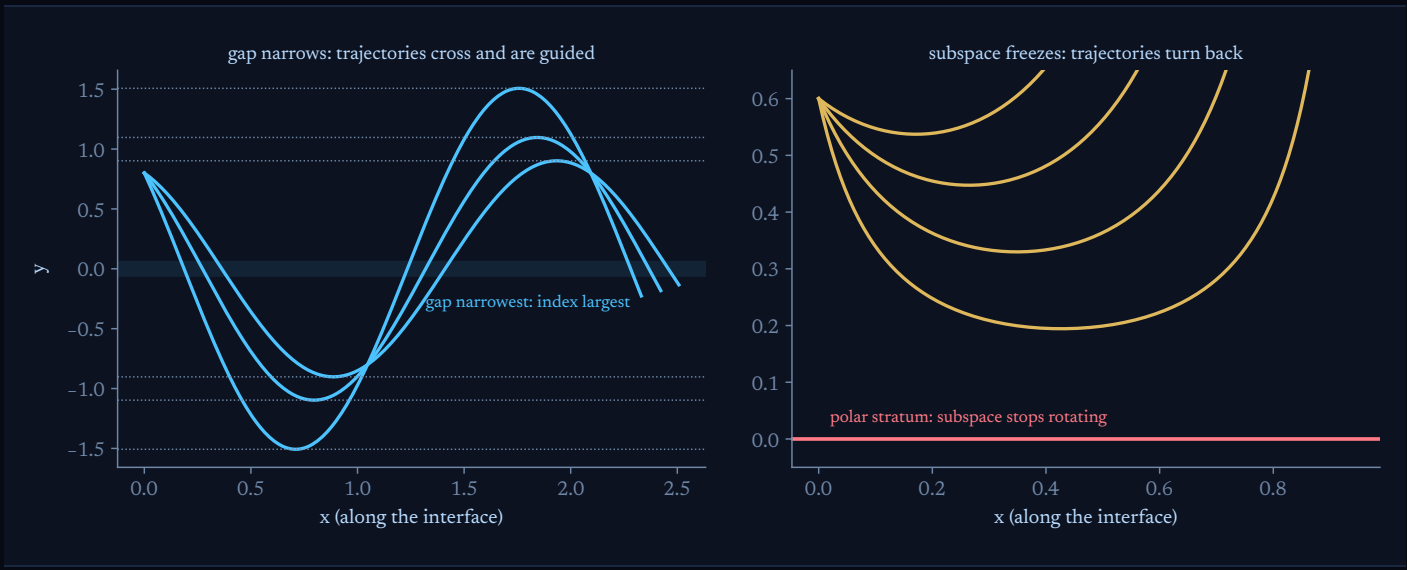


FIGURE 14. Two ways the index can vary across an interface, with the same refraction law. Left: the gap narrows along y = 0 while the subspace keeps rotating; the index is largest there, and geodesics cross it and turn back where the index falls to their conserved level (dotted, predicted). Right: the subspace stops rotating on y = 0; the index falls to zero there, and geodesics turn back before reaching it.

9 Operator Foundations for Mixed-State Manifolds

To extend this coordinate-free framework from pure states to mixed states, we pass from the Grassmannian $\mathbf{Gr}(k, \mathcal{H})$ to the space $\mathcal{S}(\mathcal{H})$ of positive-semidefinite density operators. Mixed states break the projection constraint, $\rho^2 \neq \rho$, so tangent variations are defined through the symmetric logarithmic derivative L_V, given implicitly by

$$V = \tfrac{1}{2}\big(L_V\rho + \rho L_V\big). \tag{9.1}$$

The intrinsic tensor on $\mathcal{S}(\mathcal{H})$ takes the coordinate-free form

$$Q_\rho(V, W) := \tfrac{1}{4}\,\mathrm{Tr}\big(\rho\, L_V L_W\big), \tag{9.2}$$

whose real and imaginary parts are

$$g_\rho(V, W) = \tfrac{1}{8}\,\mathrm{Tr}\big(\rho\{L_V, L_W\}\big), \qquad \Omega_\rho(V, W) = \tfrac{i}{4}\,\mathrm{Tr}\big(\rho[L_V, L_W]\big), \tag{9.3}$$

so that $Q_\rho = g_\rho - \frac{i}{2}\Omega_\rho$, the same form as (2.3). Here g_ρ is the Bures metric [Braunstein–Caves 1994] – one quarter of the quantum Fisher information, $g_\rho(V, V) = \frac{1}{4}F_Q$, and equal to the leading term of the squared Bures distance between neighbouring states (to 4×10^{-6} on a random full-rank state) – and Ω_ρ is the mean Uhlmann curvature. The geometric response is thereby carried on the bundle of

purifications over $\mathcal{S}(\mathcal{H})$, preserving the operator-algebraic architecture of the framework in open systems.

PROPOSITION 9.1 *(Exact reduction to the pure case). Let $k = 1$ and $\rho = \Pi$, so that $L_V = 2V$ solves (9.1) by the tangent constraint. Then $Q_\rho = Q_\Pi$, $g_\rho = g_\Pi$ and $\Omega_\rho = \Omega_\Pi$.*

PROOF. $Q_\rho = \tfrac{1}{4}\mathrm{Tr}(\Pi \cdot 2V \cdot 2W) = \mathrm{Tr}(\Pi V W) = Q_\Pi$;
$g_\rho = \tfrac{1}{8}\mathrm{Tr}(\Pi\{2V, 2W\}) = \tfrac{1}{2}\big(\mathrm{Tr}(\Pi V W) + \mathrm{Tr}(\Pi W V)\big) = g_\Pi$ by Remark 2.3;
$\Omega_\rho = \tfrac{i}{4}\mathrm{Tr}(\Pi[2V, 2W]) = i\,\mathrm{Tr}(\Pi[V, W]) = \Omega_\Pi$. ∎

The factor $\tfrac{1}{4}$ in (9.2) is what makes the reduction exact; without it $\mathrm{Tr}(\rho L_V L_W) = 4Q_\Pi$. For $k > 1$ the normalised state on the subspace is $\rho = \Pi/k$, and the reduction carries a factor $1/k$ throughout, as in Lemma 5.1. The same factor is the reason that any statement obtained by substituting the unnormalised Π for a density operator is off by k.

REMARK 9.2 (Why the symmetric logarithmic derivative exists here). For a projector, and more generally for any singular ρ, equation (9.1) is solvable precisely because the variation has no component in the kernel–kernel block, $(I - \Pi)\,\delta\Pi\,(I - \Pi) = 0$. Corollary 3.4 is therefore the existence condition for (9.1), not merely a property quoted here: adding a kernel–kernel block makes (9.1) unsolvable. L_V is not unique on that block, but $\mathrm{Tr}(\rho L_V L_W)$ is.

9.1 Computing the symmetric logarithmic derivative

Numerically, (9.1) is a Sylvester equation. With column-stacking vectorisation,
$\mathrm{vec}(\rho Y) = (I \otimes \rho)\,\mathrm{vec}\,Y$ and $\mathrm{vec}(Y\rho) = (\rho^{\mathsf{T}} \otimes I)\,\mathrm{vec}\,Y$, so the regularised equation
$\tfrac{1}{2}(\rho Y + Y\rho) + \gamma Y = X$ becomes the linear system

$$\Big[\tfrac{1}{2}\big(I \otimes \rho + \rho^{\mathsf{T}} \otimes I\big) + \gamma\, I \otimes I\Big]\,\mathrm{vec}\,Y = \mathrm{vec}\,X, \tag{9.4}$$

which is invertible for every $\gamma > 0$, including singular ρ, and returns the symmetric logarithmic derivative in the limit $\gamma \to 0$ whenever Remark 9.2's condition holds. Differentiating (9.3) along a third parameter gives the Uhlmann curvature dipole,
$\partial_k \Omega_\rho = \tfrac{i}{4}\mathrm{Tr}\big(\partial_k\rho\,[L_i, L_j] + \rho[\partial_k L_i, L_j] + \rho[L_i, \partial_k L_j]\big)$, with $\partial_k L_i$ obtained from (9.4) applied to the differentiated (9.1); its systematic evaluation is not carried out here.

10 From Raw Data to Intrinsic Angles

The framework makes three structural predictions that can be checked on measured data. Tangent deformations are strictly off-diagonal with respect to the instantaneous spectral decomposition. The spectral subspace keeps its identity while individual eigenvectors inside it are free to scramble. Transport angles are constrained by the metric–curvature inequality. Each of these is a statement about the projector, not about a basis. Data, however, arrive in a basis: the instrument's own. This section sets out how to get from one to the other, then applies it to two independent public datasets.

10.1 A standard for extracting angles

Laboratories process their data before publishing it. Readout axes are rotated and rescaled, correction matrices are applied, populations are fitted. Every such step changes raw angles while leaving the physics unchanged. An angle read directly off processed data therefore mixes two things: the system's geometry and the instrument's point of view. To separate them, every extracted angle is assigned one of three levels.

1. **Level 0, raw angle θ_{raw}.** The angle as the instrument's frame sees it, computed from the measured object with no model imposed.
2. **Level 1, geometric angle θ_G.** What remains once the point of view is removed: the angle inside the intrinsic metric of the Grassmannian,

$$\cos\theta_G = \frac{g_\Pi(V,W)}{\sqrt{g_\Pi(V,V)\,g_\Pi(W,W)}}. \tag{10.1}$$

3. **Level 2, physical interpretation.** A physical meaning is assigned only after θ_G has been obtained unambiguously.

The test that separates the first two levels is a change of frame. Under a unitary change of basis $\Pi \mapsto U\Pi U^\dagger$, $V \mapsto UVU^\dagger$, the geometric angle (10.1) does not move: every quantity in it is a trace. A raw angle between two readout vectors moves as soon as the readout axes are rescaled; rescaling one axis by a factor of two shifts a raw angle of $52°$ by $9.7°$. Only what survives every change of view is a property of the system.

To make extractions comparable across datasets and across competing frameworks, every reported transport tensor should carry at least the four quantities of Table 10.1.

TABLE 10.1. Standard reporting for geometric transport extractions.

QUANTITY	SYMBOL	MEANING
Transport capacity	$C(V) = g_\Pi(V, V)$	Total available geometric transport strength
Transport alignment	$\cos\theta_G$	Mutual orientation of two transport channels
Transport chirality	$\sin\theta_G$	Fraction of capacity convertible to curvature
Resulting curvature	$\Omega_\Pi(V, W)$	Measured or derived Berry or Uhlmann curvature

With these four numbers in hand, Theorem 6.1 becomes directly testable on data. Since $g_\Pi(V,V)g_\Pi(W,W) - g_\Pi(V,W)^2 = C(V)C(W)\sin^2\theta_G$, inequality (6.1) reads

$$|\Omega_\Pi(V, W)| \;\le\; 2\sqrt{C(V)\,C(W)}\,\sin\theta_G. \tag{10.2}$$

This is (6.1) exactly, not a weaker form, and it is sharp on the same extremal partner. It says that curvature can only be produced from the part of the capacity that is perpendicular in the intrinsic metric. Any proposed alternative transport metric can then be compared with the intrinsic one on the same footing.

One further condition decides whether a Level 1 angle exists at all. A direction extracted as the leading eigenvector of an estimated operator is defined only as sharply as its eigenvalue is separated from the next. For a two-dimensional covariance with eigenvalues $\lambda_1 > \lambda_2$ estimated from N samples, the leading direction scatters by

$$\delta\theta \;\approx\; \frac{\sqrt{\lambda_1\lambda_2}}{(\lambda_1 - \lambda_2)\sqrt{N}} \tag{10.3}$$

radians. At $N = 9{,}949$ this predicts $0.38°$, $3.15°$ and $29°$ for eigenvalue ratios 4, 1.2 and 1.02; direct simulation gives $0.38°$, $3.13°$ and $30.8°$. As the ratio approaches one the direction becomes uniformly random and carries no information (Figure 15, left). A direction is reported only where its gap is resolved.

10.2 The tip and the tail: why the subspace is the observable

Formula (10.3) is one instance of a general law, and Section 3 already contains its infinitesimal form: by Proposition 3.5 the projector moves by at most $\sqrt{2k}\|\delta\Sigma\|/\Delta$, where Δ is the gap that separates the whole subspace from the rest. The finite form follows from one identity.

PROPOSITION 10.1 *(Gap identity). Let A and $A' = A + E$ be self-adjoint, let P be a spectral projector of A and Q' a spectral projector of A', with $AP = PA$ and $A'Q' = Q'A'$. Then*

$$(Q'A'Q')(Q'P) - (Q'P)(PAP) = Q'EP. \tag{10.4}$$

PROOF. Using $Q'^2 = Q'$ and $A'Q' = Q'A'$, $(Q'A'Q')(Q'P) = Q'A'P$. Using $P^2 = P$ and $AP = PA$, $(Q'P)(PAP) = Q'AP$. Subtract. ∎

The block $Q'P$ measures how far the subspace has moved: its operator norm is the sine of the largest principal angle between $\mathbf{Ran}\, P$ and $\mathbf{Ran}\, P'$, where $P' = I - Q'$. Equation (10.4) is a Sylvester equation for that block. If the spectrum of A' on $\mathbf{Ran}\, Q'$ and the spectrum of A on $\mathbf{Ran}\, P$ are separated by $\delta > 0$, it gives

$$\|\sin\Theta(P, P')\| \;=\; \|Q'P\| \;\le\; \frac{\|Q'EP\|}{\delta}, \tag{10.5}$$

the $\sin\Theta$ theorem of Davis and Kahan [Davis–Kahan 1970; Bhatia 1997]. It is a known result and is stated here because the rest of the section rests on it. On 2,000 random instances of sizes 3 to 8, (10.5) held every time, with the largest ratio of left to right side 0.998.

What (10.5) makes visible is that two different gaps govern two different objects.

- **The tip.** A single eigenvector is a rank-one projector. Its δ is the distance to its nearest neighbour, which inside a cluster of close eigenvalues can be arbitrarily small.
- **The tail.** The projector onto the whole cluster has its δ set by the distance from the cluster to the rest of the spectrum, which can be large while the internal splittings vanish.

REMARK 10.2 (The tip is free at a tie). If $Av = cv$ and $Aw = cw$, then $A(av + bw) = c(av + bw)$ for all scalars a, b.

This is linearity, and it is all the remark needs.

At an exact tie no single vector is selected; any perturbation, however small, picks one. In a six-level test with a doubly degenerate ground level, a perturbation of size 10^{-3} turns the selected vector by a median of $40.7°$, while the projector onto the pair stays within its bound (10.5), never exceeding it. This is the mechanism behind the framework's central choice: *the eigenvectors vote, and they can trade identity; the subspace keeps it*. Geometry built on the projector inherits the stability of the tail. Geometry built on eigenvectors inherits the freedom of the tip.

Near a tie the freedom has a definite shape. For two levels with splitting Δ and coupling g, the upper eigenvector sits at mixing angle θ with

$$\tan 2\theta = \frac{2g}{\Delta}. \tag{10.6}$$

An eigenvector angle is a coupling-to-gap ratio. As Δ is swept through zero, the two eigenvectors do not pass through each other: they trade places, and the upper one turns through a full $90°$ inside a window of width about g. Sampled at 29 points, as the heat-engine drive amplitudes are, a wide crossing never turns by more than $5°$ in one step, while a crossing narrower than the sampling step shows up as a single jump of $87°$. A large single-step jump of a leading eigenvector is therefore the signature of a crossing or of a near-tie decided by noise. An independent split of the data separates the two: a real crossing jumps in both halves, while a noise-decided tie jumps at random.

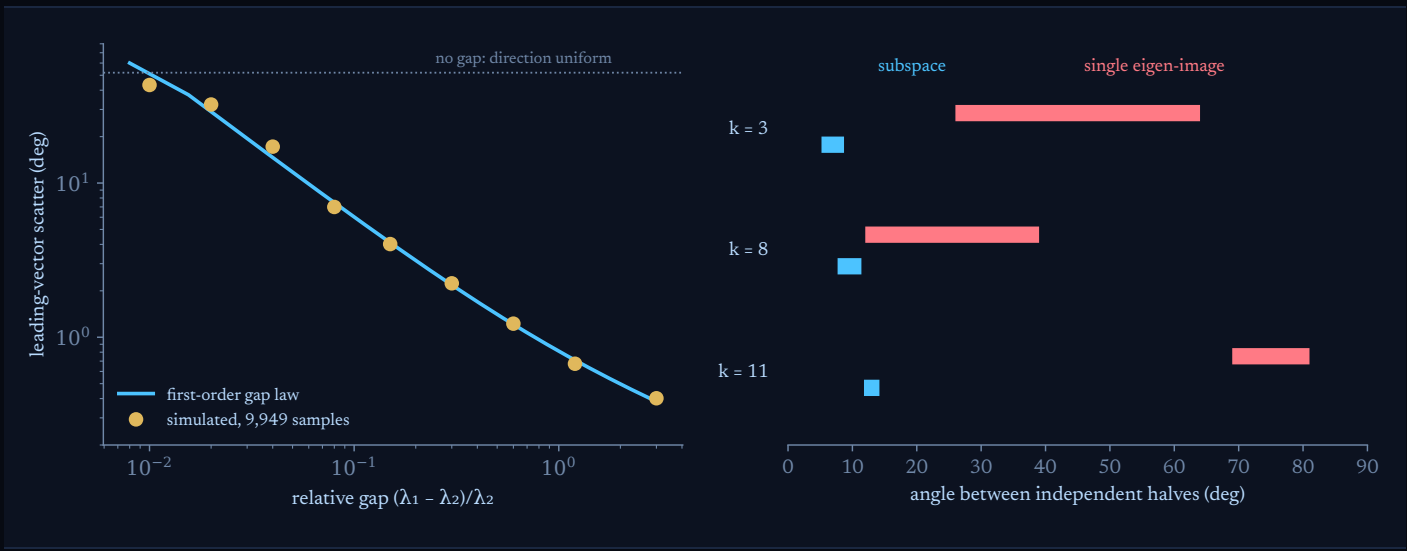


FIGURE 15. The tip and the tail. Left: scatter of a leading eigenvector estimated from 9,949 samples, against the relative gap; simulation follows the first-order law (10.3) until the direction becomes uniform. Right: dark-soliton images split into independent halves; at each cut that closes a near-degenerate pair, a single eigen-image (rose) turns by tens of degrees while the subspace (blue) moves by a few. Bars span three splits.

10.3 Superconducting quantum heat engine readout

The first dataset is the public single-shot readout of a transmon quantum Otto engine [Uusnakki et al. 2025]. In the calibration set, 19,899 shots were recorded for each prepared state g, e, f, each shot a point in the in-phase/quadrature (I–Q) readout plane. These are Level 0 objects: the plane is the instrument's.

Number of populations. A Gaussian mixture fitted to the pooled calibration shots is scored by the Bayesian information criterion. The score is lowest at four components (**−329,740.6**), against **−328,497.6** at three and **−329,614.2** at five. Three states were prepared and four populations are resolved. The fourth is a small component (weight **0.069**) with the most elongated covariance of the four, eigenvalue ratio 2.49. What it is physically is not tested here.

The core and the tail of one cloud. Within the ground-state cloud alone, split by distance from its mean:

- The inner half (9,950 shots) is almost round, with eigenvalue ratio 1.076. By (10.3) its long axis is defined only to about **8°**, and because of the radial cut the realised scatter is larger still.
- The outer tenth (1,990 shots) is clearly elongated, with ratio 2.216.

A noise model with the core's covariance shows which of these is real. Three hundred synthetic clouds were cut exactly as the data were.

- **Angle between the two axes.** In the data it is **75.4°**. In the synthetic clouds it ranges from **0°** to **87°** (median **17.8°**, 99.9th percentile **86.8°**). The angle is therefore not a Level 1 quantity: the core has no direction to compare against.
- **Elongation of the tail.** In the synthetic clouds it never exceeds 1.27. The measured 2.22 lies far outside that band.

The tail has a shape that the noise does not have (Figure 16). That shape is a property of the data. Its direction is still read in the readout frame, so it remains Level 0 until a frame-free version is built.

What the tail is made of, and how it moves with the cycle. The same core/tail split applied to the engine shots, about the centre of all shots at one drive setting, separates the populations sharply. Early in the first stroke, while f is high, the inner 90% of shots read as f 0.722, g 0.218, e 0.059; the outer 10% read almost entirely as e 0.623 and h 0.369, with f only 0.006. Late in the same stroke, as f drains, the core moves to f 0.569, g 0.251, e 0.173, and the tail's balance reverses to e 0.387, h 0.602. With about a thousand tail shots at each point, the shift of the e share is close to eleven standard errors. The tail therefore tracks the engine: it is where the e and h populations live, and its make-up changes as the stroke proceeds. Reading these shots as ones caught mid-transition is a Level 2 interpretation and is not tested here.

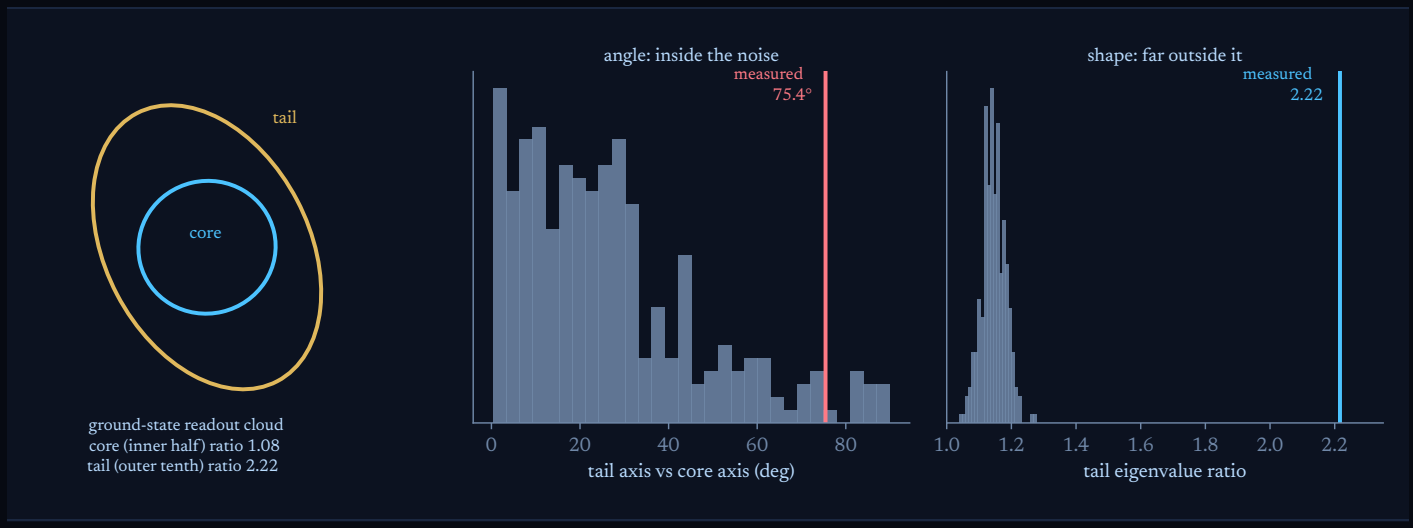


FIGURE 16. A direction exists only where a gap exists. Left: covariance shapes of the ground-state readout cloud of the heat engine (not to scale); the inner half is nearly round, the outer tenth elongated. Centre: the angle between their long axes, against 300 noise clouds cut the same way; the measured 75.4° lies inside the noise band. Right: the elongation of the outer tenth against the same noise clouds; the measured 2.22 lies far outside it.

The leading eigenvector along the cycle. Counting every shot of the three-cycle run into the four populations, and correcting the counts with the authors' own overlap-correction matrix, gives a clean 128-point $g/e/f/h$ trajectory through the engine strokes. In the first stroke it shows f draining from 0.620 to 0.487 while e rises from 0.111 to 0.182. The leading eigenvector of the population covariance over a sliding five-point window turns by $\mathbf{23.1}°$ per step on average on the corrected trajectory. On the uncorrected counts it averages $\mathbf{21.6}°$, with isolated single-step jumps of $\mathbf{64.6}°$ and $\mathbf{85.9}°$. By (10.6), single jumps near $\mathbf{90}°$ are what a crossing between samples produces, but they are also what a near-tie decided by shot noise produces.

Crossing or tie: the split-half test. The two readings make opposite predictions. A crossing is in the system, so if every setting's shots are divided into two independent halves, each half must jump at the same step. A tie decided by noise jumps at random places. Each half was given its own 128-point trajectory, both with and without the correction matrix, and the halves were formed twice: odd against even shots, and after shuffling the shot order. Before the run the test was checked on synthetic shots of the same layout: planted structure repeated in both halves, and pure shot noise did not. The kill condition, written in advance, was that fewer than half of one half's jumps (steps over $\mathbf{45}°$) recur in the other within one step, or that the number of recurrences does not exceed the 99th percentile of the count obtained by sliding one half along the other.

- The full-data trajectories reproduce the notebook: a mean step of $\mathbf{23.1}°$ corrected and $\mathbf{21.6}°$ uncorrected, with the same largest jumps.

- Odd/even, corrected: 20 and 14 jumps, 6 recurring, against 5.4 expected by chance (99th percentile 10). Uncorrected: 19 and 17, 5 recurring, against 6.7 (10.8).
- Shuffled split, corrected: 23 and 12, 3 recurring, against 6.3 (12). Uncorrected: 21 and 17, 8 recurring, against 8.2 (12).
- The step sizes of the two halves are uncorrelated: $+0.05$, $+0.03$, -0.03, $+0.23$.

The big jumps do not repeat beyond chance in any of the four comparisons. They are ties decided by shot noise, not crossings in the engine. That is itself the law of Section 10.2: a five-point window of populations has no resolved gap between its top two directions, so the leading eigenvector is the free tip, and noise casts its vote.

10.4 Dark solitons in a Bose–Einstein condensate

The second dataset is a different experiment entirely: the NIST dark-soliton image set [Fritsch et al. 2022]. It contains 16,478 absorption images of ^{87}Rb condensates (11,913 from July and 4,565 from August), binned 2×2 to 5,412 pixels and normalised by atom number. The images are split into two halves, either odd against even or at random, and the covariance of each half is diagonalised independently. The prediction for how far each rank-k projector should move between halves is the first-order formula of Lemma 3.1 applied to the covariance difference $E = C_A - C_B$:

$$\sin\theta \;\approx\; \Big\| \sum_{j} \frac{u_j^{\mathsf{T}} E U}{\lambda_i - \lambda_j} \Big\|. \tag{10.7}$$

Before use, this predictor was calibrated on synthetic data and returned a ratio of 1.00.

Results.

- **Agreement.** The median ratio of measured to predicted motion is 1.00 for the odd/even split, and 1.08 and 0.99 for two random splits. A pure-noise control gives projectors that disagree by $88.9°$ at rank 3.
- **Vector against subspace, rank 3, odd/even split.** The third eigen-image turns by $40.69°$ between halves. The rank-3 subspace that contains it moves by $5.19°$.
- **The gap law.** The 36 rank cuts (12 ranks, 3 splits) were ordered by relative gap $(\lambda_k - \lambda_{k+1})/\lambda_k$.
  - At every cut with relative gap at least 0.10, 18 of 18, the measured motion lies between 0.93 and 1.12 of the prediction.

- Below 0.10 the ratio scatters from 0.10 to 7.82, because the halves' own noise reorders near-tied modes. At rank 10 the formula expects $10°$–$18°$ and the measurement is $70°$–$81°$: modes 10 and 11 exchanged order.

Locked pairs. The spectrum carries near-degenerate pairs $(2, 3)$, $(7, 8)$ and $(10, 11)$ in every split. At the cut that closes each pair, the single vectors are loose while the subspace holds (Table 10.2; Figure 15, right).

TABLE 10.2. Angle between independent halves at the cut closing each pair (range over three splits).

CLOSING CUT	SINGLE EIGEN-IMAGE	SUBSPACE
$k = 3$	$26°$–$64°$	$5.2°$–$8.7°$
$k = 8$	$12°$–$39°$	$7.7°$–$11.4°$
$k = 11$	$69°$–$81°$	$11.8°$–$14.2°$

The eigen-images make the pairs readable. Images 2–5, 7, 8 and 10–12 are stripe patterns across the long axis of the cloud, recording where the soliton sits. Each pair is two stripe patterns offset by a quarter period: a sine–cosine pair for one translation family, which is how principal components describe a feature that moves. Inside such a pair the two members are nearly tied, so the tip is free. The pair as a whole is separated from its neighbours, so the tail holds.

The months themselves differ. Between July and August the top eigen-image differs by $88.6°$, the eigenvalues differ by 9–60% at every rank, and first order holds at only 5 of 12 ranks. They are two different clouds, not two samples of one, and the analysis above is therefore made within the pooled set and its splits rather than across months.

Nothing in this subsection is new as a law: it is the gap law of Section 10.2, which is the Davis–Kahan theorem. What the data add is a measurement of it, in an experiment unrelated to the heat engine, with the vector and the subspace measured side by side on the same images: *the vectors trade identity, the subspace keeps it*.

10.5 What remains open

One test is specified but not yet run: a frame-free version of the readout-tail direction, obtained by rescaling the readout plane to equal noise in both quadratures before extracting the axis, so that a Level 1 angle can be stated. Its kill condition is written in advance.

11 Discussion and Outlook

The central achievement of this work is the construction of a fully coordinate-free quantum geometric tensor on the Grassmannian manifold of spectral projectors. Starting from the Riesz–Kato theory of isolated spectral subspaces, both the quantum metric and the Berry curvature emerge as natural components of a single intrinsic Hermitian form defined directly on the tangent bundle of $\mathbf{Gr}(k, \mathcal{H})$. The formulation needs no local eigenvector gauge, no Berry connection and no coordinate chart on parameter space.

A key conceptual point is that the real and imaginary parts of the tensor are not independent objects. Both are generated by the same cross-gap virtual transitions between the spectral subspace of $\mathbf{\Pi}_k$ and its complement. The tangent-space constraint

$$V = \mathbf{\Pi}_k V (I - \mathbf{\Pi}_k) + (I - \mathbf{\Pi}_k) V \mathbf{\Pi}_k$$

makes this mechanism explicit and leads directly to the metric–curvature inequality of Section 6. That the metric bounds the curvature is not a coincidence of two quantities but a consequence of their common origin: they are the symmetric and antisymmetric parts of one object built from the same amplitudes. The inequality explains why Berry curvature and the topological phenomena built on it are suppressed when interband mixing is weak or the gap is large, and its control is relative: near a node both sides grow together, so it limits the curvature per unit of metric, not the curvature itself.

The framework also yields a lower bound on non-adiabatic dissipation, and the way it does so is part of the result. The intrinsic metric is the quantum Fisher information of the moving subspace, its information-geometric speed; on its own it measures how much the driving must change, not what that change costs. A bath turns speed into cost. For relaxation to the instantaneous Gibbs state, and equally for a detailed-balance bath, the friction carried by the pairs that move the projector is the intrinsic metric reweighted pair by pair by twice the population difference times the gap times a relaxation factor. At zero temperature this confines the cost to a band of the metric, and the excess work of a protocol is bounded below by the squared intrinsic length divided by the time allowed. The geometric response is thereby connected to thermodynamic irreversibility through an explicit, tested bath weight rather than an assumed proportionality.

From a broader perspective, the construction clarifies the geometric foundations of transport. Higher-order coefficients arise as coordinate-dependent pullbacks of structures intrinsic to the Grassmannian, not as primitive features of the parameter manifold. Section 7 makes this concrete:

one sum over cross-gap pairs, read with different powers of the gap, gives the quantum metric, the Berry connection polarizability that drives the intrinsic nonlinear Hall effect, the static response, the rotation friction, the Berry curvature and the orbital-moment kernel. At first order every such quantity factors through the gap. At second order the factorisation acquires a remainder from the rotation of the eigenvectors inside each sector, which is what the exact dipole (7.3) contains, and a direction that does not move the subspace at all can still change how far the other directions move it.

The refraction results of Section 8 deserve a closing comment, since a Snell-type law arising in an operator-theoretic setting invites more interpretation than it can carry. The law is Clairaut's relation for geodesics of the pullback metric, and its content lies in the identification of the index: what refracts a geodesic is a change in the rate at which the occupied subspace rotates. The sine squared that appears is not an additional structure; it is inherited from the fact that the Hilbert–Schmidt distance between projectors is a sum of squared sines of principal angles. The sharpest statements are the two that fix the scope. Where the subspace stops rotating along an interface, the stratum is one projector, it is repulsive, and trajectories turn back by total internal reflection. Where instead the gap narrows while the subspace keeps rotating, the index rises, and trajectories cross and are guided. A fold, where the collapse is transverse and the projector keeps moving along the locus, is crossed freely, and there the degenerate metric does not by itself determine the connection along which transport is computed. The two kinds are separated by order (Proposition 8.6): a polar stratum is silent at every order, a fold only at the first. The bending itself is a metric dipole (Remark 8.4), so the refraction of Section 8 and the transport tensors of Section 7 are two readings of one derivative of the metric.

Each bound in the paper is also rigid, and the equality cases (Theorem 6.3) are as informative as the bounds. The gap bound is met only by couplings of one strength across one gap; the speed limit only by a rigid rotation at fixed gap; the dissipation floor only by motion at constant intrinsic speed, so that the cheapest protocol follows a geodesic at constant speed; the curvature bound only on a complex line, through the exact deficit identity (6.3); and the permutation sandwich only by involutions. In each case the configuration that meets the bound is reconstructed from the fact that it meets it. This is the pattern of rigidity theorems in geometric analysis. A recent manuscript on spacetime Penrose inequalities, posted as an unrefereed preprint, follows the same pattern at much greater depth: equality there forces Schwarzschild data, and in the anti-de Sitter case a nonnegative second-order deficit has a finite-dimensional kernel that is resolved at fourth order [OpenAI 2026a]. The parallel here is one of structure only, and nothing in this paper depends on that manuscript. Here the second-order deficit (6.3) is exact and its kernel is the complex line, so no higher order is needed for the curvature bound; the step to higher order appears instead at the silent directions of a

degenerate stratum (Proposition 8.6), where the kernel of the metric is resolved at second order in the projector and fourth order in the squared distance.

The data of Section 10 close the loop with experiment. In two unrelated systems — the readout of a superconducting heat engine and absorption images of a Bose–Einstein condensate — single leading eigenvectors are free wherever their gap is not resolved: in the engine their large jumps do not repeat across independent halves of the data, and in the condensate a single eigen-image turns by tens of degrees between halves while the subspace containing it moves by what the gap formula predicts, within 0.93–1.12 of it at every resolved gap. That is the reason, measured rather than assumed, for building the geometry on the projector: the vectors trade identity, the subspace keeps it.

Several directions are left open. The non-abelian sector (Remark 4.2) requires lifting the construction from $\mathrm{Gr}(k, \mathcal{H})$ to the Stiefel bundle, where the traceless part of the curvature lives. Transport through a fold requires a choice of kernel vector field or Ehresmann connection before the geodesic problem can be posed; the three-body example of Section 8.5 shows one natural choice, the round metric of the shape sphere, but no general rule is given here. The refraction tests of Section 8 hold the gap open or let it narrow; whether the index law survives when the gap closes exactly, where the projector itself ceases to be defined, is not determined here. The mean Uhlmann curvature dipole of Section 9 is written as an operator identity but not evaluated spectrally, and the full nonlinear orbital magnetisation tensor is not derived. On the experimental side, a frame-free version of the readout-tail direction would turn the one remaining Level 0 observation of Section 10 into a Level 1 angle.

12 Separately and as a Whole

Every object in this paper is a subspace held in place by a gap. The quantum metric measures how fast that subspace moves; the curvature measures how it turns; the friction measures what its motion costs; the refractive index is the rate at which it rotates; the data of Section 10 measure whether it holds still between independent halves of an experiment. Read together, the sections say one thing at two grains, and this closing section draws it out.

12.1 One law at two grains

Section 10.2 stated the gap identity and its consequence, the $\sin\Theta$ theorem of Davis and Kahan. Read as a statement about what can be looked at, it says the following. Take a cluster of levels

separated from the rest of the spectrum by an *outer gap* Δ_{out}, and let the levels inside the cluster be separated from each other by *inner gaps* δ_{in}. Under a perturbation E,

$$\angle(\text{cluster subspace}) \lesssim \frac{\|E\|}{\Delta_{\text{out}}}, \qquad \angle(\text{one vector inside it}) \lesssim \frac{\|E\|}{\delta_{\text{in}}}. \tag{12.1}$$

The cluster as a whole is held by the outer gap; each member separately is held only by its nearest neighbour. When $\delta_{\text{in}} \ll \Delta_{\text{out}}$ the members swing while the whole stays put. We call the single vector the *tip* and the cluster subspace the *tail*: tips swing, the tail holds, and the outer gap sets the width of the corridor the tail can move in.

Figure 17 (left) shows this on a seven-level operator with a three-level cluster, the outer gap fixed at 3 and a perturbation of norm 0.005. As the inner gap falls from 1 to 0.01, the median turn of one vector inside the cluster rises from $0.08°$ to $7.8°$, following $\|E\|/\delta_{\text{in}}$, while the cluster subspace stays at $0.04°$–$0.05°$, below its bound $\|E\|/\Delta_{\text{out}} = 0.096°$ in every one of 900 trials. Levels that sit together fuse into one object; the gap decides where one object ends and the next begins.

The same law governs motion in time. Theorem 5.2 bounds the speed of a subspace by the drive divided by the gap, $\sqrt{g} \le \sqrt{k}\,\|\partial_t \Sigma\|/\Delta$. Figure 17 (right) follows a driven three-level path: the metric speed stays under the ceiling everywhere, and the two approach where the gap is narrowest. A two-level off-diagonal drive reaches the ceiling exactly. Whether the question is how far a subspace can be pushed by a perturbation or how fast it can be driven, the answer is set by the gap that encloses it.

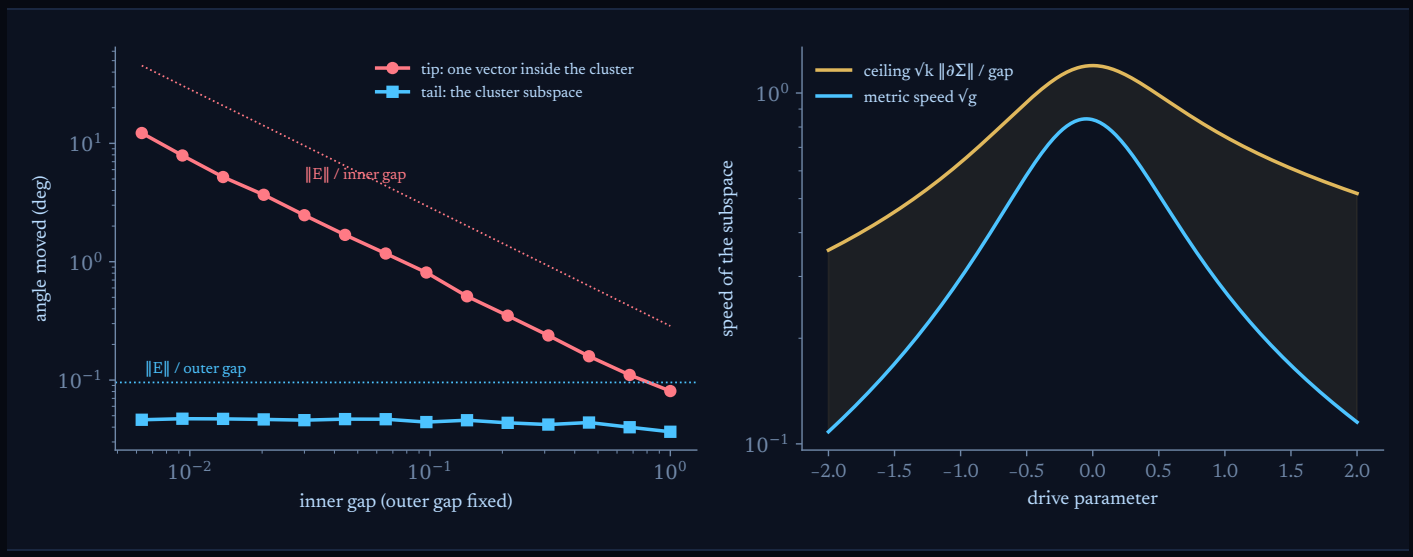


FIGURE 17. One law at two grains. Left: a three-level cluster inside a seven-level spectrum, outer gap fixed. As the inner gap closes, one vector inside the cluster turns further and further, following the perturbation over the inner gap; the cluster subspace stays below the perturbation over the outer gap. Right: along a driven path, the speed of the occupied subspace in the intrinsic metric stays under the ceiling set by the drive and the gap.

REMARK 12.1 (Two rhythms that become one). The same threshold appears when two oscillators lock. In the Kuramoto model of two phase oscillators with natural frequencies differing by $\Delta\omega$ and coupling K, the phase difference obeys $\dot{D} = \Delta\omega - 2K\sin D$, and the pair locks into one rhythm exactly when $|\Delta\omega| \le 2K$ [Kuramoto 1984]; integrated directly, $\Delta\omega/2K = 0.995$ locks and 1.005 does not. The two-level operator with detuning $\Delta\omega$ and coupling K has mixing angle $\tfrac{1}{2}\arctan(2K/\Delta\omega)$, which equals $22.5°$ at the same threshold. Both are governed by the single ratio of coupling to splitting: below one the two members are separate, above it they act as one. This is a correspondence of one ratio, not a derivation of one model from the other.

EXAMPLE 12.2 (Light going both ways round a ring). In a ring cavity, light at one longitudinal order runs both ways, and the forward and backward modes span one two-dimensional subspace: the pair is the tail, and a single mode inside it is a tip. The outer gap is the free spectral range F to the next order. The inner gap is set by two things that compete. Rotation splits the two directions by the Sagnac frequency Ω; scattering couples them with strength κ. The pair's own levels are then $\pm\tfrac{1}{2}\delta$, $\delta = \sqrt{\Omega^2 + 4\kappa^2}$, and the modes inside it mix at the angle $\tfrac{1}{2}\arctan(2\kappa/\Omega)$: standing waves when scattering wins, travelling waves when rotation wins, crossing $22.5°$ exactly at $\Omega = 2\kappa$. Both mirrors moving together only slides the standing-wave pattern, the tip running while the tail holds, as the condensate's stripes do in Section 12.3.

In a four-mode model of two orders ($F = 10$, $\kappa = 0.05$) under random perturbations of norm 0.002, the single lowest mode turns by $0.25°$ when scattering dominates and by $0.01°$ when rotation dominates, falling as $1/\delta$ (log-log slope -0.93), while the pair turns by $0.006°$ at every rotation rate, below its bound $\|E\|/F$ in all 720 draws (Figure 18). In an active ring laser the same competition is nonlinear: the phase difference between the two beams obeys Adler's equation $\dot{\psi} = \Omega - \Omega_L \sin\psi$, and the beams lock to one frequency when $|\Omega| \le \Omega_L$; integrated directly, $0.995\,\Omega_L$ locks and $1.005\,\Omega_L$ does not. This is the lock-in of the ring-laser gyroscope [Aronowitz 1971], and it is Remark 12.1 again: below the threshold the two directions act as one, and only the pair can be read; above it the directions separate, and each tip is measurable. The example reproduces known optics in the language of this section; it adds no new result about ring lasers.

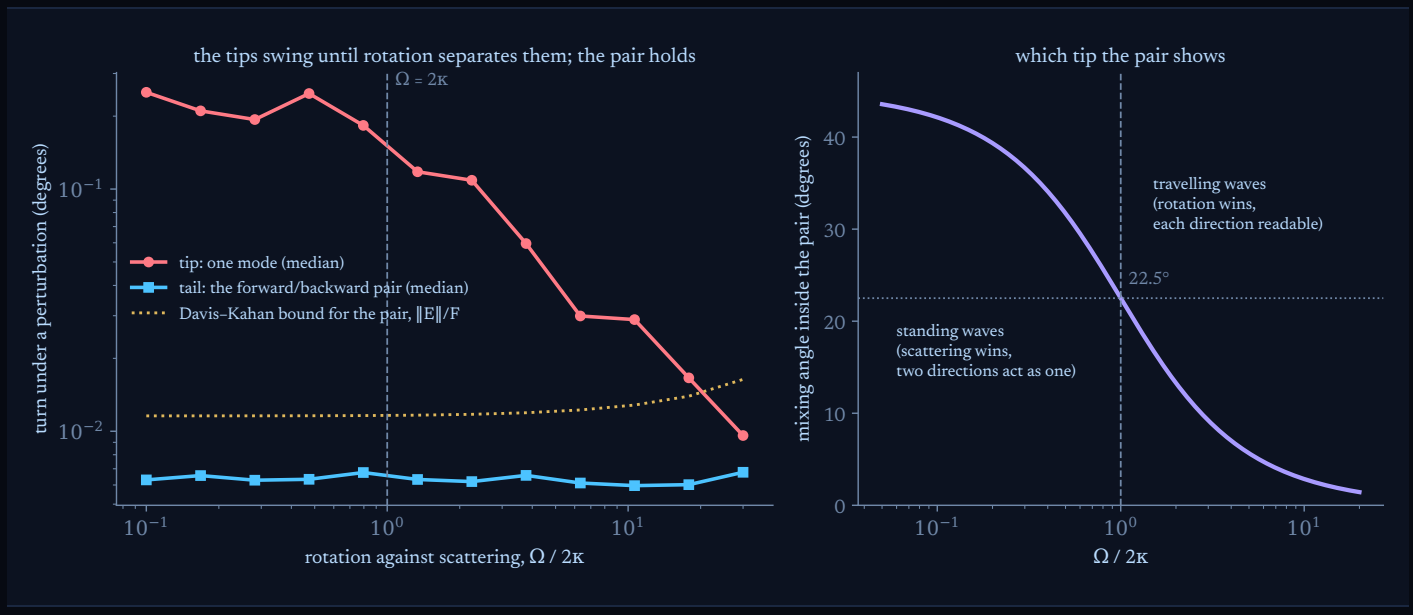


FIGURE 18. Light going both ways round a ring (Example 12.2). Left: under random perturbations, a single mode (the tip) turns by an amount falling as one over the inner gap, from scattering-dominated to rotation-dominated, while the forward/backward pair (the tail) holds below its Davis–Kahan bound at every rotation rate. Right: the mixing angle inside the pair, from standing waves, where the two directions act as one, through 22.5° at Ω = 2κ, to travelling waves, where each direction can be read.

12.2 How each part reads under it

- **The tangent law (Section 2).** A subspace can only move by mixing across its own boundary: V is block off-diagonal. Motion inside the cluster does not move the tail at all.
- **The gap bound (Section 3).** Proposition 3.5 is (12.1) at first order: the tail moves by at most $\sqrt{2k}\|\delta\Sigma\|/\Delta$.
- **Pullbacks and silent directions (Section 4).** A direction that only reshuffles levels inside the cluster is silent: it leaves the tail unmoved at first order, though at second order it changes how far the other directions move it.
- **Dissipation (Section 5).** The friction is a sum over pairs straddling the outer gap, each weighted by its own gap; the speed limit is the ceiling of Figure 17.
- **The curvature bound (Section 6).** Curvature lives in the sine between two directions of the tail's motion; it cannot exceed what the metric allows.
- **Transport tensors (Section 7).** One sum over the same cross-gap pairs, read with different powers of the gap.
- **Degenerate strata (Section 8).** The refractive index is the rotation rate of the tail. Where the tail stops rotating, trajectories turn back; where the gap narrows and the tail turns faster, they are guided.

- **Mixed states (Section 9).** The symmetric logarithmic derivative exists exactly when the variation has no block inside the kernel: the same boundary that decides what moves.

12.3 Looking separately, then as a whole

This gives a working rule for any data set, and it is the rule Section 10 followed. First look at each part separately: compute every eigenvector and every gap on independent halves of the data, and ask of each direction whether its own gap is resolved. Then look at the whole: cut the spectrum only where a gap is resolved, and treat everything above the cut as one subspace. A part is reported only where its own gap holds it; a whole is reported at every resolved gap. Nothing is lost by this. What cannot be named separately is still measured, as part of the whole that contains it.

The condensate data show the rule at every cut (Figure 19). On the odd/even split, the single eigen-image and the rank-k subspace move together wherever the cut falls on a resolved gap. Inside each near-degenerate pair they part company:

- at $k = 3$, the cut that closes the pair $(2, 3)$, the third eigen-image turns by $40.7°$ and the subspace by $5.2°$;
- at $k = 8$ it is $29.9°$ against $11.4°$;
- at $k = 11$, $78.9°$ against $12.8°$.

At the cut inside a pair ($k = 2, 7, 10$) the subspace inherits the free vector and turns with it, to within a degree. Across all three splits, every cut with relative gap at least 0.10 lands within 0.93–1.12 of the gap prediction, 18 of 18; every cut that misses it sits at a near-tie, with relative gap below 0.10.

The pairs themselves say what "the whole" is. The relative gap inside the pair $(2, 3)$ is 0.03; at its outer edge it is 0.30. Eigen-images 2 and 3 are two stripe patterns offset by a quarter period, the sine and cosine of one moving pattern, and the soliton position oscillates in hold time at 2.70 Hz. A combination $\cos\varphi \cdot \text{image } 2 + \sin\varphi \cdot \text{image } 3$ is the stripe at phase φ of its swing. The two directions inside the pair are nearly tied, so each half of the data picks its own combination of them, and the tip is free; the plane the two images span is the motion itself, and it holds. Sharp gaps make sharp subspaces: ten eigen-images carry half of all the variation in 16,478 images, and the participation of the spectrum is 26.5 against 1,911 for noise of the same spread.

The same rule, applied where no gap is resolved, gives the other half of the answer. In the heat-engine populations a five-point window has no resolved gap between its top two directions, and the

leading eigenvector's large jumps do not repeat across independent halves of the shots. There the honest report is that no single direction exists at that grain. That is a measurement too.

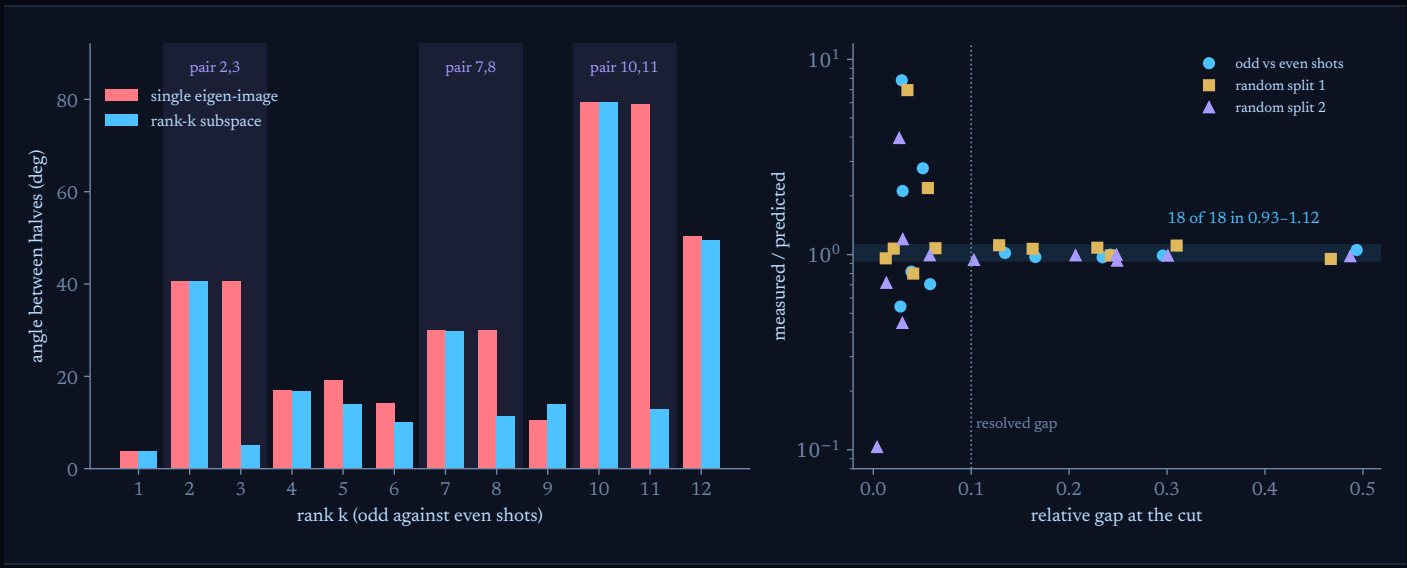


FIGURE 19. The condensate, every cut. Left: angle between independent halves (odd against even shots) for the single eigen-image and the rank-k subspace; shaded bands mark the near-degenerate pairs. Inside a pair the vector is free and the subspace that cuts the pair turns with it; the cut that closes a pair holds. Right: measured over predicted subspace motion against the relative gap at the cut, for three independent splits.

12.4 The same geometry in a deck of cards

The rule applies wherever a distance is read off subspaces, and one such place is the Hamming metric on permutations that underlies the soficity question [Chapman–Lubotzky 2025; Beasley 2026]. For a permutation matrix U of degree n, let P_U be the orthogonal projector onto its graph, the column space of $\frac{1}{\sqrt{2}}\binom{I}{U}$. Then

$$d_H(U,V) = \frac{1}{n}\|P_U - P_V\|_F^2 = \frac{2}{n}\sum_{j=1}^{n} \sin^2\theta_j, \tag{12.2}$$

where θ_j are the principal angles between the two graphs and $\sin^2\theta_j = \sin^2(\phi_j/2)$ over the eigenphases ϕ_j of U^*V. The proof is two lines: $P_U - P_V$ has off-diagonal blocks $\frac{1}{2}(U-V)$ and its adjoint, so $\|P_U - P_V\|_F^2 = \frac{1}{2}\|U-V\|_F^2 = n\,d_H$; and $Q_U^* Q_V = \frac{1}{2}(I + U^*V)$ has singular values $|\cos(\phi_j/2)| = \cos\theta_j$. The card count is therefore a chordal distance between two points of the Grassmannian.

The intrinsic metric adds the geodesic. Along $U(t) = \exp(t \log U)$ the graph projector moves with constant speed in the metric of Section 2, so its intrinsic length from P_I to P_U is

$$\mathcal{L} = \Big(\sum_j \theta_j^2 \Big)^{1/2}, \tag{12.3}$$

the Grassmannian geodesic distance (agreement 6×10^{-10} on 20 random permutations, with the speed constant along each path). The count reads $\sum \sin^2 \theta_j$; the intrinsic length reads $\sum \theta_j^2$. They see the same angles weighted differently, and the difference is the arrangement the count forgets. On twenty points, five disjoint swaps and one 10-cycle both move ten points and both give $\sum \sin^2 \theta_j = 5$; the swaps place five angles at $90°$, the cycle spreads its angles from $0°$ to $90°$ in steps of $18°$, and the intrinsic lengths are 3.512 and 2.896 (Figure 20). Per point the two are tied together by $d_H/2 \le \frac{1}{n}\sum \theta_j^2 \le \frac{\pi^2}{8} d_H$ [Beasley 2026], which held on 2,000 random pairs. So the geodesic separates arrangements, but it still vanishes with the defect, and like every native reading of the Hamming metric it obstructs nothing. A recently claimed obstruction to soficity is of a different kind: an unrefereed manuscript constructs a finitely presented torsion-free group whose group algebra over $\mathbb{F}_2$ is not directly finite [OpenAI 2026b], and sofic groups always have directly finite group algebras [Elek–Szabó 2004], so that group would not be sofic. The obstruction there is algebraic, not a reading of a distance.

The tip and the tail are visible here too. The displacement Gram $G = I - \frac{1}{2}(U + U^{*})$ of an ℓ-cycle has eigenvalues $2\sin^2(\pi j/\ell)$, which coincide for j and $\ell - j$: a cyclic motion comes in sine–cosine pairs, exactly as the condensate's moving stripes do in Section 12.3. Five swaps give the eigenvalue 2 five times over (and 0 five times), a complete tie in which no single mode is selected; the cycle resolves its modes into pairs. Each pair is a tail held by its gap; the vectors inside it are tips.

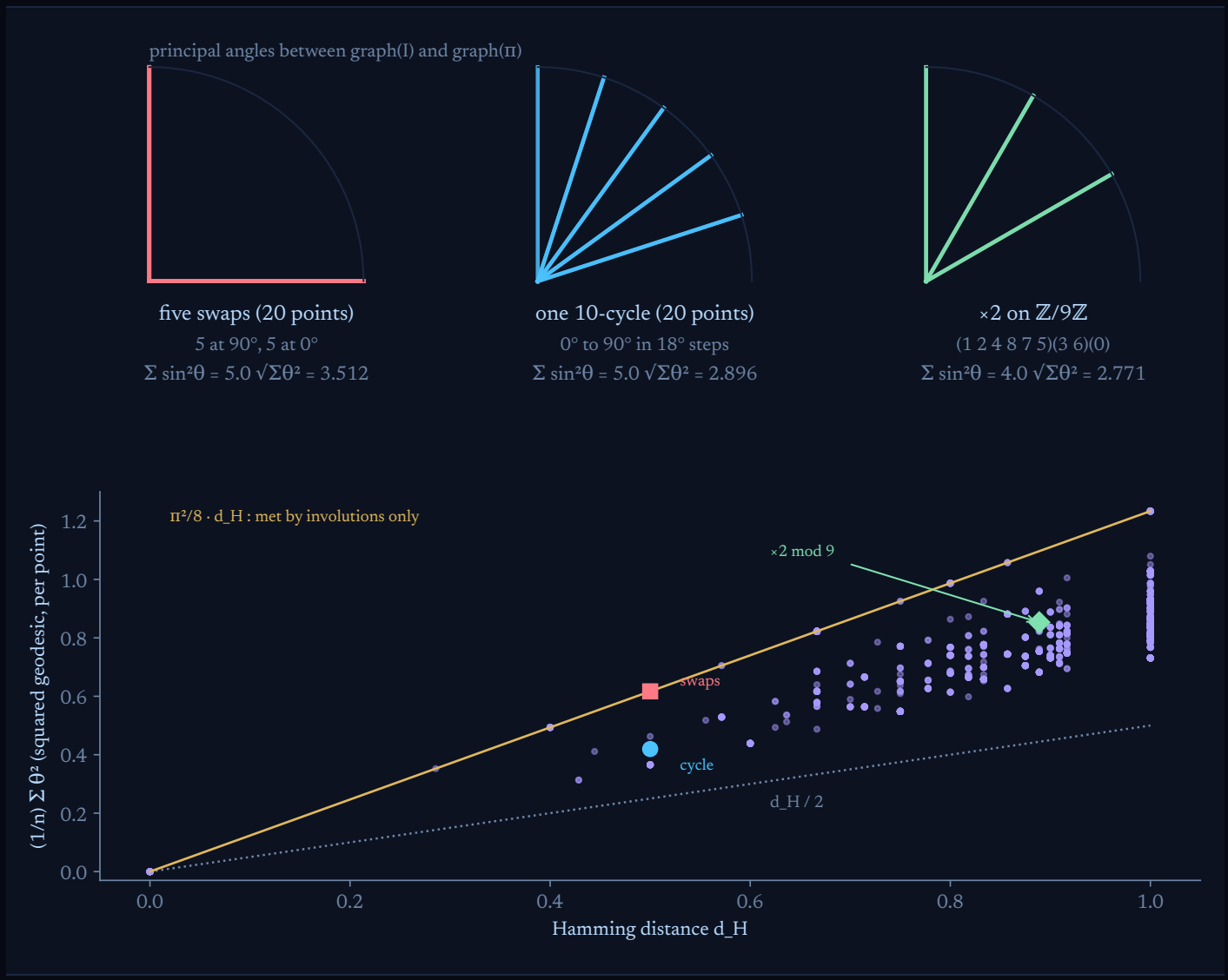


FIGURE 20. The card count and the intrinsic length. Top: principal angles between the graph of the identity and the graph of a permutation: five disjoint swaps and one 10-cycle on twenty points, which have the same count and different lengths, and multiplication by 2 on $\mathbb{Z}/9\mathbb{Z}$. Bottom: per-point squared geodesic against Hamming distance for 1,500 random pairs, between the two bounds of the sandwich; the upper bound is met by involutions only (Proposition 12.3), which lie on it, while the cycle and the doubling map lie strictly below.

PROPOSITION 12.3 *(When the sandwich is tight). The lower bound $\mathbf{d_H/2 \le \frac{1}{n}\sum\theta_j^2}$ is attained only for $\boldsymbol{U = V}$. The upper bound $\frac{1}{n}\sum\theta_j^2 \le \frac{\pi^2}{8}\boldsymbol{d_H}$ is attained if and only if every principal angle is $\mathbf{0}$ or $\boldsymbol{\pi/2}$, if and only if $\boldsymbol{U^*V}$ is an involution, $\boldsymbol{(U^*V)^2 = I}$.*

PROOF. The sandwich is the sum over j of $\sin^2\theta_j \le \theta_j^2 \le \frac{\pi^2}{4}\sin^2\theta_j$ on $[0, \pi/2]$. The first is strict for $\theta_j > 0$, since $\sin\theta < \theta$. The second is Jordan's inequality $\sin\theta \ge 2\theta/\pi$, which is strict strictly between 0 and $\pi/2$. So the upper bound is tight exactly when every $\theta_j \in \{0, \pi/2\}$. Since $\theta_j = |\phi_j|/2$ with $\phi_j \in (-\pi, \pi]$, $\theta_j = \pi/2$ exactly when $\phi_j = \pi$; so every eigenvalue $e^{i\phi_j}$ of the unitary U^*V is ± 1, which for a diagonalisable operator is $(U^*V)^2 = I$. ∎

A shuffle meets the upper bound exactly when it is made of swaps. The five swaps of Figure 20 sit on it (ratio 1.00000000), as does any involution; the 10-cycle sits at 0.680 of it. For a given card count, swaps are the longest motion, and any cycle of length three or more moves the cards along a strictly shorter path.

EXAMPLE 12.4 (Doubling on nine residues). Multiplication by 2 on $\mathbb{Z}/9\mathbb{Z}$ is the permutation $r \mapsto 2r \bmod 9$, with cycles $(1\,2\,4\,8\,7\,5)(3\,6)(0)$. Because 2 is invertible mod 9, $\gcd(2r, 9) = \gcd(r, 9)$, so the map carries each of the three families $\{1, 2, 4, 5, 7, 8\}$, $\{3, 6\}$, $\{0\}$ to itself and they never mix; $2^6 = 64 \equiv 1$, so six doublings return every residue. The eigenphases are the sixth roots of unity on the 6-cycle, ± 1 on the 2-cycle and 1 on the fixed point, so the nonzero principal angles are $30°, 30°, 60°, 60°, 90°, 90°$. The count is $\sum \sin^2 \theta_j = 4$, half the eight points moved, and the intrinsic length is 2.7706. Four disjoint swaps also move eight points and also give $\sum \sin^2 \theta_j = 4$, with all four nonzero angles at $90°$ and length π. The doubling map sits at 0.778 of the upper bound and the swaps on it: the same count, a shorter motion, because the doubling map moves its cards in cycles longer than two.

12.5 What the paper adds to the law

The $\sin \Theta$ bound is classical. What this paper adds is the reading of it as the organising principle of quantum geometry, and its use as a measuring rule:

- the gap identity (10.4) written as a measuring rule, with its tie case (Remark 10.2), both checked in Lean;
- the observation that every intrinsic quantity of Sections 2–9 is a property of the tail, held by the outer gap, and so inherits its stability;
- the three-level standard of Section 10.1, which asks of every reported angle whether a gap holds it;
- two independent public data sets in which the vector and the subspace were measured side by side, showing the tips trade identity while the tail keeps it.

Looked at separately, the parts of a quantum system are only as definite as the gaps between them. Looked at as a whole, the subspace above a resolved gap is definite, stable and measurable, and every geometric quantity in this paper is a quantity of that whole.

13 Conclusion

We have constructed a fully coordinate-free formulation of the Quantum Geometric Tensor on the Grassmannian manifold of rank-k spectral projectors associated with a gapped self-adjoint operator. By building directly on the Riesz–Kato spectral projection formalism, we have shown that the geometry of an isolated spectral subspace can be expressed entirely at the level of the orthogonal projector, without reference to local eigenvector bases, Berry connections, coordinate charts or parameter-space constructions.

The intrinsic tensor

$$Q_\Pi(V, W) = \mathrm{Tr}(\Pi V W)$$

provides a unified geometric object whose symmetric component is the Hilbert–Schmidt quantum metric and whose antisymmetric component is the Berry curvature 2-form. Both arise from the canonical Hermitian form on the Grassmannian tangent bundle. The fundamental tangent-space constraint makes explicit that all geometric response is generated exclusively by virtual cross-gap excitations between the occupied and complementary spectral subspaces.

The real and imaginary parts of the tensor are therefore intertwined rather than independent. The same cross-gap matrix elements that determine the quantum metric also control the Berry curvature, which yields the sharp metric–curvature inequality of Section 6. Under an explicit bath model the intrinsic metric, reweighted by the spectrum, governs the minimal dissipative cost of finite-time driving, and gives a lower bound on the excess work in terms of the intrinsic length of the path. Every one of these bounds is sharp, and each is met only by one rigid configuration: one gap and one coupling strength, a rigid rotation, constant intrinsic speed, a complex line.

When applied to physical systems, the framework shows that parameter spaces do not generate geometric properties; they select finite-dimensional trajectories through a pre-existing Grassmannian geometry by pullback. This is why geometric transport is robust under smooth deformations of the Hamiltonian as long as the spectral gap stays open, and why, where the pullback degenerates, what happens next depends on whether the subspace has stopped rotating or the gap has narrowed.

By anchoring the construction in the functional-analytic foundations of analytic perturbation theory, the framework remains consistent across bounded matrix models and unbounded differential operators. It unifies perturbation theory, topological invariants, quantum

thermodynamic geometry and the extraction of geometric angles from raw data within a single coordinate-free operator architecture.

A Unbounded Operators: Relative Boundedness and Domain Stabilisation

Let Σ be a (possibly unbounded) self-adjoint operator on the separable Hilbert space $\mathcal{H}$ with dense domain $D(\Sigma)$. For the Riesz projector Π_k of (2.4) to be well defined, the contour Γ_k must lie entirely in the resolvent set $\rho(\Sigma)$.

When Σ is unbounded, stability under perturbations requires the notion of relative boundedness. An operator B is relatively bounded with respect to Σ if there exist constants $a, b \geq 0$ such that

$$\|B\psi\| \leq a\|\Sigma\psi\| + b\|\psi\| \qquad \text{for all } \psi \in D(\Sigma). \tag{A.1}$$

The constant a is the relative bound of B with respect to Σ.

The fundamental tool for controlling perturbations is the second resolvent identity. With $A = zI - \Sigma$ and $B = zI - \Sigma - \delta\Sigma$, the algebraic identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ and $B - A = -\delta\Sigma$ give

$$(zI - \Sigma)^{-1} - (zI - \Sigma - \delta\Sigma)^{-1} = -(zI - \Sigma)^{-1}\,\delta\Sigma\,(zI - \Sigma - \delta\Sigma)^{-1}. \tag{A.2}$$

Solving (A.2) for the perturbed resolvent gives the Neumann series $(zI - \Sigma - \delta\Sigma)^{-1} = (zI - \Sigma)^{-1} \sum_{n \geq 0} \big(\delta\Sigma (zI - \Sigma)^{-1}\big)^n$, which converges uniformly on the contour under the condition

$$\sup_{z \in \Gamma_k} \big\| \delta\Sigma\,(zI - \Sigma)^{-1} \big\| < 1. \tag{A.3}$$

Because Γ_k runs through the gap, (A.3) asks that the perturbation be small compared with the gap, not merely with the distance to far-away spectrum. For a relatively bounded $\delta\Sigma$ it holds whenever $a \sup_{z \in \Gamma_k, \lambda \in \sigma(\Sigma)} |\lambda/(z - \lambda)| + b/\operatorname{dist}(\Gamma_k, \sigma(\Sigma)) < 1$. Under (A.3) the perturbed resolvent exists on Γ_k, the spectral gap persists, and Π_k depends analytically on the perturbation. In a six-level test with $\sup_\Gamma \|\delta\Sigma (zI - \Sigma)^{-1}\| = 0.44$, the perturbed projector computed by contour quadrature has trace 2.00000000; the identity (A.2) holds to 2×10^{-16}, and the same identity with a plus sign fails by 0.26.

REMARK A.1 (Rank is preserved). That Γ_k continues to enclose the same number of eigenvalues follows because $\operatorname{Tr} \Pi_k$ is continuous along the perturbation and integer-valued, hence locally constant. The

projector itself is bounded whatever Σ is; no domain question arises for it.

PROPOSITION A.2 *(Gap bound, unbounded case). Let $\delta\Sigma$ be self-adjoint and relatively bounded with constants a, b. Then*

$$\|V\|_{\mathrm{HS}} \le \frac{\sqrt{2k}\,\big(a \max_{m\in\mathcal{I}} |\lambda_m| + b\big)}{\Delta}. \tag{A.4}$$

PROOF. $\operatorname{Ran}\Pi_k$ is spanned by eigenvectors with eigenvalues in a compact set, so $\|\delta\Sigma\,\Pi_k\psi\| \le a\|\Sigma\Pi_k\psi\| + b\|\Pi_k\psi\| \le (a\max_{m\in\mathcal{I}}|\lambda_m| + b)\|\psi\|$. Since $\delta\Sigma$ is self-adjoint, $\|\Pi_k\delta\Sigma(I-\Pi_k)\| = \|(I-\Pi_k)\delta\Sigma\,\Pi_k\| \le \|\delta\Sigma\,\Pi_k\|$. Now argue as in Proposition 3.5. ∎

On 500 random instances with $\delta\Sigma = a\Sigma + bB$, $\|B\| \le 1$, the left side of (A.4) never exceeded 0.39 of the right. Relative boundedness alone therefore suffices: the framework extends to unbounded Σ with an explicit bound. The hypothesis holds for a wide class of physically relevant perturbations, including vector potentials entering differential operators and many-body interaction terms; the standard instance is the Kato–Rellich theorem, by which the Coulomb potential is relatively bounded with respect to the Laplacian with relative bound zero [Kato 1995; Reed–Simon 1975].

B Two-Band Model: Explicit Mapping and Singular Angular Stability

Consider the canonical two-band Dirac operator parameterised by a vector $\mathbf{d} = (d_x, d_y, d_z)$,

$$\Sigma(\mathbf{d}) = \mathbf{d}\cdot\boldsymbol{\sigma},$$

with eigenvalues $\lambda_\pm = \pm|\mathbf{d}|$ and gap $\Delta = 2|\mathbf{d}|$. The rank-one projector onto the lower band is

$$\Pi_- = \tfrac{1}{2}\big(I - \hat{\mathbf{d}}\cdot\boldsymbol{\sigma}\big), \qquad \hat{\mathbf{d}} = \mathbf{d}/|\mathbf{d}|,$$

so Π_- depends on $\mathbf{d}$ only through its direction on the Bloch sphere.

Let $V = \delta_1\Pi_-$ and $W = \delta_2\Pi_-$ be generated by variations $\delta_1\mathbf{d}$, $\delta_2\mathbf{d}$, and write $\delta\mathbf{d}_\perp = \delta\mathbf{d} - (\delta\mathbf{d}\cdot\hat{\mathbf{d}})\hat{\mathbf{d}}$ for the part tangent to the sphere. The intrinsic metric and curvature are

$$g_{\Pi_-}(V,W) = \frac{\delta_1\mathbf{d}_\perp\cdot\delta_2\mathbf{d}_\perp}{4|\mathbf{d}|^2}, \qquad \Omega_{\Pi_-}(V,W) = \frac{\mathbf{d}\cdot(\delta_1\mathbf{d}\times\delta_2\mathbf{d})}{2|\mathbf{d}|^3}. \tag{B.1}$$

Both forms agree with the trace definitions (2.3) on 200 random configurations to 2×10^{-10}, with the sign convention of Section 2.

Substituting (B.1) into Theorem 6.1, the subtraction term inside the radical becomes, by $|\mathbf{A}|^2|\mathbf{B}|^2 - (\mathbf{A}\cdot\mathbf{B})^2 = |\mathbf{A}\times\mathbf{B}|^2$,

$$g_{\Pi_-}(V,V)\,g_{\Pi_-}(W,W) - g_{\Pi_-}(V,W)^2 = \frac{|\delta_1\mathbf{d}_\perp \times \delta_2\mathbf{d}_\perp|^2}{16|\mathbf{d}|^4},$$

so that $2\sqrt{gg - g^2} = |\delta_1\mathbf{d}_\perp \times \delta_2\mathbf{d}_\perp|/(2|\mathbf{d}|^2)$. Because $\mathbf{d}\cdot(\delta_1\mathbf{d}\times\delta_2\mathbf{d}) = |\mathbf{d}|\,\hat{\mathbf{d}}\cdot(\delta_1\mathbf{d}_\perp\times\delta_2\mathbf{d}_\perp)$ and the cross product of two tangent vectors is normal to the sphere, the curvature has exactly the same modulus:

$$|\Omega_{\Pi_-}(V,W)| = 2\sqrt{g_{\Pi_-}(V,V)\,g_{\Pi_-}(W,W) - g_{\Pi_-}(V,W)^2}. \tag{B.2}$$

The bound is saturated for every pair of directions (to 3×10^{-11} on 200 random pairs). This saturation is forced: with $k = 1$ and a one-dimensional complement, the blocks X and Y of (2.1) are complex numbers, $|Q_\Pi(V,W)|^2 = |X|^2|Y|^2 = Q_\Pi(V,V)Q_\Pi(W,W)$, and the Gram determinant in the proof of Theorem 6.1 vanishes identically. Every two-band pair is an extremal partner in the sense of Corollary 6.2.

This extremal case has been realised on a device. Gianfrate et al. measured the polarisation pseudospin of an exciton-polariton microcavity over the whole (k_H, k_V) plane and computed the Berry curvature and all three metric components from its two angles [Gianfrate et al. 2020; Bleu et al. 2017]. Their maps are a direct measurement of (B.1) and (B.3) on a two-band system. They cannot, however, test the strict inequality: any curvature and metric computed from a unit pseudospin through its two angles satisfy $4\det g = \Omega^2$ at every point, whatever the field (to 10^{-16} on an arbitrary smooth field with no Hamiltonian behind it). A device test of the strict inequality needs a measured tensor on three or more levels, where generic bands fall inside the bound (Appendix C). Coupled spins measured through their Rabi response, as in the nitrogen-vacancy experiment [Yu et al. 2020], are a setting that can supply one.

Introducing the angle θ between the tangent variations, $|\delta_1\mathbf{d}_\perp \times \delta_2\mathbf{d}_\perp| = |\delta_1\mathbf{d}_\perp||\delta_2\mathbf{d}_\perp|\sin\theta$, exposes the directional safeguard. As the node $|\mathbf{d}| \to 0$ is approached, the metric components grow as $|\mathbf{d}|^{-2}$ and the curvature grows at the same rate, so the curvature per unit of metric stays fixed by $\sin\theta$: control is relative, as in Section 6.1. For collinear variations ($\theta = 0$) the curvature vanishes identically however close the node is.

In spherical coordinates (θ, ϕ) on the Bloch sphere the components reduce to the monopole values

$$g_{\theta\theta} = \tfrac{1}{4}, \qquad g_{\phi\phi} = \tfrac{1}{4}\sin^2\theta, \qquad \Omega_{\theta\phi} = \tfrac{1}{2}\sin\theta, \tag{B.3}$$

and the first Chern number of the lower band is $\frac{1}{2\pi}\int_{S^2}\Omega_{\theta\phi}\,d\theta\,d\phi = 1$ (quadrature: 1.0000026). The coordinate-free construction therefore recovers the conventional parametric geometry and its topological integer exactly.

C Three-Band Model: Flat-Band Projections and Particle–Hole Duality

Consider the spin-1 operator

$$\Sigma(\mathbf{d}) = \mathbf{d}\cdot\mathbf{S},$$

with eigenvalues $\lambda_\pm = \pm|\mathbf{d}|$ and a flat middle band $\lambda_0 = 0$. For the rank-one projector Π_1 onto the lowest band and the rank-two projector Π_2 onto the lowest two bands, explicit computation gives identical intrinsic metrics:

$$g_{\Pi_1}(V,V) = g_{\Pi_2}(V,V) = \frac{|\delta\mathbf{d}_\perp|^2}{2|\mathbf{d}|^2}. \tag{C.1}$$

The coefficient $\frac{1}{2}$ holds on 100 random configurations to 6×10^{-9} for both ranks.

This establishes an exact particle–hole duality on the Grassmannian, and its mechanism has two parts. The first holds for every projector: $g_\Pi = g_{I-\Pi}$, because $-V$ is the tangent vector of $I-\Pi$ and the metric is quadratic (Section 2; on 100 random rank-two projectors of size six with no flat band, the two agree to 10^{-12}). Hence $g_{\Pi_2} = g_{\Pi_+}$, the metric of the top band. The second part is a property of the spin-1 spectrum: $\Pi_+(\mathbf{d}) = \Pi_1(-\mathbf{d})$, so the top and bottom bands carry the same geometry. Together they give (C.1). The second part is what the flat, symmetric middle band supplies: adding a term $0.8\,S_z^2$, which breaks the $\pm|\mathbf{d}|$ symmetry, separates the two metrics (their ratio ranges from 0.11 to 11 over 50 configurations), while $g_\Pi = g_{I-\Pi}$ continues to hold exactly.

The lowest spin-1 band also saturates Theorem 6.1, exactly as the two-band model does (to 2×10^{-12} on 200 random pairs). Saturation is not a two-band accident, but it is not generic either: for a three-band family $\Sigma = H_0 + p_1H_1 + p_2H_2$ with random Hermitian H_i, the lowest band falls strictly inside the bound, with $|\Omega|$ as small as one five-hundredth of the bound (Figure 21). The reason is holomorphy. The lowest band of $\mathbf{d}\cdot\mathbf{S}$ is a spin coherent state: with $z = \tan(\theta/2)e^{i\phi}$ and $w = -1/\bar{z}$, it

is proportional to $(1, \sqrt{2}\,w, w^2)$ (to 6×10^{-16} on 200 random directions), the Veronese embedding of the sphere into $\mathbb{CP}^2$. A band that depends on one complex coordinate of the parameter sphere in this way pulls the Fubini–Study form back to a Kähler form, and Kähler bands meet the determinant inequality with equality [Mera–Ozawa 2021]. A generic three-band family has no such coordinate, and falls strictly inside.

The same holds for every spin. The lowest band of $\mathbf{d} \cdot \mathbf{S}$ in the spin-j representation is the spin-j coherent state, proportional to $\big(\binom{2j}{r}^{1/2} w^r\big)_{r=0}^{2j}$, the degree-$2j$ Veronese embedding [Perelomov 1986]. Against the round metric h of the parameter sphere it has $g = \frac{j}{2} h$ and $|\Omega_{\theta\phi}| = j \sin\theta$, so it meets Theorem 6.1 and the trace condition (6.4) with equality at every point, and its Chern number is $2j$. For $j = \frac{1}{2}, 1, \frac{3}{2}, 2, 3, 4$ the computed metric is $\frac{j}{2}$ to 10^{-6}, both equalities hold to 3×10^{-14}, and the Chern number is $2j$ to 10^{-3}; adding $0.8\, S_z^2$ lifts the trace condition strictly for every $j \geq 1$, by 0.03 to 0.27. These bands are physical: the spin-j representation is the one-particle lowest Landau level on a sphere carrying $2j$ flux quanta [Haldane 1983], so the lowest Landau level is the extremal case of the curvature bound in every flux.

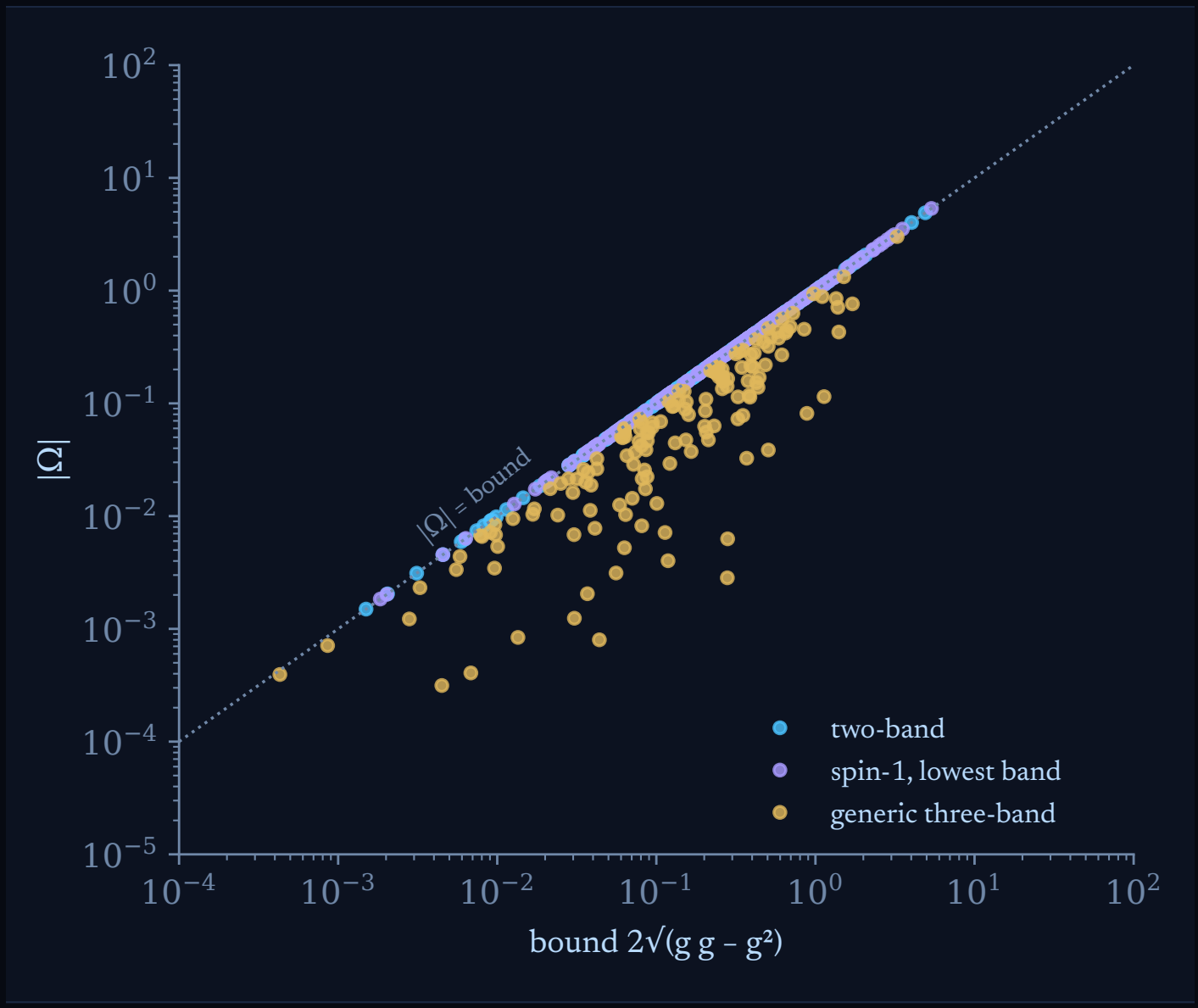


FIGURE 21. Curvature against its bound, for random pairs of directions. Two-band systems and the lowest spin-1 band sit on the line: every pair saturates. The lowest band of a generic three-band family falls inside it.

D Numerical Benchmarking on Lattice Models and Krylov Methods

To validate the coordinate-free framework on lattice systems, the geometric construction is implemented for a two-dimensional tight-binding Hamiltonian $H_L(\mathbf{a})$ with Peierls phases controlled by parameters $\mathbf{a}$. The spectral projector onto the lowest k bands is computed by trapezoidal quadrature of the Riesz integral over a contour Γ_k in the gap,

$$\Pi_k \approx \frac{1}{2\pi i} \sum_j w_j \, (z_j I - H_L)^{-1}, \tag{D.1}$$

which converges exponentially in the number of nodes on a circle that stays away from the spectrum. The tangent vectors are obtained by differentiating the Riesz resolvent, $V_i = \partial_{a_i} \Pi_k$, and the gauge-invariant metric and curvature markers are

$$g_{ij} = \tfrac{1}{2}\,\mathrm{Tr}(V_i V_j), \qquad \Omega_{ij} = i\,\mathrm{Tr}\big(\Pi_k [V_i, V_j]\big), \tag{D.2}$$

with the sign of Section 2. Along a finite-time protocol the excess work is the Riemann sum $W_{\mathrm{ex}} \approx \sum_m \Delta t\, \zeta(\dot{\mathbf{a}}_m)$ of the friction functional of Section 5.3, which by Theorem 5.3 is bounded below by the band-weighted intrinsic metric.

For large sparse lattices, full diagonalisation and dense inversion are bypassed by a Krylov iteration. Let $|\psi_{m,i}\rangle = \partial_{a_i} H_L |m\rangle$ for each occupied state $|m\rangle$, $m \in \mathcal{I}$. The regularised cross-gap response vector solves the shifted system

$$\big[(H_L - \lambda_m I)^2 + \eta^2 I\big]\, |X^{\eta}_{m,i}\rangle = (I - \Pi_k)|\psi_{m,i}\rangle. \tag{D.3}$$

The operator on the left is Hermitian and positive definite, so conjugate gradients (or MINRES) is the solver to use. The same solutions give both halves of the tensor:

$$g^{\eta}_{ij} = \sum_{m \in \mathcal{I}} \mathrm{Re}\,\langle \psi_{m,i} | X^{\eta}_{m,j} \rangle, \qquad \Omega^{\eta}_{ij} = -2 \sum_{m \in \mathcal{I}} \mathrm{Im}\,\langle \psi_{m,i} | X^{\eta}_{m,j} \rangle, \tag{D.4}$$

which reduce to the spectral sums (3.3)–(3.4) as $\eta \to 0$. The error is quadratic in the regulator. On a 14×14 lattice (196 sites, $k = 3$) the relative error against the exact pair sums is 8.3×10^{-4}, 8.3×10^{-6} and 8.3×10^{-12} for the metric, and 1.4×10^{-3}, 1.4×10^{-5} and 1.4×10^{-11} for the curvature, at $\eta = 10^{-2}, 10^{-3}, 10^{-6}$ (Figure 22), with 153 conjugate-gradient iterations per solve. The cost is k solves per parameter direction, each of order (non-zeros)$\times$(iterations): linear in the occupied rank k, and still growing with the lattice through the number of non-zeros. Higher-order tensors such as the quantum metric dipole are obtained either by differentiating g^{η}_{ij} along $\mathbf{a}$ or from the exact second-order formula (7.3).

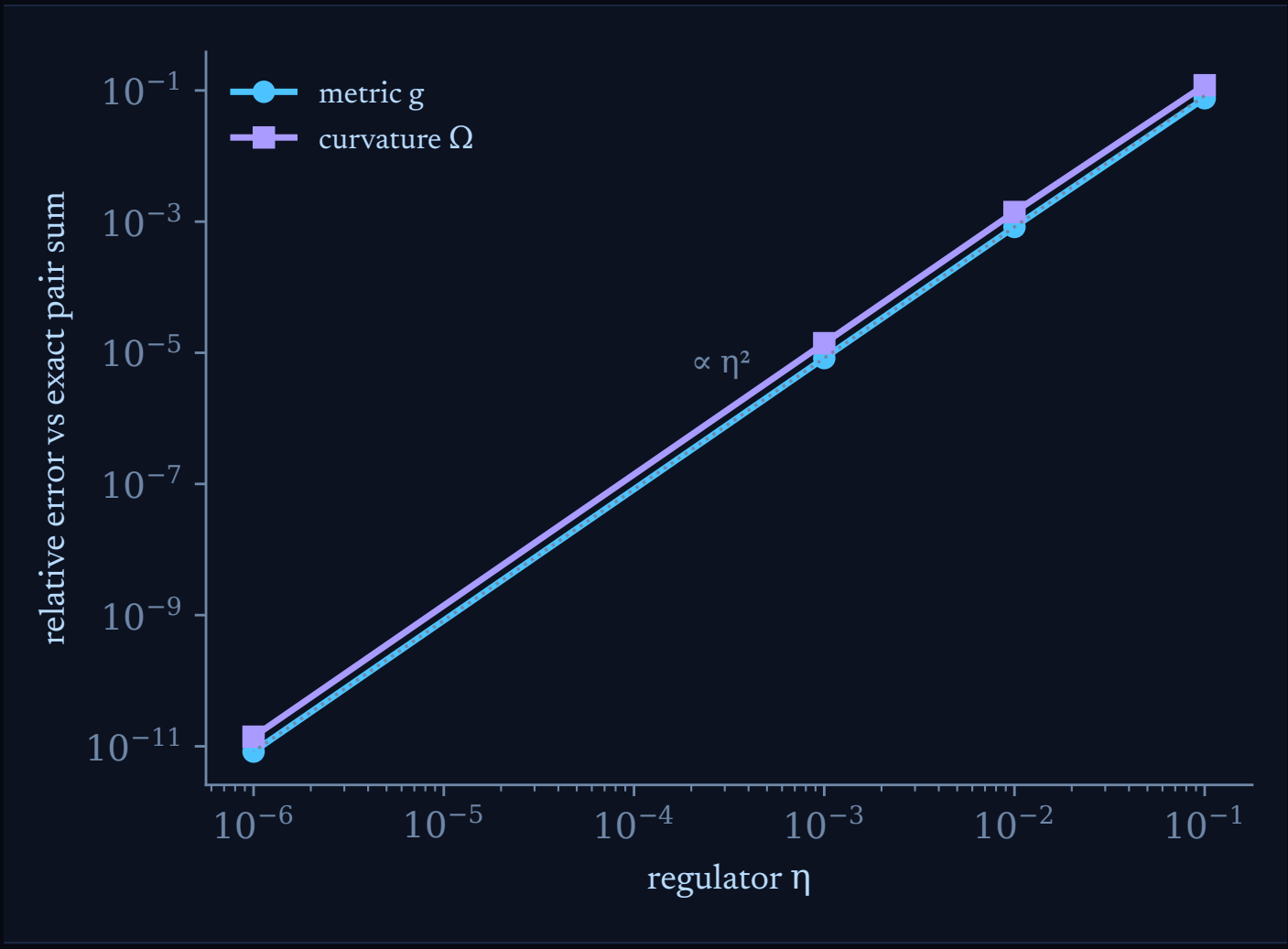


FIGURE 22. Krylov evaluation of the regularised metric and curvature on a 196-site lattice, against the exact cross-gap pair sums. The error falls as the square of the regulator.

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